CPE334 SOFTWARE ENGINEERING

LAB 3 REPORT

TokTickIT Users, Roles, IT Staff Ticketing, and Admin Screens

Student
Kirakit Kingkaew

Student ID
67070503460

Section
131

Repository
https://github.com/songt888/toktickit

Final main commit
f6dc6f2

Earlier report/screenshot snapshot
b2ed779 (after PR #63; not its merge commit)

Answer Part 1

Git Use with Engineering Workflow

The repository records feature delivery into lab3-staging and release integration into main. PR #63 was approved and merged on 2 October 2026 at 14:37 Asia/Bangkok; final main is f6dc6f2. The later b2ed779 snapshot contains report/screenshot artifacts and was not merged by PR #63. Relevant application, test and documentation files match between those snapshots. Historical screenshots keep their original commit labels.

Links

Repository: https://github.com/songt888/toktickit

GitHub Project: https://github.com/users/songt888/projects/2

Reviewer record: https://github.com/songt888/toktickit/blob/e3bf402/docs/lab-03/reviewer.md

Release PR: https://github.com/songt888/toktickit/pull/58

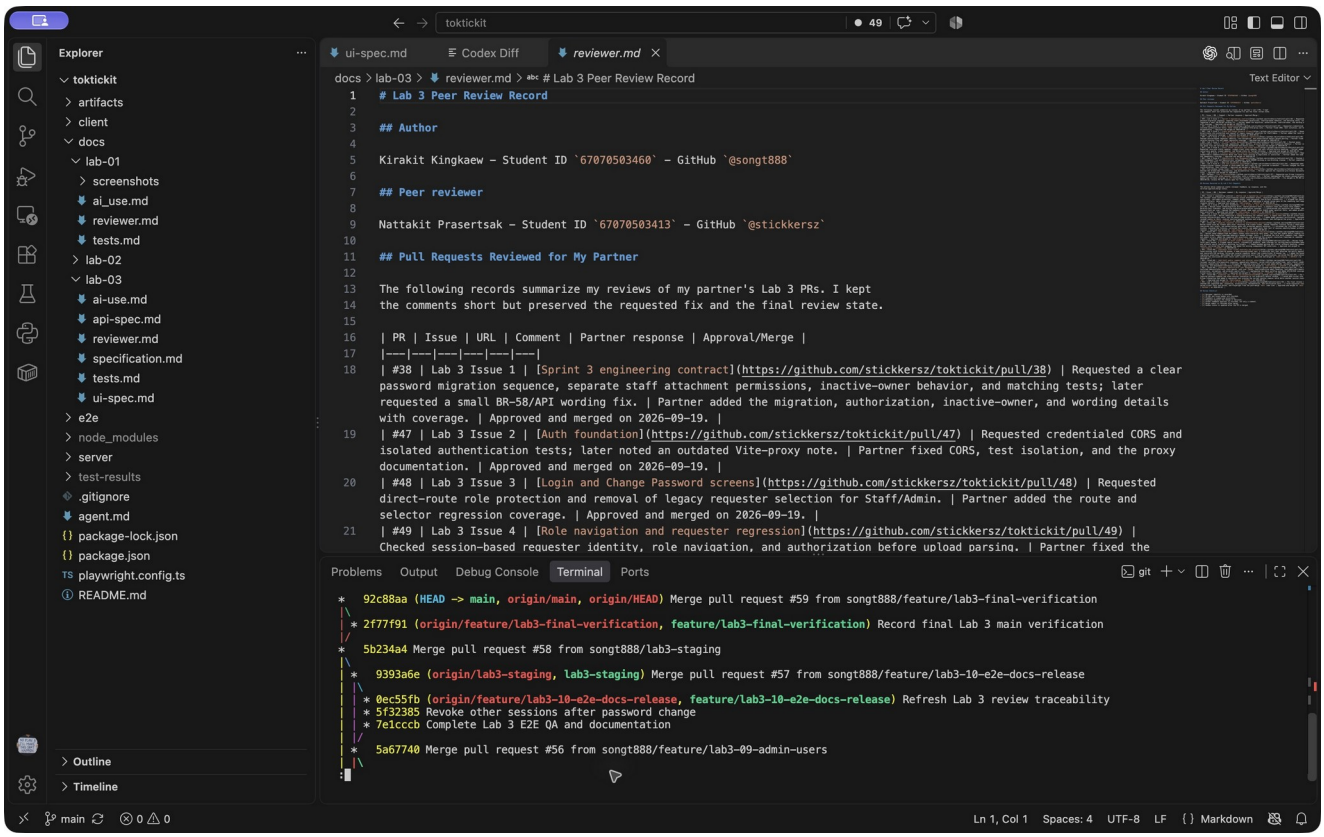
Final verification PR: https://github.com/songt888/toktickit/pull/63

Staff Download fix PR: https://github.com/songt888/toktickit/pull/60

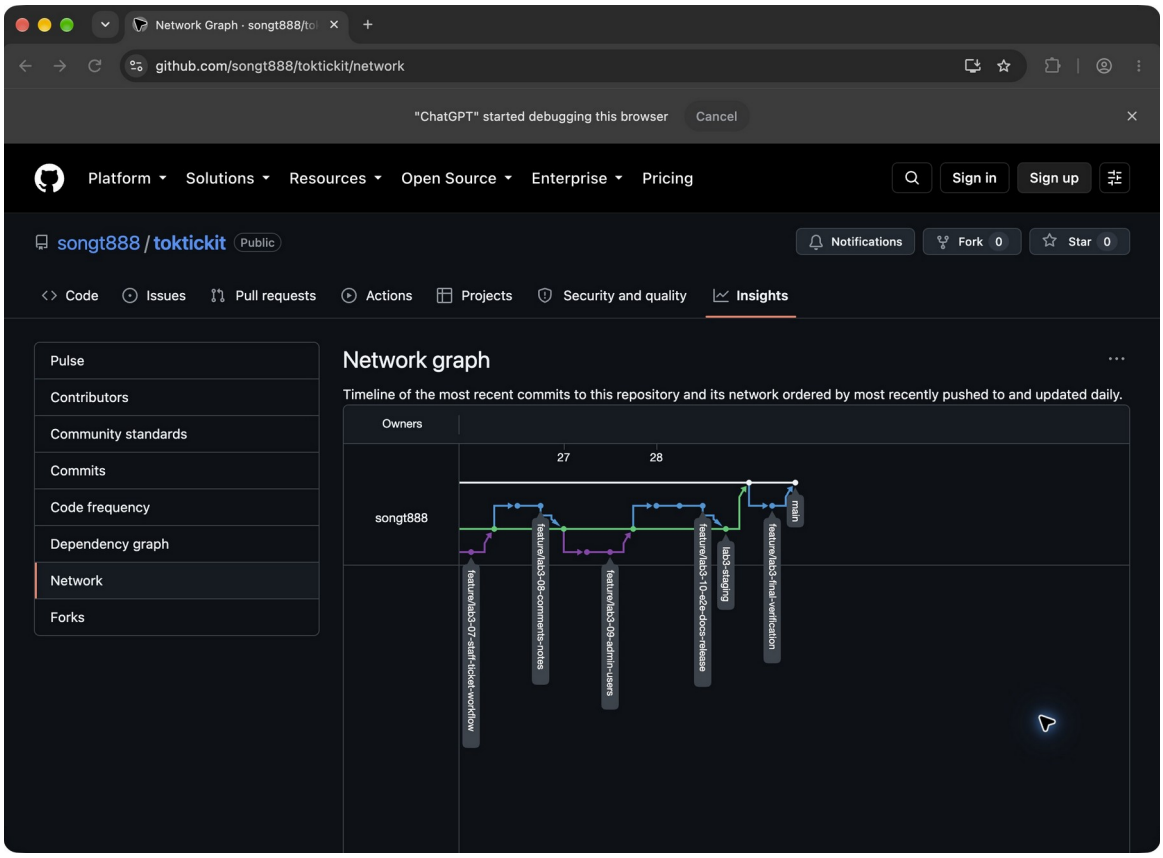
README: https://github.com/songt888/toktickit/blob/f6dc6f2/README.md

.gitignore: https://github.com/songt888/toktickit/blob/f6dc6f2/.gitignore

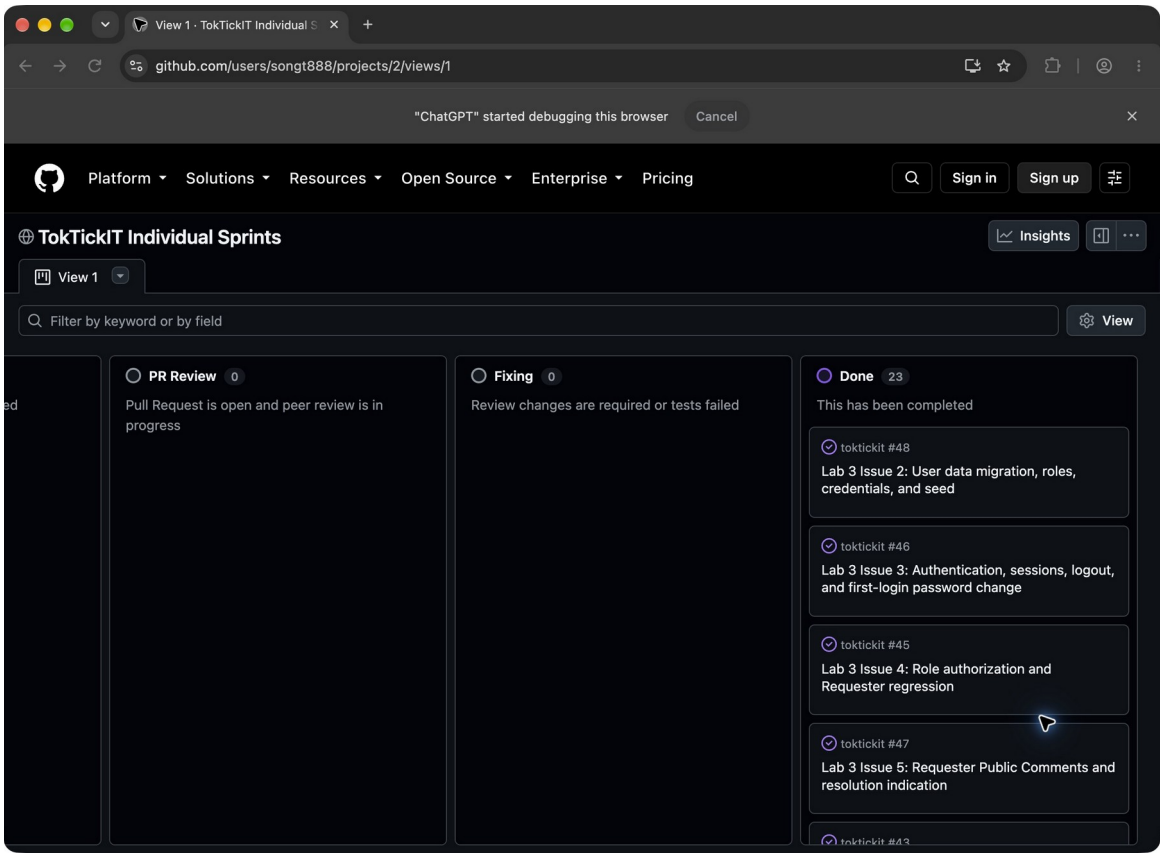
Screenshots



feature branches → lab3-staging → main, with branch and merge names readable



GitHub Network graph showing the Sprint 3 branch flow



public Lab 3 Project with all Sprint 3 Issues in Done

The following native VS Code Markdown Preview captures show every row of both reviewer.md tables. Each group shows the left columns followed by the corresponding response and approval/merge columns. Partner PR numbers refer to stickkersz/toktickit; reviews received refer to songt888/toktickit.

Lab 3 Peer Review Record

Author

Kirakit Kingkaew - Student ID `67070503460` - GitHub `@songt888`

Peer reviewer

Nattakit Prasertsak - Student ID `67070503413` - GitHub `@stickkersz`

reviewer.md — author and peer reviewer identity

PR	Issue	URL	Comment
#38	Lab 3 Issue 1	Sprint 3 engineering contract	Requested a clear password migration sequence, separate staff attachment permissions, inactive-owner behavior, and matching tests; later requested a small BR-58/API wording fix.
#47	Lab 3 Issue 2	Auth foundation	Requested credentialed CORS and isolated authentication tests; later noted an outdated Vite-proxy note.
#48	Lab 3 Issue 3	Login and Change Password screens	Requested direct-route role protection and removal of legacy requester selection for Staff/Admin.
#49	Lab 3 Issue 4	Role navigation and requester regression	Checked session-based requester identity, role navigation, and authorization before upload parsing.
#50	Lab 3 Issue 5	IT Staff Ticket Queue	Checked queue authorization, filters, sorting, pagination, owner markers, priority backfill, and responsive behavior.
#51	Lab 3 Issue 6	Staff Ticket operations and Ticket Detail	Requested race-safe status updates, scoped client state updates, and safe refresh-failure handling.

Partner reviews: #38, #47, #48, #49, #50, #51 — PR, issue, link and comment

Partner response	Approval/Merge
Partner added the migration, authorization, inactive-owner, and wording details with coverage.	Approved and merged on 2026-09-19.
Partner fixed CORS, test isolation, and the proxy documentation.	Approved and merged on 2026-09-19.
Partner added the route and selector regression coverage.	Approved and merged on 2026-09-19.
Partner fixed the selector/session flow and added regression coverage.	Approved and merged on 2026-09-19.
Partner completed the queue flow and verified the documented test/build counts.	Approved and merged on 2026-09-19.
Partner added the concurrency guards, scoped updates, and Reload state for failed refreshes.	Approved and merged on 2026-09-19.

Partner reviews: #38, #47, #48, #49, #50, #51 — response and approval/merge

#52	Lab 3 Issue 7	Public Comments and Internal Notes	Requested seeded Public Comments/Internal Notes and rerun tests proving no duplicates or overwrites.
#53	Lab 3 Issue 8	Administrator User Management	Checked the user-management flow and Administrator safeguards; noted README wording as non-blocking cleanup.
#54	Lab 3 Issue 9	E2E and responsive pass	Requested name-based category/system lookups instead of hard-coded IDs and a wait for the resolved screenshot.
#56	Pre-release review fixes	Pre-release code review fixes	Checked the breadcrumb, traceability, and documentation fixes.
#57	Release	Lab 3 release	Requested atomic password/session writes and rollback coverage; the next pass requested reviewer.md and ai-use.md.
#59	Issue #58: atomic password change	Atomic password and session writes	Rechecked e5507ab: password update and session revocation share one transaction; API-06 proves rollback. Build and 330 server tests passed locally.

Partner reviews: #52, #53, #54, #56, #57, #59 — PR, issue, link and comment

Partner added the seed examples and idempotency coverage.	Approved and merged on 2026-09-24.
Partner completed the Admin flow and tests.	Approved and merged on 2026-09-24.
Partner changed the lookups and synchronization, then retested.	Approved and merged on 2026-09-27.
Partner applied the requested pre-release documentation fixes.	Approved and merged on 2026-09-27.
Partner merged the transaction fix in PR #59 and required documents in PR #61.	Approved at c7d40de; merged to main (3179394) on 2026-09-28.
Partner made both writes atomic and added the safe-failure/retry regression requested on release PR #57.	Approved by @songt888 at e5507ab on 2026-09-28; merged (122ed70) the same day.

Partner reviews: #52, #53, #54, #56, #57, #59 — response and approval/merge

#61	Release documentation	Reviewer and AI-use records	Requested accurate wording about the docs-only commit after approval on partner PR #57 and fix commits only when changes were requested.
#63	Issue #62: final-main record	Record final main verification	Rechecked 9c7d6da: documentation satisfies Issue #62. Server 330/330 and client 253/253 passed locally; Playwright 19/19 was recorded in the PR.
#65	Issue #64: README update	Bring README up to date for Lab 3	Rechecked 041934d: README covers Lab 3, correct docs/ADRs/CONTEXT and test/screenshot paths, seeded-account wording, and punctuation cleanup.
#66	Post-release review	Post-release corrections	Requested clearing stale Close/Cancel confirmations after a 409 refresh and clearing old password-change success before a new attempt.

Partner reviews: #61, #63, #65, #66 — PR, issue, link and comment

Partner corrected both statements in 9007da4.	Approved at 9007da4; merged (c7d40de) on 2026-09-28.
Partner recorded results from main at 3179394 and completed the reviewer entries for #57/#61; no code or test changes.	Approved by @songt888 at 9c7d6da on 2026-09-28; merged (18c4ec6) the same day.
Partner updated README only; no code or test changes. The documentation-only diff had no blockers.	Approved by @songt888 at 041934d on 2026-09-28; merged (bc40edd) the same day.
Partner fixed both states in e97ca59 and added UI-40/UI-41 regressions.	Approved at e97ca59; merged (ee4cb2d) on 2026-10-01.

Partner reviews: #61, #63, #65, #66 — response and approval/merge

PR	Issue	URL	Reviewer comment
#38	Issue 1: engineering contract	Define Lab 3 engineering contract	The reviewer found contract inconsistencies around attachment access, resolution fields, safe errors, logout, optimistic concurrency, last-admin protection, comment safety, seed passwords, and AC/test traceability.
#49	Lab 3 Issue 2: user data foundation	Lab 3 Issue 2: User data foundation	Review found schema/migration drift, shared password salts, a sequence reset that could reuse numbers, parallel-test flakiness, and missing migration-preservation coverage.
#50	Lab 3 Issue 3: authentication	Implement Lab 3 Issue 3 authentication and sessions	Review identified a timing difference for unknown users, a logout test that did not prove revocation, missing session/origin cases, and the absent same-origin Vite proxy.

Reviews received: #38, #49, #50 — PR, issue, link and comment

My response	Approval/Merge
I aligned the specification, API/UI contracts, data rules, and traceability table, then addressed a second review pass on the remaining test mappings and response details.	Approved and merged to `lab3-staging` (`b6a5f87`) on 2026-09-19.
I synchronized the migration and schema, generated a password hash per user, removed the sequence rewind, made seed reruns stable under parallel tests, and added preservation/hash assertions.	Approved and merged to `lab3-staging` (`401a944`) on 2026-09-19.
I added dummy password verification, replayed the revoked cookie in tests, covered inactive/expired sessions and origin checks, and configured the proxy.	Approved and merged to `lab3-staging` (`4fffd75`) on 2026-09-19.

Reviews received: #38, #49, #50 — response and approval/merge

#51	Issue #45	Migrate requester workflow to authenticated sessions	Review found stale My Tickets data after returning from Create Ticket, shared requester fixtures racing a seed-count test, inaccurate test totals, and authorization paths not protected against mutation.
#52	Issue #47	Add requester public comments and resolution indication	Review found comment-load and submit states could overwrite each other, the form was usable before loading finished, and author/order/length/timestamp behaviors needed stronger tests.
#53	Issue #43	Implement IT Staff ticket queue	Review found missing int32 query bounds, a clipped tablet control, traceability mismatch, weak coverage for sorting/search/forbidden behavior, and wildcard search characters matching too broadly.

Reviews received: #51, #52, #53 — PR, issue, link and comment

I refreshed the list when it becomes visible, isolated the fixtures, corrected the results, and added tests that fail if session identity/header protections are removed.	Approved and merged to `lab3-staging` (`eb13c09`) on 2026-09-19.
I disabled the form until comments load, separated load and submit errors, added the requested API assertions, and preserved the original resolution timestamp on repeated requests.	Approved and merged to `lab3-staging` (`717c8f0`) on 2026-09-19.
I added bounded parsing and literal wildcard handling, widened the control, corrected the test mapping, and added the missing independent API assertions.	Approved and merged to `lab3-staging` (`c25168b`) on 2026-09-24.

Reviews received: #51, #52, #53 — response and approval/merge

#54	Issue #44	Implement Staff ticket ownership and workflow	Review found missing int32 ticket-ID bounds, a weak missing-ticket test, a transition test that asserted the matrix against itself, and inaccurate E2E wording; follow-ups covered requester detail and create-ticket ID bounds too.
#55	Issue #41	Add Staff public comments and internal notes	The reviewer verified role/ownership responses, append-only behavior, server-controlled author/time, safe literal rendering, and distinct comment/note styling.
#56	Issue #42	Implement Administrator user management	The reviewer confirmed Administrator-only route guards, safe user fields, case-insensitive email handling, last-admin/self-deactivation protections, validation, and wildcard search tests.

Reviews received: #54, #55, #56 — PR, issue, link and comment

I added the bounds and regression assertions, hard-coded the allowed transition pairs from the specification, corrected the E2E description, and verified the oversized-ID cases return safe client errors.	Approved and merged to `lab3-staging` (`d9578c6`) on 2026-09-24.
I followed the approved plain-text policy and added API/UI, validation, append-only, ownership, and attachment-continuity coverage.	Approved and merged to `lab3-staging` (`b75ef06`) on 2026-09-27.
I implemented the Admin workflow and added API/UI/E2E coverage for the listed access and safety rules.	Approved and merged to `lab3-staging` (`5a67740`) on 2026-09-27.

Reviews received: #54, #55, #56 — response and approval/merge

#57	Issue #40	Complete Lab 3 E2E QA and documentation	The reviewer required password updates and other-session revocation to run atomically before release.
#58	Lab 3 release	Release Lab 3 to main	The final release review checked the completed E2E, responsive, accessibility, documentation, and verification work.
#60	Staff attachment download	Fix Staff active attachment downloads	Reviewer checked authenticated download authorization, removed-file hiding, concurrent-download guards, safe errors and URL cleanup.
#63	Verification and E2E configuration	Final-main verification evidence	Requested matching verification headers, configurable E2E API URLs, matching Vite proxy/server PORT, and an honest pending-rerun note.

Reviews received: #57, #58, #60, #63 — PR, issue, link and comment

I moved both writes into one Prisma transaction and added a regression test proving the current session remains valid while another session receives `401`.		Approved and merged to `lab3-staging` (`9393a6e`) on 2026-09-28.
I reran migration status, server/client tests and builds, and Playwright from the post-merge `main` code line.		Approved and merged to `main` (`5b234a4`) on 2026-09-28.
I reused downloadAttachment, added success/failure regression coverage and replaced the README password example with a placeholder.		Approved by @stickkersz; merged to main (ac8a3dd) on 2026-10-01.
I aligned the headers and E2E_API_URL across specs, health check, proxy and server; I preserved the historical patched-copy result and rerun limitation.		Approved by @stickkersz; merged to main (f6dc6f2) on 2026-10-02.

Reviews received: #57, #58, #60, #63 — response and approval/merge

#64	Report and verification records	Report corrections	Reviewer checked logs/counts and requested complete reviewer tables, corrected PR #57 status, later review rounds, durable verification wording and accurate PR-body pages/links.

Reviews received: #64 — PR, issue, link and comment

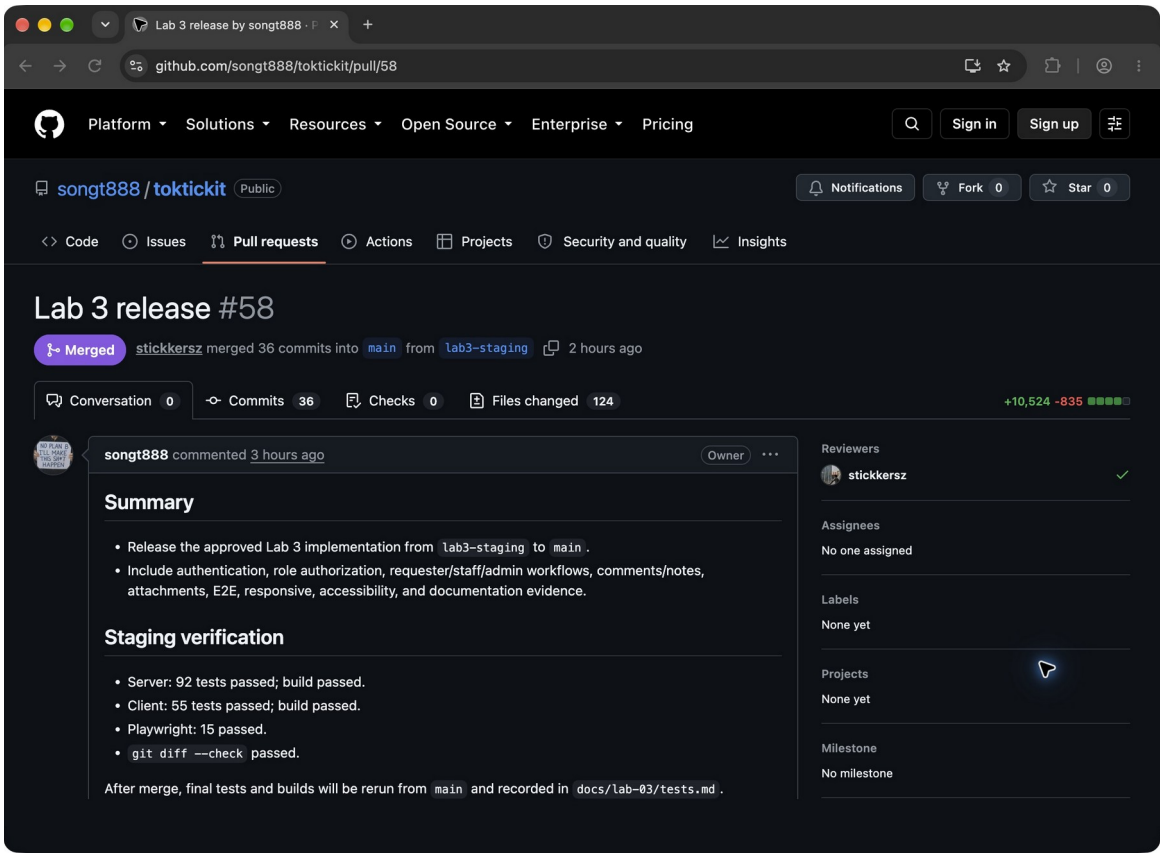
I updated these records, captured the complete Markdown Preview tables and synchronized the PDF and PR description.	Changes requested by @stickkersz on 2026-10-04 (Asia/Bangkok). Approval and merge, if later granted, are recorded on the linked PR.

Reviews received: #64 — response and approval/merge

Review Checklist

- [x] Reviewer identity is recorded.
- [x] PR URL and Issue number are recorded.
- [x] Feedback is summarized accurately.
- [x] Response explains the change or decision.
- [x] Formal teammate approval is recorded, not only a comment.
- [x] Merge commit is recorded after merge.
- [x] Kanban status is updated after the PR is merged.

reviewer.md — complete Review Checklist



release PR approval and merge into main

TokTickIT

TokTickIT is an IT service desk application with a React + TypeScript frontend and an Express + TypeScript backend. The backend uses Prisma and PostgreSQL for requester-specific tickets, categories, related systems, and attachment metadata.

Requirements

- Node.js 20 or newer
- npm
- Docker Desktop for local PostgreSQL

Project structure

```
client/   React + TypeScript + Vite frontend
server/   Express + TypeScript + Prisma backend
docs/     Lab documentation and evidence
```

Actual README Markdown Preview: requirements and repository structure

```
cd server
npm install

cd ../client
npm install
```

Start PostgreSQL with Docker. Create the container once:

```
docker run --name toktickit-postgres \
  -e POSTGRES_USER=toktickit \
  -e POSTGRES_PASSWORD=toktickit \
  -e POSTGRES_DB=toktickit \
  -p 5432:5432 \
  -d postgres:16
```

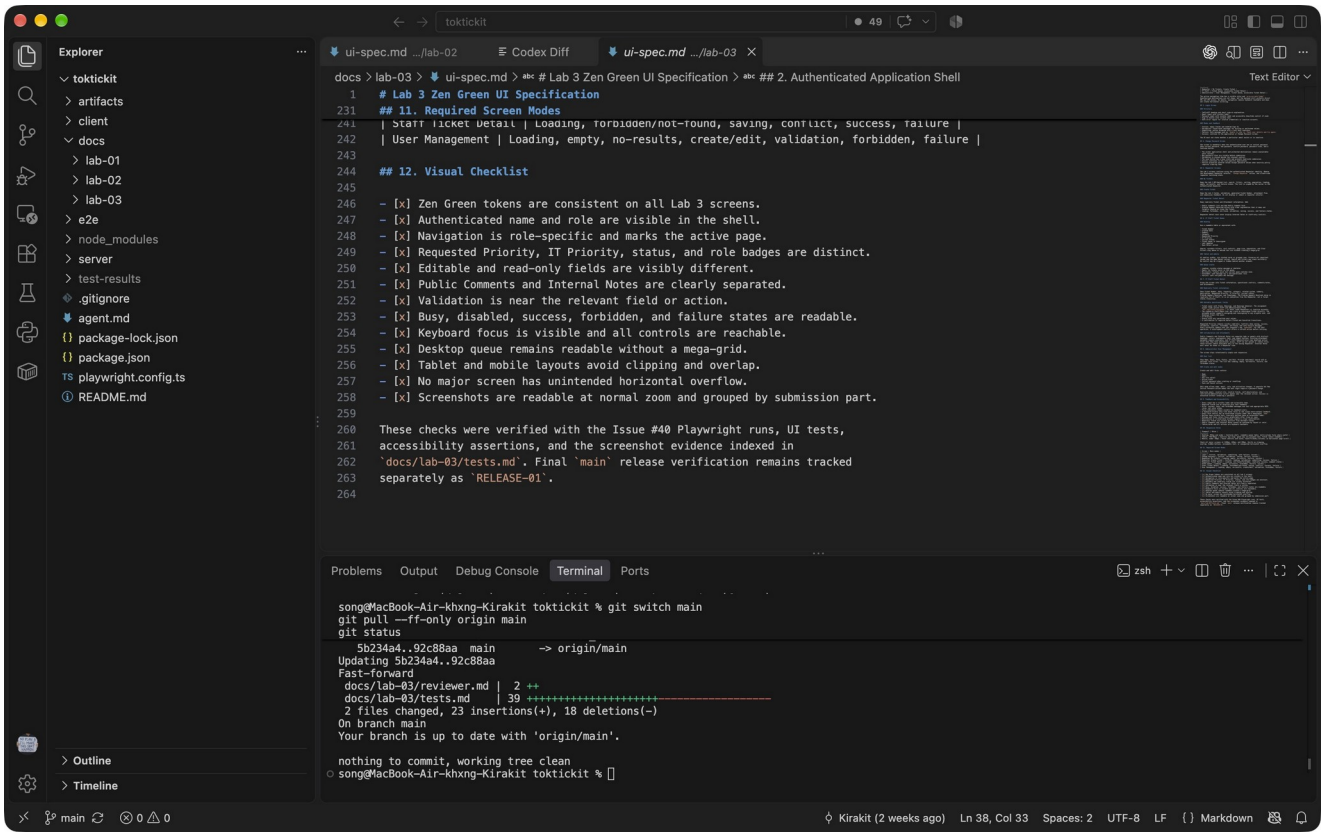
If the container already exists, start it with:

```
docker start toktickit-postgres
```

Actual README Markdown Preview: dependency installation and local PostgreSQL setup

```
# dependencies
node_modules/
# env & secrets
.env
*.env
!.env.example
# build output
dist/
build/
*.tsbuildinfo
.vite/
# TypeScript compiler output next to source files
client/src/**/*.js
client/tests/**/*.js
server/src/**/*.js
server/tests/**/*.js
server/prisma/**/*.js
# prisma
server/prisma/*.db
# runtime attachment files; keep only an empty directory marker if needed
server/uploads/*
!server/uploads/.gitkeep
# Playwright generated output; committed visual evidence lives in screenshots/.
test-results/
artifacts/lab-02/playwright-report/
# logs & OS
*.log
.DS_Store
```

Actual VS Code screenshot: all 28 .gitignore lines, including secrets, generated files, uploads and test results



VS Code screenshot showing the full visible TokTickIT repository tree and the main checkout

Answer Part 2

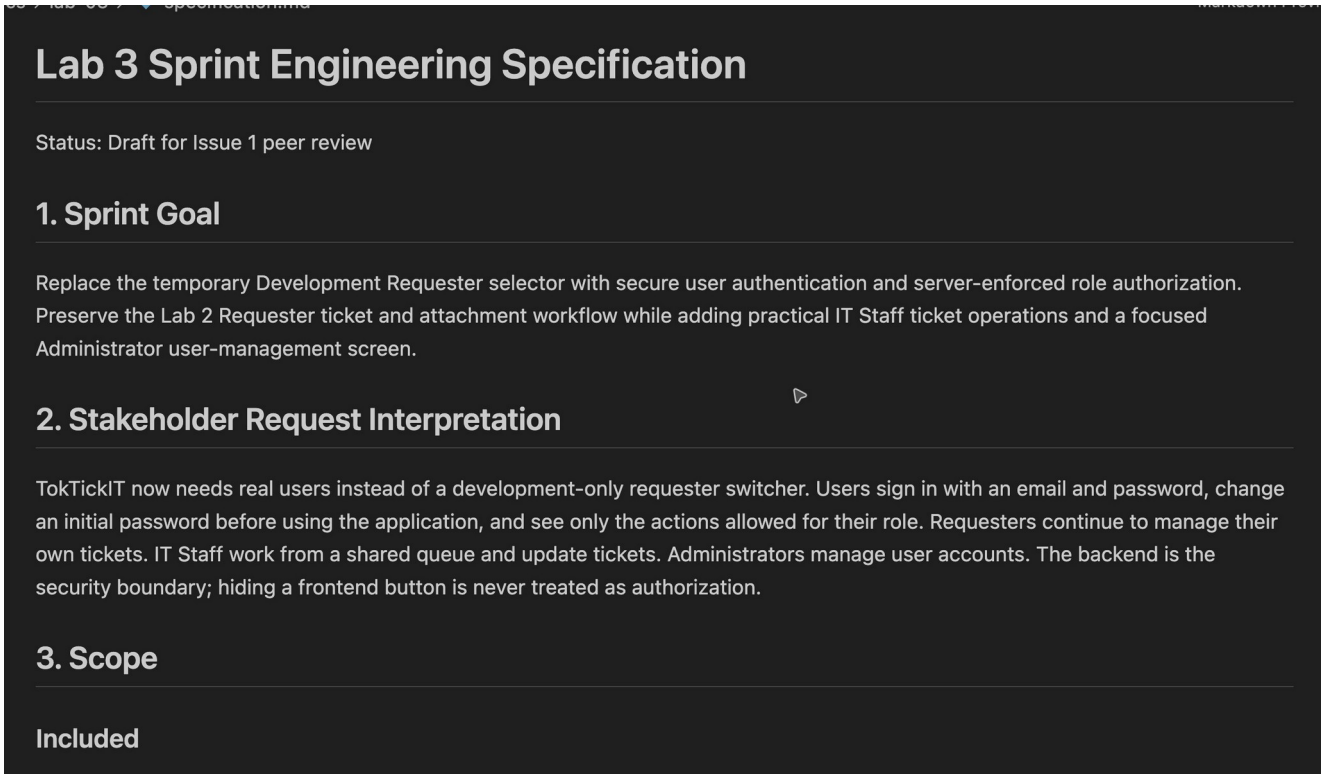
Specification-Driven Development

The Sprint 3 engineering contract defines authentication, authorization, role behavior, migration, APIs, acceptance criteria, and Product Definition of Done before the implementation branches were completed. Initial specification commit b173f40 is dated 2026-09-14.

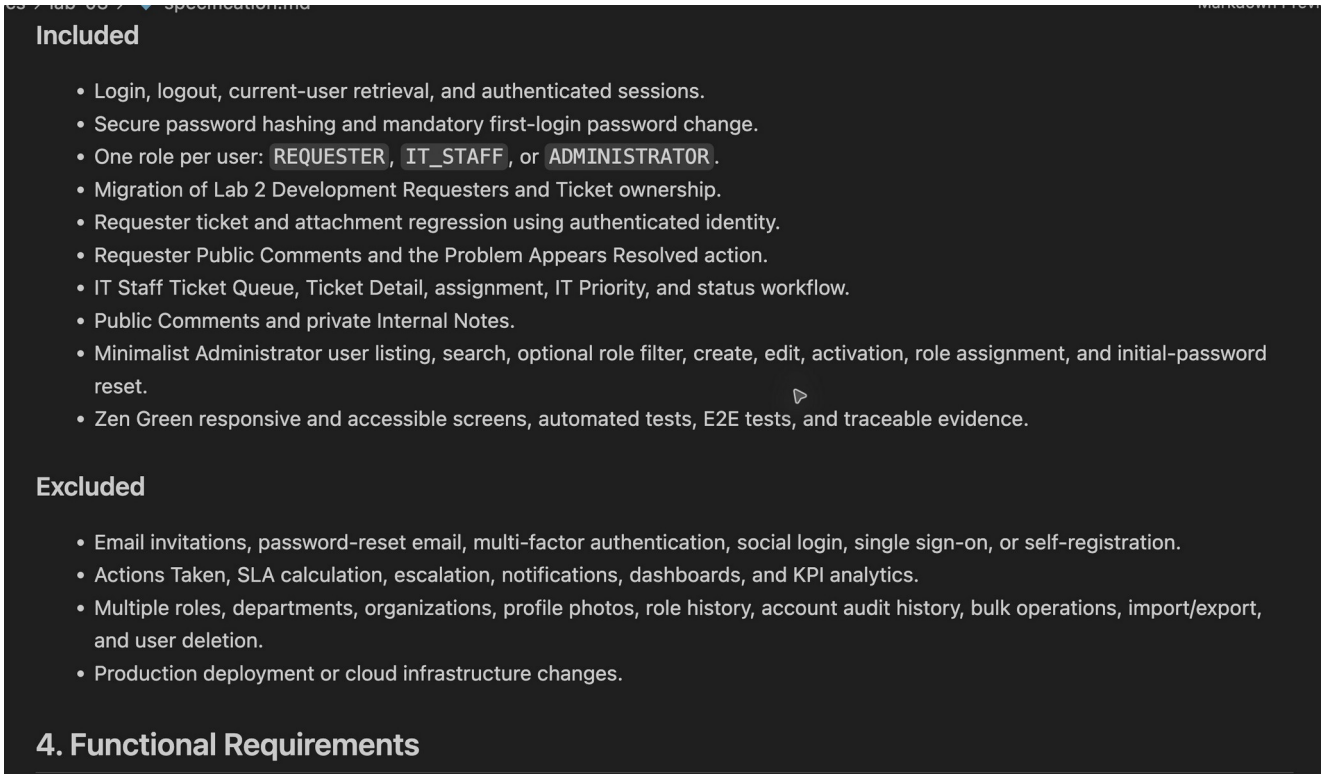
Links

Specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/specification.md
Initial specification commit: https://github.com/songt888/toktickit/commit/b173f40
Contract PR: https://github.com/songt888/toktickit/pull/38
API specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/api-spec.md

Screenshots



Actual VS Code Markdown Preview: Sprint Goal and stakeholders



Included

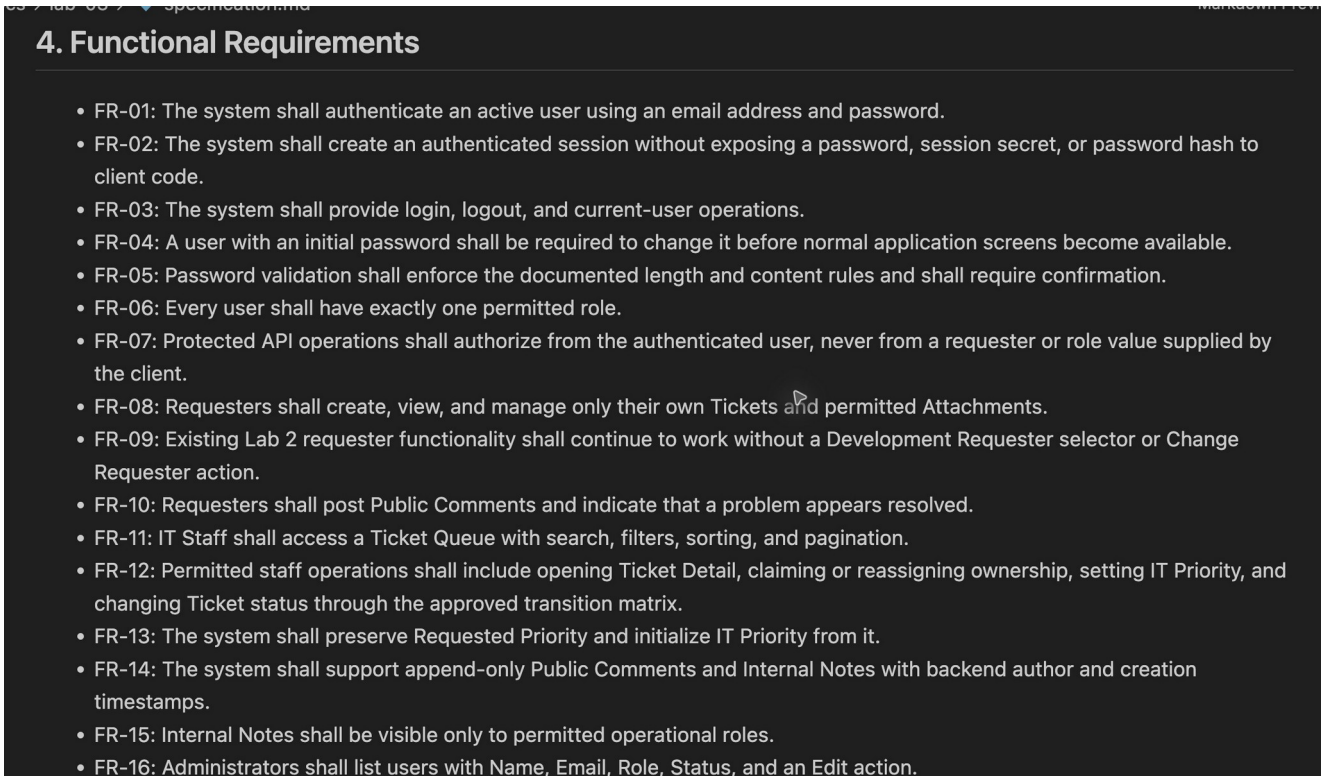
- Login, logout, current-user retrieval, and authenticated sessions.
- Secure password hashing and mandatory first-login password change.
- One role per user: `REQUESTER`, `IT_STAFF`, or `ADMINISTRATOR`.
- Migration of Lab 2 Development Requesters and Ticket ownership.
- Requester ticket and attachment regression using authenticated identity.
- Requester Public Comments and the Problem Appears Resolved action.
- IT Staff Ticket Queue, Ticket Detail, assignment, IT Priority, and status workflow.
- Public Comments and private Internal Notes.
- Minimalist Administrator user listing, search, optional role filter, create, edit, activation, role assignment, and initial-password reset.
- Zen Green responsive and accessible screens, automated tests, E2E tests, and traceable evidence.

Excluded

- Email invitations, password-reset email, multi-factor authentication, social login, single sign-on, or self-registration.
- Actions Taken, SLA calculation, escalation, notifications, dashboards, and KPI analytics.
- Multiple roles, departments, organizations, profile photos, role history, account audit history, bulk operations, import/export, and user deletion.
- Production deployment or cloud infrastructure changes.

4. Functional Requirements

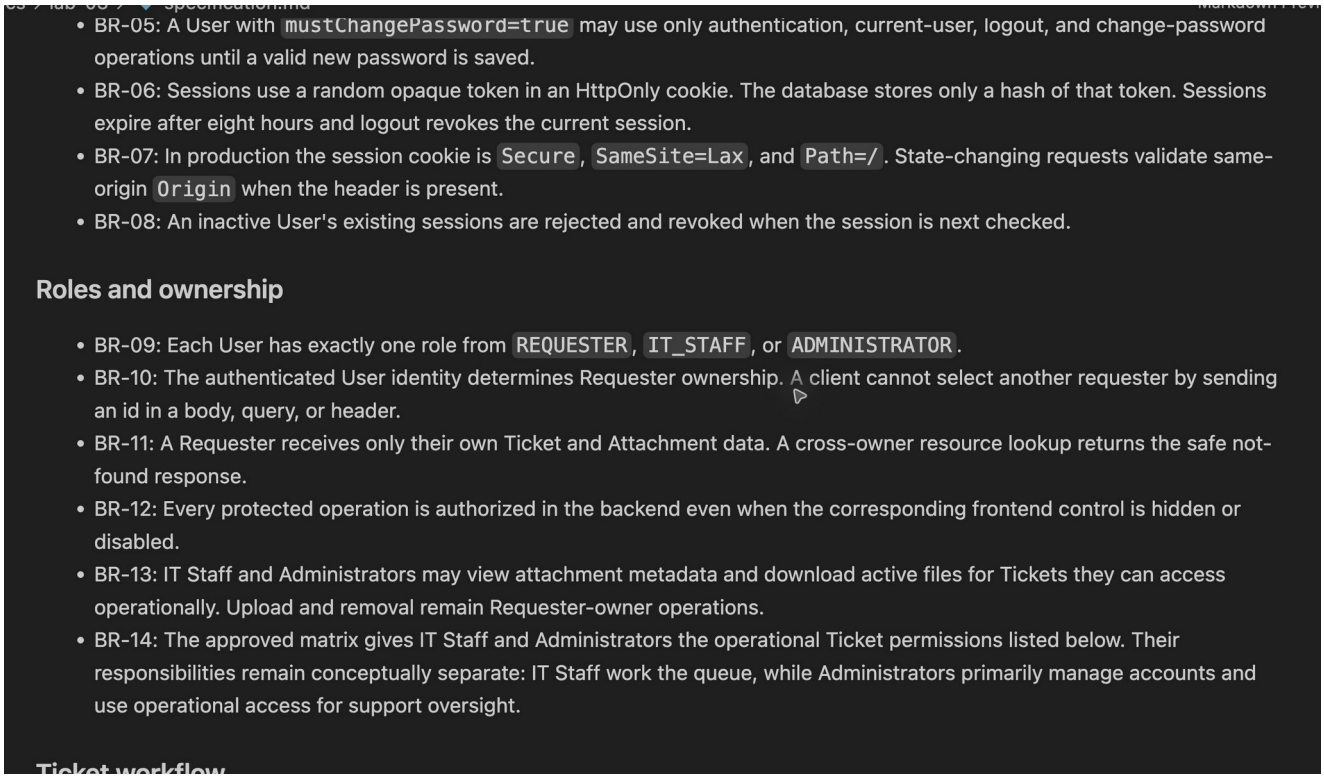
Actual VS Code Markdown Preview: Included and excluded scope



4. Functional Requirements

- FR-01: The system shall authenticate an active user using an email address and password.
- FR-02: The system shall create an authenticated session without exposing a password, session secret, or password hash to client code.
- FR-03: The system shall provide login, logout, and current-user operations.
- FR-04: A user with an initial password shall be required to change it before normal application screens become available.
- FR-05: Password validation shall enforce the documented length and content rules and shall require confirmation.
- FR-06: Every user shall have exactly one permitted role.
- FR-07: Protected API operations shall authorize from the authenticated user, never from a requester or role value supplied by the client.
- FR-08: Requesters shall create, view, and manage only their own Tickets and permitted Attachments.
- FR-09: Existing Lab 2 requester functionality shall continue to work without a Development Requester selector or Change Requester action.
- FR-10: Requesters shall post Public Comments and indicate that a problem appears resolved.
- FR-11: IT Staff shall access a Ticket Queue with search, filters, sorting, and pagination.
- FR-12: Permitted staff operations shall include opening Ticket Detail, claiming or reassigning ownership, setting IT Priority, and changing Ticket status through the approved transition matrix.
- FR-13: The system shall preserve Requested Priority and initialize IT Priority from it.
- FR-14: The system shall support append-only Public Comments and Internal Notes with backend author and creation timestamps.
- FR-15: Internal Notes shall be visible only to permitted operational roles.
- FR-16: Administrators shall list users with Name, Email, Role, Status, and an Edit action.

Actual VS Code Markdown Preview: Functional Requirements



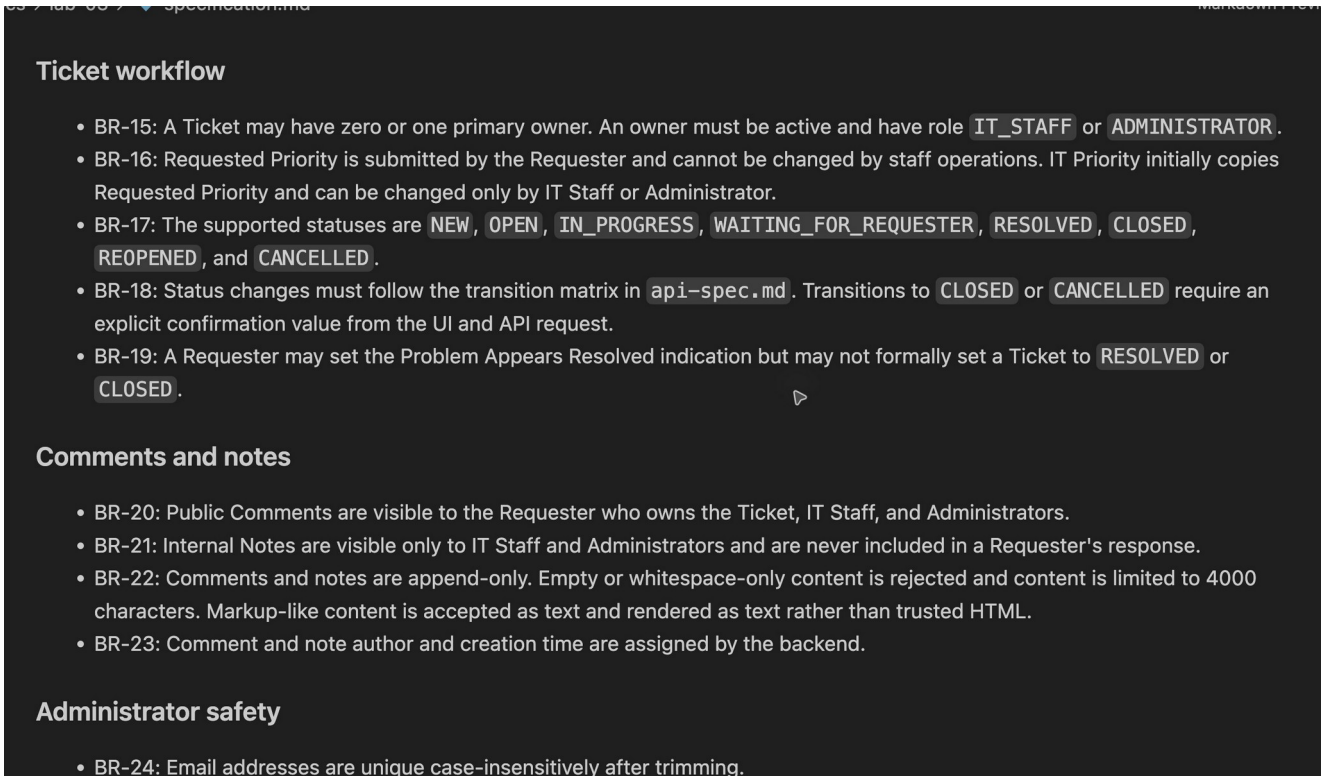
- BR-05: A User with `mustChangePassword=true` may use only authentication, current-user, logout, and change-password operations until a valid new password is saved.
- BR-06: Sessions use a random opaque token in an HttpOnly cookie. The database stores only a hash of that token. Sessions expire after eight hours and logout revokes the current session.
- BR-07: In production the session cookie is `Secure`, `SameSite=Lax`, and `Path=/`. State-changing requests validate same-origin `Origin` when the header is present.
- BR-08: An inactive User's existing sessions are rejected and revoked when the session is next checked.

Roles and ownership

- BR-09: Each User has exactly one role from `REQUESTER`, `IT_STAFF`, or `ADMINISTRATOR`.
- BR-10: The authenticated User identity determines Requester ownership. A client cannot select another requester by sending an id in a body, query, or header.
- BR-11: A Requester receives only their own Ticket and Attachment data. A cross-owner resource lookup returns the safe not-found response.
- BR-12: Every protected operation is authorized in the backend even when the corresponding frontend control is hidden or disabled.
- BR-13: IT Staff and Administrators may view attachment metadata and download active files for Tickets they can access operationally. Upload and removal remain Requester-owner operations.
- BR-14: The approved matrix gives IT Staff and Administrators the operational Ticket permissions listed below. Their responsibilities remain conceptually separate: IT Staff work the queue, while Administrators primarily manage accounts and use operational access for support oversight.

Ticket workflow

Actual VS Code Markdown Preview: Authentication and role Business Rules



Ticket workflow

- BR-15: A Ticket may have zero or one primary owner. An owner must be active and have role `IT_STAFF` or `ADMINISTRATOR`.
- BR-16: Requested Priority is submitted by the Requester and cannot be changed by staff operations. IT Priority initially copies Requested Priority and can be changed only by IT Staff or Administrator.
- BR-17: The supported statuses are `NEW`, `OPEN`, `IN_PROGRESS`, `WAITING_FOR_REQUESTER`, `RESOLVED`, `CLOSED`, `REOPENED`, and `CANCELLED`.
- BR-18: Status changes must follow the transition matrix in `api-spec.md`. Transitions to `CLOSED` or `CANCELLED` require an explicit confirmation value from the UI and API request.
- BR-19: A Requester may set the Problem Appears Resolved indication but may not formally set a Ticket to `RESOLVED` or `CLOSED`.

Comments and notes

- BR-20: Public Comments are visible to the Requester who owns the Ticket, IT Staff, and Administrators.
- BR-21: Internal Notes are visible only to IT Staff and Administrators and are never included in a Requester's response.
- BR-22: Comments and notes are append-only. Empty or whitespace-only content is rejected and content is limited to 4000 characters. Markup-like content is accepted as text and rendered as text rather than trusted HTML.
- BR-23: Comment and note author and creation time are assigned by the backend.

Administrator safety

- BR-24: Email addresses are unique case-insensitively after trimming.

Actual VS Code Markdown Preview: Ticket workflow, comments and Administrator Business Rules

- BR-24: Email addresses are unique case-insensitively after trimming.
- BR-25: An Administrator cannot deactivate their own account.
- BR-26: The system cannot deactivate or change the role of the last active Administrator. Users are deactivated rather than deleted.
- BR-27: A reset initial password marks `mustChangePassword=true` and never returns the password in an API response.

6. Authorization Matrix

Operation	Requester	IT Staff	Administrator
Login, logout, current user	Own account	Own account	Own account
Change own required password	Yes	Yes	Yes
Create and list own Tickets	Yes	No	No
Upload/remove own Attachments	Yes	No	No
View/download operational Attachments	No	Yes	Yes
View operational Ticket Queue	No	Yes	Yes
View operational Ticket Detail	No	Yes	Yes
Claim/reassign, IT Priority, status	No	Yes	Yes

Actual VS Code Markdown Preview: Administrator safeguards and authorization matrix

Read/write Public Comments	Own Tickets	Queue Tickets	Permitted Tickets
Read/write Internal Notes	No	Yes	Yes
List/search users	No	No	Yes
Create/edit/deactivate users	No	No	Yes
Set another user's initial password	No	No	Yes

The backend applies this matrix to every endpoint. A `401` means no valid session exists. A `403` means the session is valid but the role or password state does not permit the operation.

Actual Markdown Preview: remaining authorization matrix rows for comments, Internal Notes and Administrator actions

All protected endpoints distinguish unauthenticated access, forbidden roles, invalid input, missing resources, conflicts, and unexpected failures without leaking protected data.

10. Acceptance Criteria

- AC-01: An active user with valid credentials receives an authenticated session and safe current-user data.
- AC-02: Invalid credentials and inactive accounts receive the same safe login failure and no session.
- AC-03: A user with an initial password cannot enter normal application screens until a valid new password is saved.
- AC-04: Logout revokes the current session and direct protected access after logout is rejected.
- AC-05: The authenticated identity and role are displayed in the shell and protected navigation is role-specific.
- AC-06: A Requester cannot select another identity and sees only their own Lab 2 Tickets and Attachments.
- AC-07: Lab 2 Create Ticket, My Tickets, Ticket Detail, and Attachment flows continue to work after authentication migration.
- AC-08: Requester Public Comments and Problem Appears Resolved work only for owned Tickets; Requesters cannot formally resolve or close Tickets.
- AC-09: IT Staff can load a Queue with documented search, filters, sorting, pagination, owner data, status, and both priorities.
- AC-10: Queue query validation and direct role checks return documented safe responses.
- AC-11: IT Staff/Admin can open Ticket Detail and perform permitted claim/reassign, IT Priority, and status operations.
- AC-12: Invalid status transitions, inactive owners, and missing Tickets are rejected without changing data.
- AC-13: Public Comments are visible to the Requester owner, IT Staff, and Administrator with safe rendering and backend author/time.

Actual Markdown Preview: Acceptance Criteria and Product Definition of Done - segment 1/3

- AC-13: Public Comments are visible to the Requester owner, IT Staff, and Administrator with safe rendering and backend author/time.
- AC-14: Internal Notes are visible only to IT Staff and Administrator, including through direct API requests.
- AC-15: Empty, whitespace-only, and oversized comment/note content is rejected; markup-like content is stored and rendered as plain text without executing.
- AC-16: Administrator can list/search/filter users and view Name, Email, Role, Status, and Edit actions.
- AC-17: Administrator can create a user with one role and an initial password; duplicate email and invalid role are rejected.
- AC-18: Administrator can edit name, email, role, and activation state.
- AC-19: Administrator can set a new initial password and the target user must change it on the next login.
- AC-20: Self-deactivation and deactivation or role demotion of the last active Administrator are rejected.
- AC-21: Data migration preserves existing Ticket and Attachment ownership and deploys without data loss.
- AC-22: Seed creates all required active/inactive roles and realistic workflow data and is safe to run repeatedly.
- AC-23: All major screens provide relevant loading, saving, success, empty, no-results, forbidden, not-found, conflict, and safe-failure states.
- AC-24: Desktop, tablet, and mobile screens have no clipping, overlap, or unintended horizontal overflow and remain keyboard accessible.
- AC-25: All required tests, builds, migrations, seed checks, and E2E flows pass from the final `main` branch with no required test skipped.

Every criterion maps to at least one planned test in `tests.md`.

11. Definition of Done

Actual Markdown Preview: Acceptance Criteria and Product Definition of Done - segment 2/3

Every criterion maps to at least one planned test in `tests.md`.

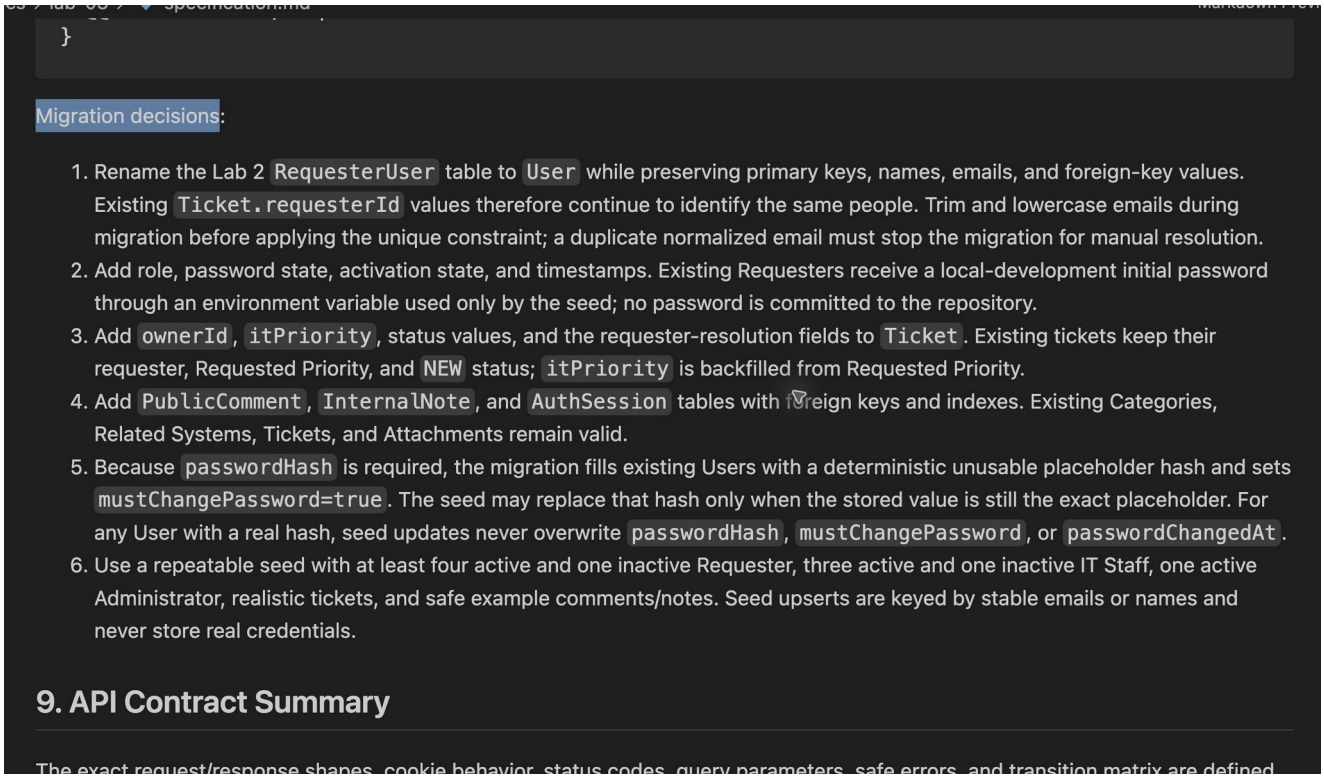
11. Definition of Done

- The approved specification, API contract, UI specification, and test plan are committed before the main implementation PRs are completed.
- User migration, session handling, password hashing, and role authorization are implemented and tested on the backend.
- Lab 2 Requester behavior remains authenticated, owner-protected, and tested.
- IT Staff Queue, Ticket operations, comments, notes, and Admin user management meet the approved acceptance criteria.
- Prisma migration and repeatable seed preserve data and satisfy role fixtures.
- Unit, API/integration, UI, style, regression, responsive, accessibility, and E2E tests pass without required skips.
- No secrets, plaintext passwords, session tokens, uploads, or generated build output are committed.
- README, reviewer record, AI-use reflection, test evidence, screenshots, and Kanban status are current.
- Each PR has a teammate's formal approval before merge into `lab3-staging`.
- The release PR is approved and merged into `main`, and final verification is rerun from `main`.

12. Assumptions and Decisions

- Lab 3 uses a database-backed opaque session cookie because it allows logout invalidation and keeps the client free of authentication secrets.
- The application uses same-origin API calls through the existing Vite/Express setup; cross-origin credential sharing is not part of this lab.

Actual Markdown Preview: Acceptance Criteria and Product Definition of Done - segment 3/3



```
}
```

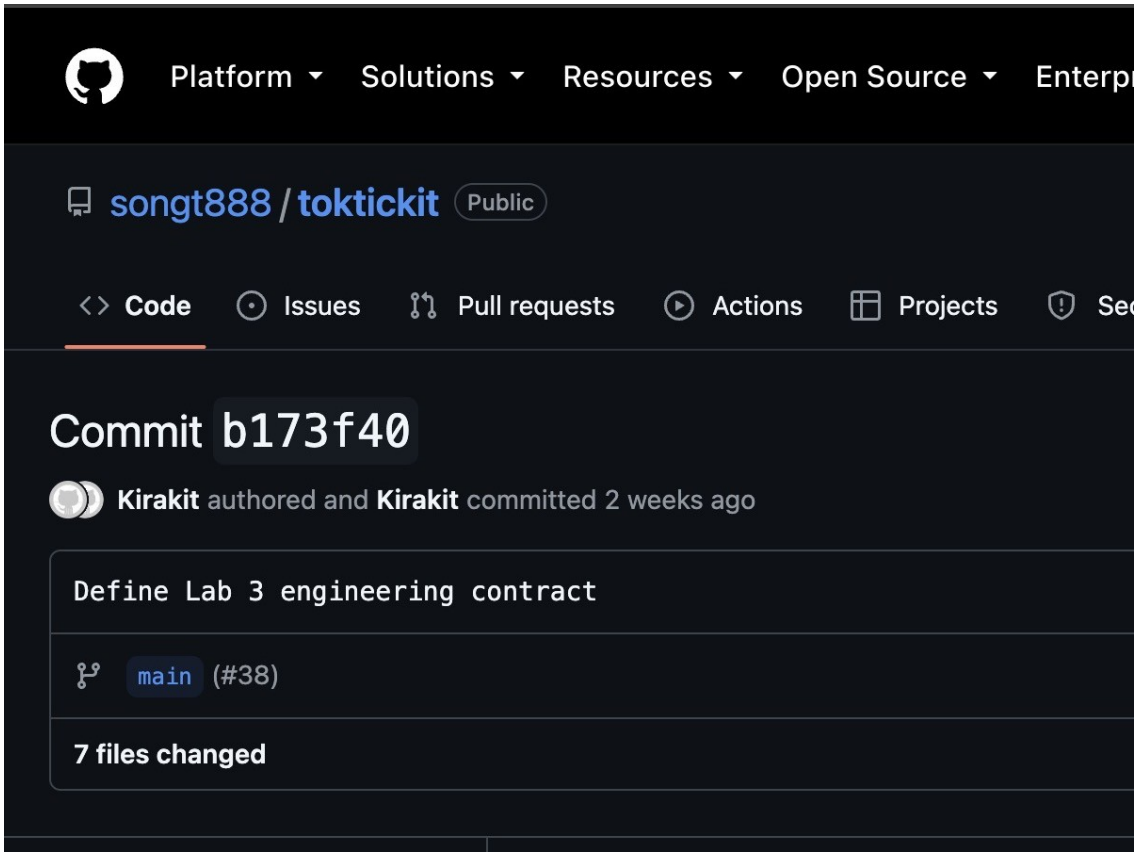
Migration decisions:

1. Rename the Lab 2 `RequesterUser` table to `User` while preserving primary keys, names, emails, and foreign-key values. Existing `Ticket.requesterId` values therefore continue to identify the same people. Trim and lowercase emails during migration before applying the unique constraint; a duplicate normalized email must stop the migration for manual resolution.
2. Add role, password state, activation state, and timestamps. Existing Requesters receive a local-development initial password through an environment variable used only by the seed; no password is committed to the repository.
3. Add `ownerId`, `itPriority`, status values, and the requester-resolution fields to `Ticket`. Existing tickets keep their requester, Requested Priority, and `NEW` status; `itPriority` is backfilled from Requested Priority.
4. Add `PublicComment`, `InternalNote`, and `AuthSession` tables with foreign keys and indexes. Existing Categories, Related Systems, Tickets, and Attachments remain valid.
5. Because `passwordHash` is required, the migration fills existing Users with a deterministic unusable placeholder hash and sets `mustChangePassword=true`. The seed may replace that hash only when the stored value is still the exact placeholder. For any User with a real hash, seed updates never overwrite `passwordHash`, `mustChangePassword`, or `passwordChangedAt`.
6. Use a repeatable seed with at least four active and one inactive Requester, three active and one inactive IT Staff, one active Administrator, realistic tickets, and safe example comments/notes. Seed upserts are keyed by stable emails or names and never store real credentials.

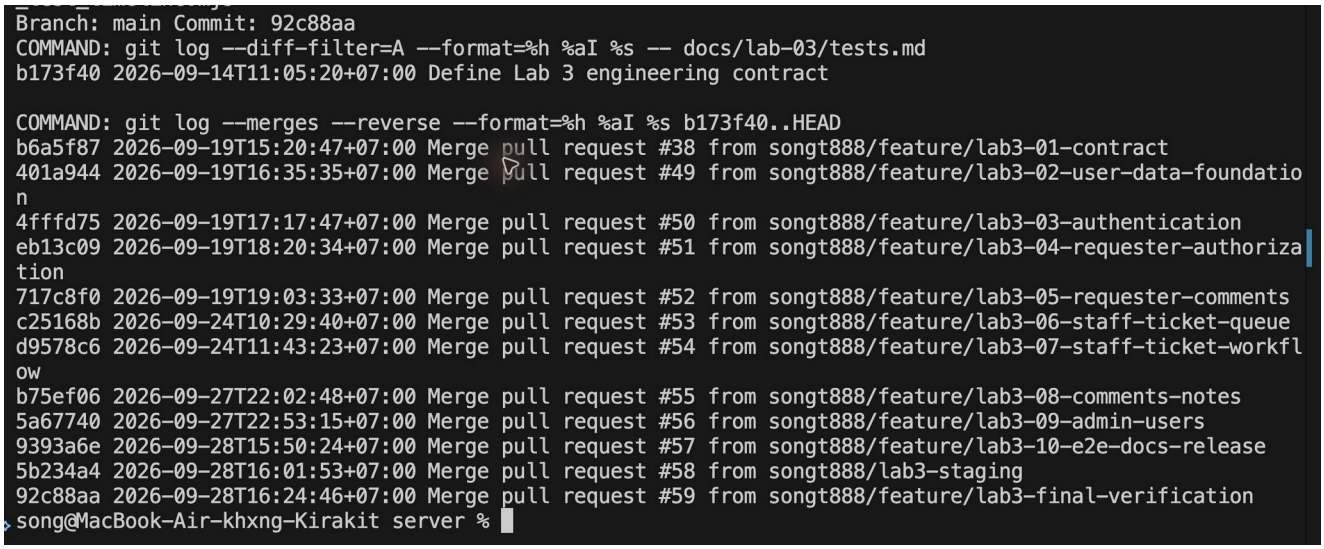
9. API Contract Summary

The exact request/response shapes, cookie behavior, status codes, query parameters, safe errors, and transition matrix are defined

Actual Markdown Preview: User, password, ownership, comments, notes and preservation migration decisions



GitHub dated commit b173f40 before the main implementation PR merges



Actual Git history: tests.md was added in b173f40 before later implementation and release merges

Answer Part 3

Test-Driven Development and Traceability

A fresh run on 3 October 2026 used an isolated main checkout at f6dc6f2: Server 92 tests / 27 files, Client 56 tests / 13 files and Playwright 15 tests passed. Four migrations were up to date, seed completed, and both production builds passed. Tracked application and test files were unchanged. Chromium opened in a normal macOS VS Code Terminal. An external runner selected unused UI/API ports; it inherited repository tests without patching their URLs. The updated verification record is submitted separately for PR review, not claimed as already merged into main.

Links

Updated test plan and verification record (review branch):
https://github.com/songt888/toktickit/blob/e3bf402/docs/lab-03/tests.md

Report and documentation corrections PR: https://github.com/songt888/toktickit/pull/64

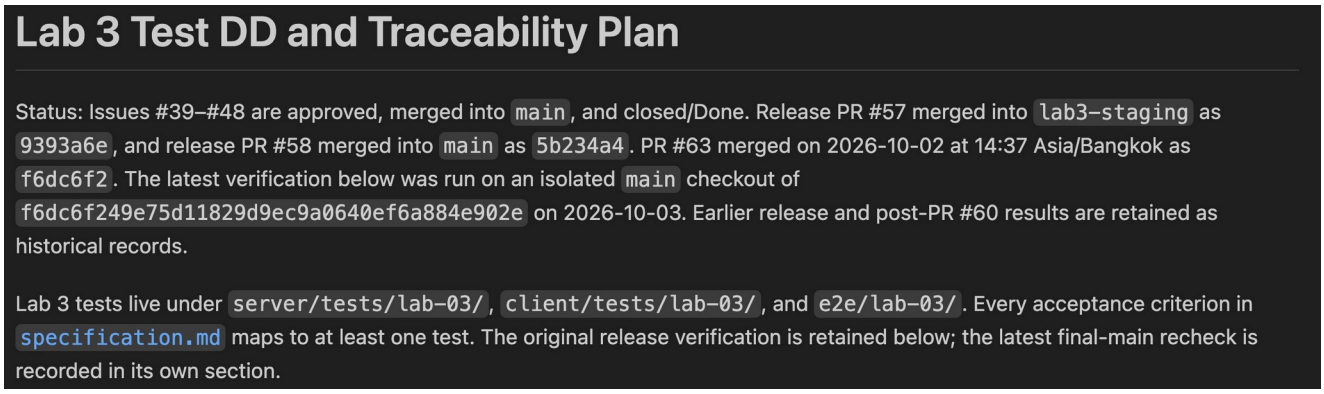
Complete raw main test/build outputs: https://github.com/songt888/toktickit/tree/e3bf402/artifacts/lab-03/verification-2026-10-03

Server Lab 3 tests: https://github.com/songt888/toktickit/tree/f6dc6f2/server/tests/lab-03

Client Lab 3 tests: https://github.com/songt888/toktickit/tree/f6dc6f2/client/tests/lab-03

E2E Lab 3 tests: https://github.com/songt888/toktickit/tree/f6dc6f2/e2e/lab-03

Screenshots

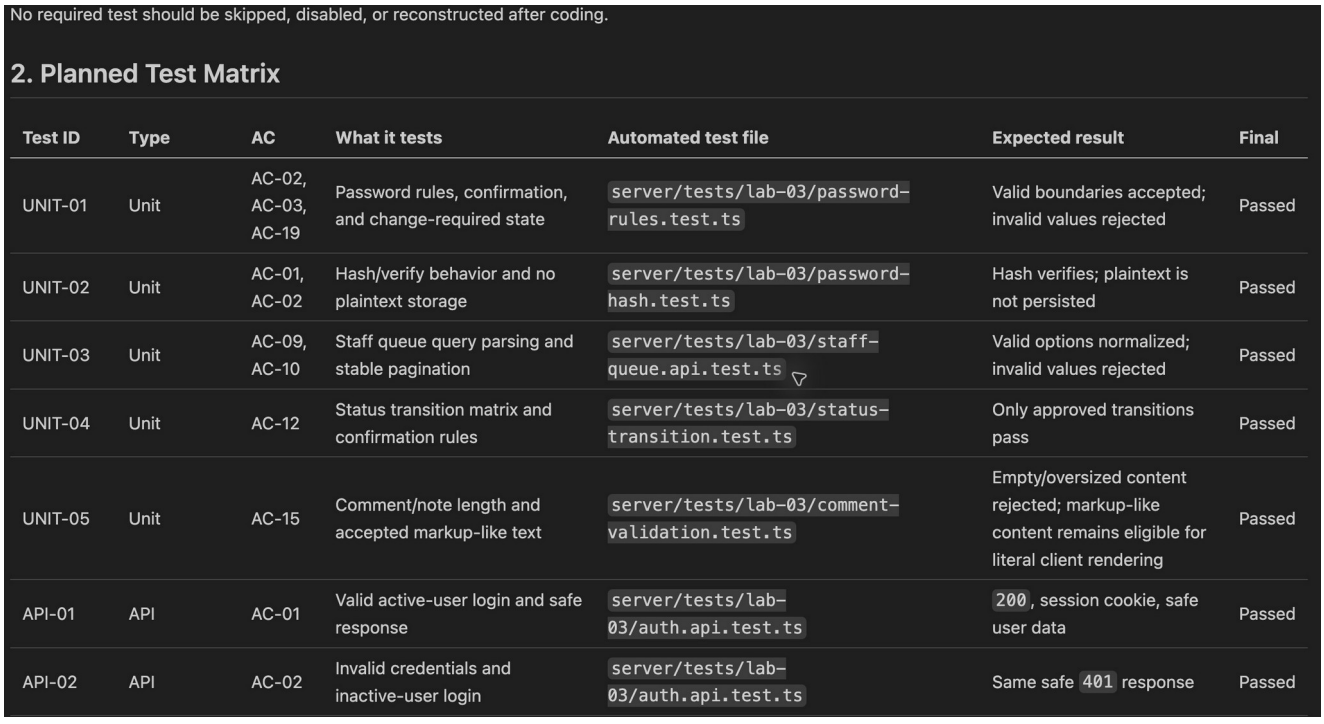


Lab 3 Test DD and Traceability Plan

Status: Issues #39–#48 are approved, merged into `main`, and closed/Done. Release PR #57 merged into `lab3-staging` as `9393a6e`, and release PR #58 merged into `main` as `5b234a4`. PR #63 merged on 2026-10-02 at 14:37 Asia/Bangkok as `f6dc6f2`. The latest verification below was run on an isolated `main` checkout of `f6dc6f249e75d11829d9ec9a0640ef6a884e902e` on 2026-10-03. Earlier release and post-PR #60 results are retained as historical records.

Lab 3 tests live under `server/tests/lab-03/`, `client/tests/lab-03/`, and `e2e/lab-03/`. Every acceptance criterion in `specification.md` maps to at least one test. The original release verification is retained below; the latest final-main recheck is recorded in its own section.

Actual updated tests.md Markdown Preview: PR #63 merge and final-main verification at f6dc6f2.



No required test should be skipped, disabled, or reconstructed after coding.

2. Planned Test Matrix

Test ID	Type	AC	What it tests	Automated test file	Expected result	Final
UNIT-01	Unit	AC-02, AC-03, AC-19	Password rules, confirmation, and change-required state	`server/tests/lab-03/password-rules.test.ts`	Valid boundaries accepted; invalid values rejected	Passed
UNIT-02	Unit	AC-01, AC-02	Hash/verify behavior and no plaintext storage	`server/tests/lab-03/password-hash.test.ts`	Hash verifies; plaintext is not persisted	Passed
UNIT-03	Unit	AC-09, AC-10	Staff queue query parsing and stable pagination	`server/tests/lab-03/staff-queue.api.test.ts`	Valid options normalized; invalid values rejected	Passed
UNIT-04	Unit	AC-12	Status transition matrix and confirmation rules	`server/tests/lab-03/status-transition.test.ts`	Only approved transitions pass	Passed
UNIT-05	Unit	AC-15	Comment/note length and accepted markup-like text	`server/tests/lab-03/comment-validation.test.ts`	Empty/oversized content rejected; markup-like content remains eligible for literal client rendering	Passed
API-01	API	AC-01	Valid active-user login and safe response	`server/tests/lab-03/auth.api.test.ts`	`200`, session cookie, safe user data	Passed
API-02	API	AC-02	Invalid credentials and inactive-user login	`server/tests/lab-03/auth.api.test.ts`	Same safe `401` response	Passed

Actual planned test matrix in Markdown Preview - selected view 1/2; complete matrix in the linked tests.md

RESP-02	Responsive	AC-24	Tablet visual and overflow checks for all major screens	`e2e/lab-03/responsive.spec.ts`, `e2e/lab-03/authentication.spec.ts`	No clipping or hidden controls	Passed
RESP-03	Responsive	AC-24	Mobile visual and overflow checks for all major screens	`e2e/lab-03/responsive.spec.ts`, `e2e/lab-03/authentication.spec.ts`	Touch/keyboard usable at 390px	Passed
A11Y-01	Accessibility	AC-24	Keyboard skip link, focus, labels, active navigation, and role states	`e2e/lab-03/accessibility.spec.ts`	Keyboard and accessible-name checks pass	Passed
E2E-01	E2E	AC-01, AC-02, AC-03, AC-04, AC-05	Login, first password change, shell, and logout/revocation	`e2e/lab-03/authentication.spec.ts`	Full auth flow passes at three viewports	Passed
E2E-02	E2E	AC-06, AC-07, AC-08	Requester regression, comments, resolution, and isolation	`e2e/lab-03/requester-flow.spec.ts`	Requester ownership flow passes	Passed
E2E-03	E2E	AC-09, AC-10, AC-11, AC-12, AC-13, AC-14, AC-15	Staff queue/workflow, cross-role comments, notes, and attachments	`e2e/lab-03/staff-ticket-flow.spec.ts`, `e2e/lab-03/requester-flow.spec.ts`	Staff and collaboration flows pass	Passed

Actual planned test matrix in Markdown Preview - selected view 2/2; complete matrix in the linked tests.md

- Administrator screenshots: desktop-administrator-user-management.png, tablet-administrator-user-management.png, and mobile-administrator-user-management.png.
- Keyboard focus evidence: accessibility-keyboard-focus.png.

`RELEASE-01` is Passed because AC-25 was verified after merge into `main`. No PDF was created, as requested.

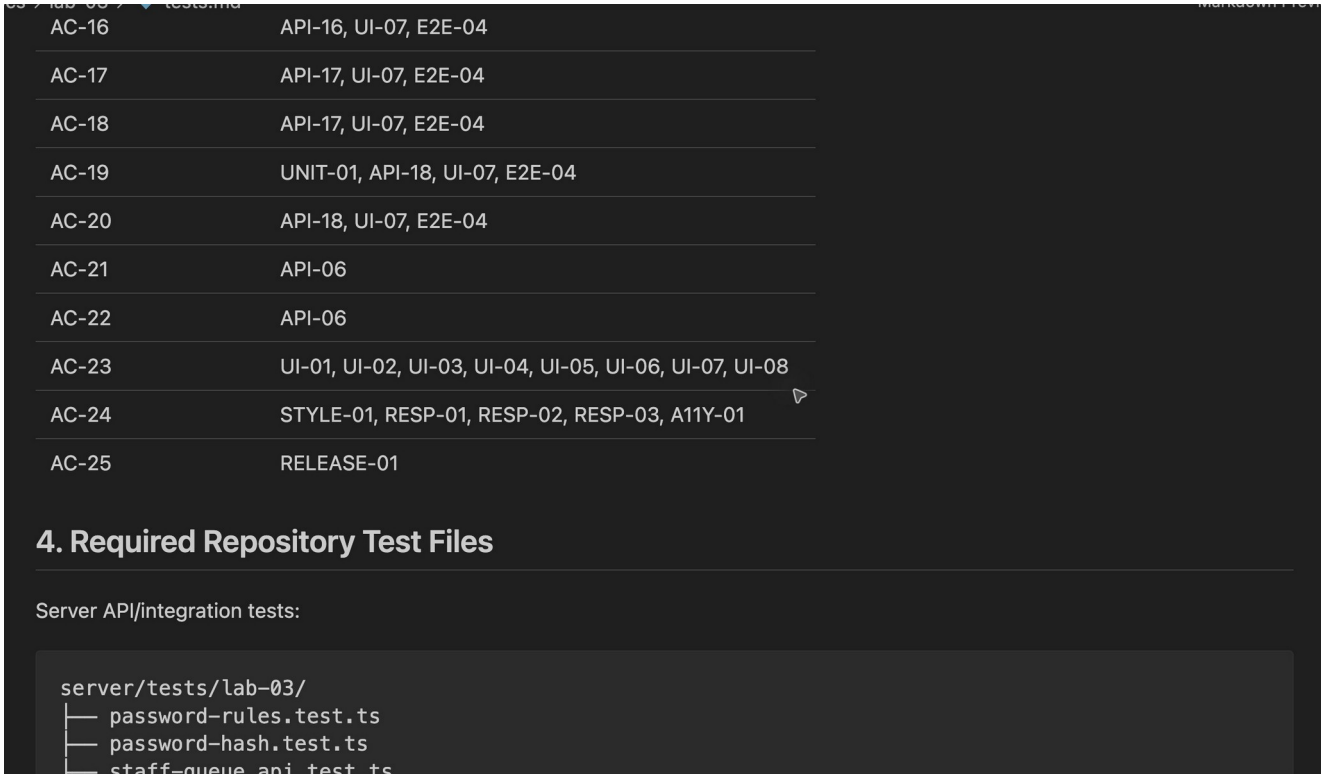
3. Acceptance-Criterion Traceability

Acceptance Criteria	Planned tests
AC-01	UNIT-02, API-01, UI-01, E2E-01
AC-02	UNIT-01, UNIT-02, API-02, UI-01, E2E-01
AC-03	UNIT-01, API-03, UI-02, E2E-01
AC-04	API-04, UI-03, E2E-01
AC-05	API-05, UI-03, STYLE-01, E2E-01
AC-06	API-07, API-08, UI-04, E2E-02
AC-07	API-07, API-08, UI-04, REG-01, REG-02, E2E-02
AC-08	API-09, UI-04, E2E-02
AC-09	UNIT-03, API-10, UI-05, E2E-03, E2E-05
AC-10	UNIT-03, API-05, API-10, UI-05, E2E-03, E2E-05

Actual Markdown Preview: AC-to-test traceability - segment 1/3

AC-06	API-07, API-08, UI-04, E2E-02
AC-07	API-07, API-08, UI-04, REG-01, REG-02, E2E-02
AC-08	API-09, UI-04, E2E-02
AC-09	UNIT-03, API-10, UI-05, E2E-03, E2E-05
AC-10	UNIT-03, API-05, API-10, UI-05, E2E-03, E2E-05
AC-11	API-11, API-19, UI-06, UI-08, E2E-03, E2E-06
AC-12	UNIT-04, API-11, API-12, UI-06, UI-08, E2E-03, E2E-06
AC-13	API-13, UI-06, E2E-03
AC-14	API-14, UI-06, E2E-03
AC-15	UNIT-05, API-15, E2E-03
AC-16	API-16, UI-07, E2E-04
AC-17	API-17, UI-07, E2E-04
AC-18	API-17, UI-07, E2E-04
AC-19	UNIT-01, API-18, UI-07, E2E-04
AC-20	API-18, UI-07, E2E-04

Actual Markdown Preview: AC-to-test traceability - segment 2/3



AC-16	API-16, UI-07, E2E-04
AC-17	API-17, UI-07, E2E-04
AC-18	API-17, UI-07, E2E-04
AC-19	UNIT-01, API-18, UI-07, E2E-04
AC-20	API-18, UI-07, E2E-04
AC-21	API-06
AC-22	API-06
AC-23	UI-01, UI-02, UI-03, UI-04, UI-05, UI-06, UI-07, UI-08
AC-24	STYLE-01, RESP-01, RESP-02, RESP-03, A11Y-01
AC-25	RELEASE-01

4. Required Repository Test Files

Server API/integration tests:

```
server/tests/lab-03/
├── password-rules.test.ts
├── password-hash.test.ts
```

Actual Markdown Preview: AC-to-test traceability - segment 3/3

Native Terminal screenshots below display the complete archived Server and Client output in consecutive original-log ranges, including every passed test, totals and exit code. The original main SHA and command are repeated for context; the exit code is copied from the same recorded run. These are log views on the documentation branch, not new test runs. The full raw logs remain linked above. This completed main f6dc6f2 run supersedes the earlier Chromium-blocked pending note and historical port-patched result.

Complete Server passing output from main f6dc6f2

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 4-16 (continuous output)

> toktickit-server@1.0.0 test
> vitest run --reporter=verbose --silent


 RUN  v2.1.9 /Users/song/Documents/Codex/2026-09-05/ca/work/lab3-final-main-check/server

 ✓ tests/lab-03/data-foundation.test.ts > Lab 3 database foundation > creates the role-based user model a
nd preserves ticket data structures
 ✓ tests/lab-03/users-admin.api.test.ts > Administrator user management API > requires Administrator role
 and returns only safe, stable user data
 ✓ tests/lab-03/data-foundation.test.ts > Lab 3 database foundation > preserves seeded user IDs, ticket o
wnership, attachments, and real hashes across reseeding
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > lists only active Staff
and Administrators as safe assignees
 ✓ tests/lab-03/staff-queue.api.test.ts > GET /api/staff/tickets > allows Staff and Administrators while
rejecting other roles
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 requester authorization > requires an authenticated ses
sion for requester-owned routes
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 1/9 - original log lines 4-16.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 17-28 (continuous output)
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > keeps operational attach
ment metadata and download available to Staff
 ✓ tests/lab-03/users-admin.api.test.ts > Administrator user management API > rejects invalid filters and
 treats wildcard characters as literal search text
 ✓ tests/lab-03/requester-comments.api.test.ts > Requester public comments and problem resolution > creat
es append-only comments with backend author/time and safe text
 ✓ tests/lab-02/attachments.api.test.ts > Attachment lifecycle API > uploads a permitted file with genera
ted storage and retrievable metadata
 ✓ tests/lab-03/comments-notes.api.test.ts > Public Comments and Internal Notes API > serves ordered Publ
ic Comments to their owner, Staff, and Administrators
 ✓ tests/lab-03/requester-comments.api.test.ts > Requester public comments and problem resolution > prote
cts comments and resolution by authenticated ownership
 ✓ tests/lab-03/staff-queue.api.test.ts > GET /api/staff/tickets > supports search, filters, stable sorti
ng, owner display, and pagination metadata
 ✓ tests/lab-03/staff-queue.api.test.ts > GET /api/staff/tickets > uses the default sort and id tiebreake
r for equal updatedAt values
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 requester authorization > enforces requester roles whil
e allowing Staff operational attachment reads
 ✓ tests/lab-03/requester-comments.api.test.ts > Requester public comments and problem resolution > rejec
ts empty, whitespace-only, and oversized comments
 ✓ tests/lab-03/requester-comments.api.test.ts > Requester public comments and problem resolution > updat
es only the requester resolution indication and never the Ticket status
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > supports claiming, assig
ning, reassigning, and unassigning with fresh versions
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 2/9 - original log lines 17-28.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 29-40 (continuous output)
 ✓ tests/lab-03/users-admin.api.test.ts > Administrator user management API > creates normalized users wi
th a hashed initial password and rejects duplicate email
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 requester authorization > derives ownership from the se
ssion and hides another requester's ticket
 ✓ tests/lab-03/authorization.api.test.ts > Lab 3 requester authorization > permits authenticated referen
ce-data reads without exposing credentials
 ✓ tests/lab-03/comments-notes.api.test.ts > Public Comments and Internal Notes API > restricts Internal
Notes to Staff and Administrators without leaking them
 ✓ tests/lab-03/staff-queue.api.test.ts > GET /api/staff/tickets > searches ticket fields and requester n
ames while treating wildcards literally
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > rejects invalid or inact
ive owners without changing the Ticket
 ✓ tests/lab-03/staff-queue.api.test.ts > GET /api/staff/tickets > rejects unsupported query values with
a safe 400 response
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > updates only IT Priority
 and rejects Requester mutations
 ✓ tests/lab-03/comments-notes.api.test.ts > Public Comments and Internal Notes API > rejects blank and o
versized public comments and internal notes without inserting rows
 ✓ tests/lab-03/users-admin.api.test.ts > Administrator user management API > validates create and edit v
alues, normalizes email, and prevents duplicate email
 ✓ tests/lab-03/staff-queue.api.test.ts > GET /api/staff/tickets > opens safe operational Ticket Detail a
nd protects the route by role
 ✓ tests/lab-03/comments-notes.api.test.ts > Public Comments and Internal Notes API > keeps comments and
notes append-only
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 3/9 - original log lines 29-40.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 41-52 (continuous output)
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > enforces allowed transit
ions and requires confirmation for terminal changes
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > rejects stale concurrent
 mutations and leaves the winner intact
 ✓ tests/lab-03/staff-ticket-detail.api.test.ts > IT Staff Ticket workflow API > returns safe missing and
 stale responses
 ✓ tests/lab-02/attachments.api.test.ts > Attachment lifecycle API > rejects unsupported, oversized, and
sixth active files safely
 ✓ tests/lab-03/users-admin.api.test.ts > Administrator user management API > resets a user's password wi
thout returning it and requires a change at next login
 ✓ tests/lab-03/users-admin.api.test.ts > Administrator user management API > rejects self-deactivation a
nd evaluates the last-active-Administrator rule
 ✓ tests/lab-02/attachments.api.test.ts > Attachment lifecycle API > allows owner download but rejects cr
oss-requester and removed-file access
 ✓ tests/lab-02/attachments.api.test.ts > Attachment lifecycle API > soft-removes an attachment with a re
ason and keeps metadata
 ✓ tests/lab-02/attachments.api.test.ts > Attachment lifecycle API > requires a session and ignores reque
ster headers before attachment access
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > validates input and returns the same safe f
ailure for invalid or inactive accounts
 ✓ tests/lab-02/query-options.test.ts > My Tickets query options > provides the documented defaults
 ✓ tests/lab-02/query-options.test.ts > My Tickets query options > parses search, filters, sorting, and p
agination
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 4/9 - original log lines 41-52.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 53-64 (continuous output)
 ✓ tests/lab-02/query-options.test.ts > My Tickets query options > rejects unsupported query values
 ✓ tests/lab-02/query-options.test.ts > My Tickets query options > always scopes the database filter to t
he selected requester
 ✓ tests/lab-02/data-foundation.test.ts > Lab 2 database foundation > creates the required tables, column
s, unique constraints, and ticket sequence
 ✓ tests/lab-02/create-ticket.api.test.ts > Create Ticket API > returns active related systems in id orde
r
 ✓ tests/lab-02/create-ticket.api.test.ts > Create Ticket API > creates one owned ticket with an official
 number and NEW status
 ✓ tests/lab-02/create-ticket.api.test.ts > Create Ticket API > rejects invalid payloads without creating
 a ticket
 ✓ tests/lab-02/create-ticket.api.test.ts > Create Ticket API > rejects reference IDs above the PostgreSQ
L integer limit with 400
 ✓ tests/lab-03/seed.test.ts > Lab 3 seed > seeds every role, active/inactive split, workflow data, and s
afe password state
 ✓ tests/lab-02/create-ticket.api.test.ts > Create Ticket API > rejects missing requester context and ina
ctive reference records safely
 ✓ tests/lab-02/validation.test.ts > Create Ticket validation > trims and accepts a valid ticket payload
 ✓ tests/lab-02/validation.test.ts > Create Ticket validation > rejects missing, short, and unsupported v
alues
 ✓ tests/lab-02/validation.test.ts > Create Ticket validation > enforces the permitted attachment type an
d size boundaries
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 5/9 - original log lines 53-64.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 65-76 (continuous output)
 ✓ tests/lab-02/ticket-detail.api.test.ts > GET /api/tickets/:id > returns read-only ticket fields and at
tachment metadata for the owner
 ✓ tests/lab-02/seed.test.ts > Lab 2 seed > seeds the required reference data and active/inactive referen
ce split
 ✓ tests/lab-03/status-transition.test.ts > Staff Ticket status transition rules > allows exactly the doc
umented transition matrix
 ✓ tests/lab-03/status-transition.test.ts > Staff Ticket status transition rules > requires explicit conf
irmation only for Closed and Cancelled
 ✓ tests/lab-02/ticket-detail.api.test.ts > GET /api/tickets/:id > does not reveal a ticket to another re
quester or an unknown ticket id
 ✓ tests/lab-02/my-tickets.api.test.ts > GET /api/tickets > returns only the selected requester's tickets
 with pagination metadata
 ✓ tests/lab-02/ticket-detail.api.test.ts > GET /api/tickets/:id > requires a session and ignores request
er headers before looking up the ticket
 ✓ tests/lab-03/requester-regression.api.test.ts > Requester session regression coverage > keeps create,
list, detail, and attachment APIs on session identity
 ✓ tests/lab-03/requester-regression.api.test.ts > Requester session regression coverage > returns a safe
 404 for another requester's session
 ✓ tests/lab-02/my-tickets.api.test.ts > GET /api/tickets > applies search, filters, sorting, and page-si
ze options
 ✓ tests/lab-02/my-tickets.api.test.ts > GET /api/tickets > rejects invalid query parameters safely
 ✓ tests/lab-02/my-tickets.api.test.ts > GET /api/tickets > requires a session and ignores a spoofed requ
ester header
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 6/9 - original log lines 65-76.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 77-88 (continuous output)
 ✓ tests/lab-02/seed.test.ts > Lab 2 seed > is safe to run repeatedly without creating duplicates
 ✓ tests/lab-03/seed.test.ts > Lab 3 seed > is safe to run repeatedly without creating duplicate seed rec
ords
 ✓ tests/lab-01/seed.test.ts > category seed > is safe to run twice without creating duplicates
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > creates a safe session, exposes current use
r, changes the initial password, and logs out
 ✓ tests/lab-03/password-rules.test.ts > Lab 3 password rules > accepts a password with the required leng
th and character classes
 ✓ tests/lab-03/password-rules.test.ts > Lab 3 password rules > rejects passwords outside the length and
character rules
 ✓ tests/lab-03/password-rules.test.ts > Lab 3 password rules > rejects non-string input
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > rejects an expired session
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > rejects inval
id or blank content: undefined
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > rejects inval
id or blank content: null
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > rejects inval
id or blank content: 42
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > rejects inval
id or blank content:
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 7/9 - original log lines 77-88.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 89-100 (continuous output)
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > rejects inval
id or blank content:
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > rejects inval
id or blank content:

 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > accepts conte
nt up to 4000 characters and rejects longer content
 ✓ tests/lab-03/comment-validation.test.ts > comment and internal-note content validation > allows markup
-like text under the approved plain-text rendering policy
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > revokes a session when its user becomes ina
ctive
 ✓ tests/lab-03/auth.api.test.ts > Lab 3 authentication API > rejects credentialed state-changing request
s from an unknown origin
 ✓ tests/lab-02/ticket-number.test.ts > Ticket Number generation > formats the database sequence value wi
th the UTC ticket date
 ✓ tests/lab-02/ticket-number.test.ts > Ticket Number generation > keeps different sequence values unique
 ✓ tests/lab-01/categories.test.ts > GET /api/categories > returns the four seeded categories in id order
 ✓ tests/lab-03/password-hash.test.ts > Lab 3 password hashing > creates salted scrypt hashes that verify
 without storing plaintext
 ✓ tests/lab-03/password-hash.test.ts > Lab 3 password hashing > does not treat the migration marker as a
 usable password hash
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 8/9 - original log lines 89-100.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --silent
Original log lines 101-111 (continuous output)
 ✓ tests/lab-02/requesters.api.test.ts > GET /api/requesters > returns only the authenticated requester f
or compatibility
 ✓ tests/lab-01/health.test.ts > GET /api/health > returns 200 with status ok and the service name
 ✓ tests/lab-01/health.test.ts > GET /api/health > returns 200 even when the client sends a cache validat
or

 Test Files  27 passed (27)
      Tests  92 passed (92)
   Start at  04:04:20
   Duration  1.40s (transform 327ms, setup 0ms, collect 3.27s, tests 4.36s, environment 3ms, prepare 1.16
s)


Exit code: 0
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Server output 9/9 - original log lines 101-111.

Complete Client passing output from main f6dc6f2

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --no-file-parallelism
Original log lines 4-16 (continuous output)

> toktickit-client@1.0.0 test
> vitest run --reporter=verbose --no-file-parallelism


 RUN  v2.1.9 /Users/song/Documents/Codex/2026-09-05/ca/work/lab3-final-main-check/client

 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > downloads active
attachments, hides removed downloads, and safely handles failure
 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > claims the Ticket
 as the signed-in user and keeps Requested Priority read-only
 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > assigns to an eli
gible user, updates IT Priority, and saves a permitted status
 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > requires confirma
tion before moving a Ticket to Closed
 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > offers a refresh
when the server rejects a stale update
 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > posts Public Comm
ents and private Internal Notes as plain text
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Client output 1/6 - original log lines 4-16.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --no-file-parallelism
Original log lines 17-28 (continuous output)
 ✓ tests/lab-03/StaffTicketDetail.test.tsx > Staff Ticket Detail operational controls > validates blank c
ollaboration forms and preserves drafts after request failures
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > shows API-backed ticket fields and the Create Ticket
 action
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > preserves list query state and active navigation aft
er returning from detail
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > refreshes the list when returning from Create Ticket
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > applies search, filter, sorting, page size, and pagi
nation controls
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > distinguishes an empty list from a filtered no-resul
ts state
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > shows a failure state and can retry the API request
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > uses the authenticated requester and does not expose
 identity switching
 ✓ tests/lab-02/MyTickets.test.tsx > My Tickets UI > ignores an older ticket response when a newer reques
t finishes first
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > shows read-only ticket fields and sep
arated attachment metadata
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > shows a loading state while the detai
l request is pending
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > keeps the public comment form disable
d until comments finish loading
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Client output 2/6 - original log lines 17-28.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --no-file-parallelism
Original log lines 29-40 (continuous output)
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > keeps comment loading errors separate
 from form validation errors
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > shows a not-found failure and can ret
ry
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > adds plain-text comments and keeps re
solution separate from ticket status
 ✓ tests/lab-02/RequesterTicketDetail.test.tsx > Ticket Detail UI > shows validation feedback for an empt
y public comment
 ✓ tests/lab-02/CreateTicket.test.tsx > Create Ticket UI > shows the selected requester and reference dat
a from the APIs
 ✓ tests/lab-02/CreateTicket.test.tsx > Create Ticket UI > shows field-level validation and does not call
 the create API for invalid input
 ✓ tests/lab-02/CreateTicket.test.tsx > Create Ticket UI > submits once, shows a busy state, and displays
 the generated ticket number
 ✓ tests/lab-02/CreateTicket.test.tsx > Create Ticket UI > uploads selected attachments after the ticket
is created
 ✓ tests/lab-02/CreateTicket.test.tsx > Create Ticket UI > preserves entered values and shows a safe erro
r when creation fails
 ✓ tests/lab-02/CreateTicket.test.tsx > Create Ticket UI > reports unsupported, oversized, and excessive
attachments clearly
 ✓ tests/lab-03/UserManagement.test.tsx > Administrator User Management UI > lists safe user data and app
lies name/email search and role filters
 ✓ tests/lab-03/UserManagement.test.tsx > Administrator User Management UI > creates and edits users, the
n sets an initial password without displaying it
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

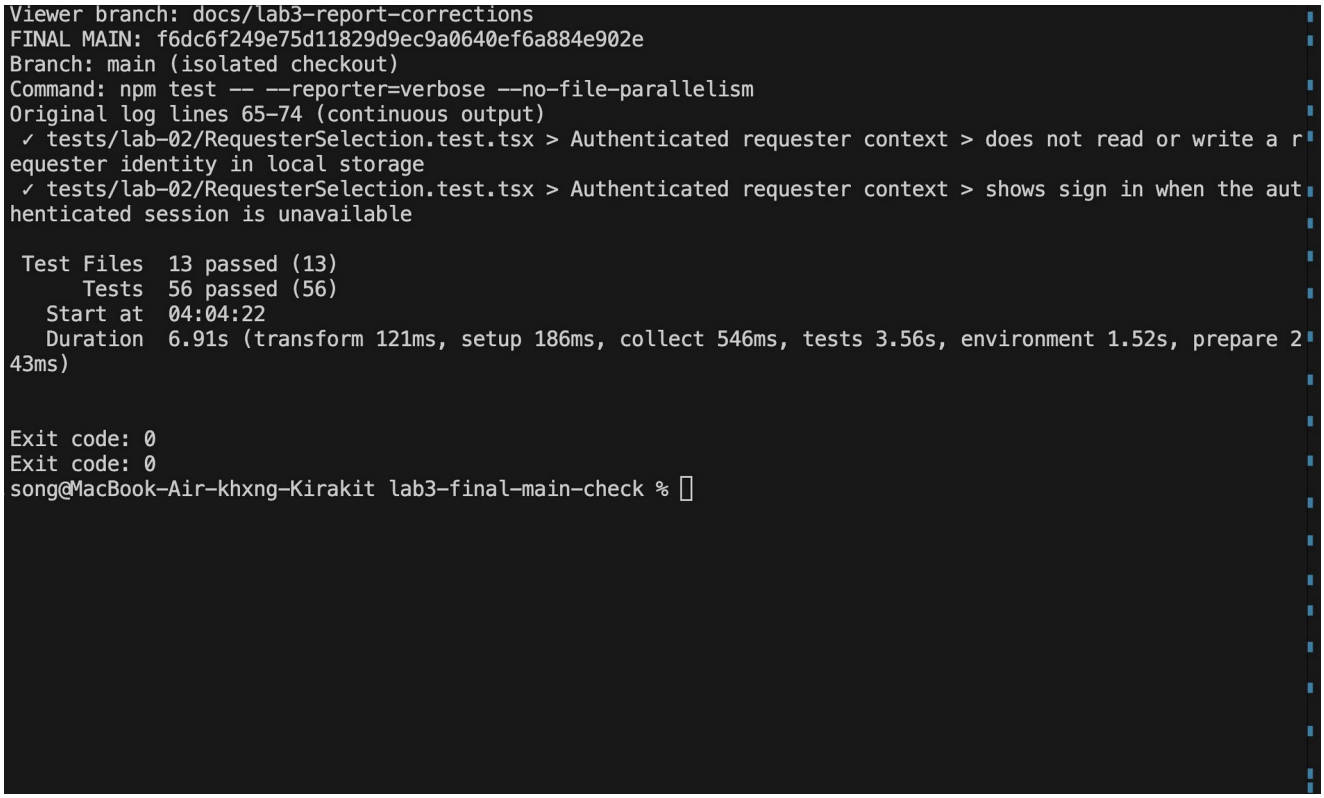
Client output 3/6 - original log lines 29-40.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --no-file-parallelism
Original log lines 41-52 (continuous output)
 ✓ tests/lab-03/UserManagement.test.tsx > Administrator User Management UI > requires confirmation before
 changing account access
 ✓ tests/lab-03/UserManagement.test.tsx > Administrator User Management UI > shows distinct empty, no-res
ults, forbidden, and retryable failure states
 ✓ tests/lab-02/AttachmentSection.test.tsx > Attachment Section UI > shows metadata, downloads active fil
es, and hides download for removed files
 ✓ tests/lab-02/AttachmentSection.test.tsx > Attachment Section UI > uploads a valid file and reports an
upload failure
 ✓ tests/lab-02/AttachmentSection.test.tsx > Attachment Section UI > requires a reason before confirming
soft removal
 ✓ tests/lab-03/RequesterRegression.test.tsx > Requester authenticated-session regression > renders ident
ity from auth/me and does not expose the old requester selector
 ✓ tests/lab-03/RequesterRegression.test.tsx > Requester authenticated-session regression > passes only t
icket identifiers to the detail API after authenticated list navigation
 ✓ tests/lab-03/RequesterRegression.test.tsx > Requester authenticated-session regression > lets the auth
enticated requester add a public comment and mark the problem resolved
 ✓ tests/lab-03/ApplicationShell.test.tsx > Authenticated application shell > shows the safe user, role,
permitted navigation, and no requester selector
 ✓ tests/lab-03/ApplicationShell.test.tsx > Authenticated application shell > keeps normal screens blocke
d when the session requires a password change
 ✓ tests/lab-03/ApplicationShell.test.tsx > Authenticated application shell > shows the staff queue navig
ation and workspace for an IT Staff session
 ✓ tests/lab-03/ApplicationShell.test.tsx > Authenticated application shell > shows user management only
for Administrators and opens it by default
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

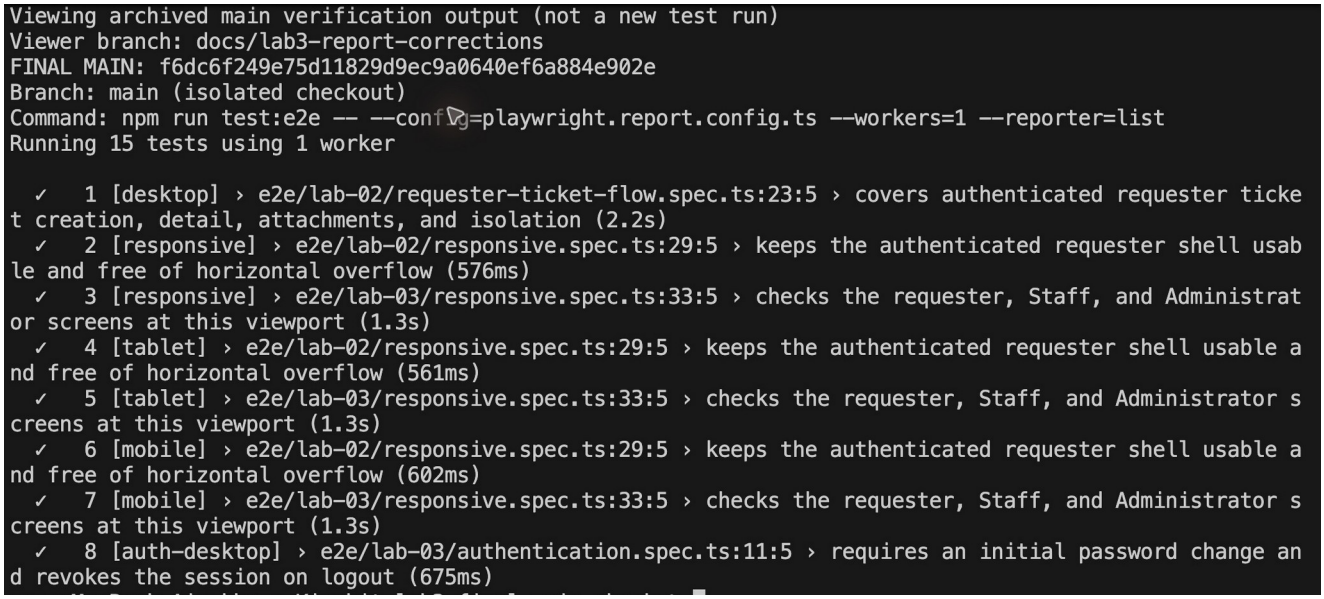
Client output 4/6 - original log lines 41-52.

```
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm test -- --reporter=verbose --no-file-parallelism
Original log lines 53-64 (continuous output)
 ✓ tests/lab-03/ApplicationShell.test.tsx > Authenticated application shell > clears the authenticated she
ll after logout
 ✓ tests/lab-03/StaffTicketQueue.test.tsx > Staff Ticket Queue UI > shows operational fields, ownership, b
adges, and opens Ticket Detail
 ✓ tests/lab-03/StaffTicketQueue.test.tsx > Staff Ticket Queue UI > sends search, filter, sort, and pagina
tion options to the API
 ✓ tests/lab-03/StaffTicketQueue.test.tsx > Staff Ticket Queue UI > shows no-results and retryable failure
 states
 ✓ tests/lab-03/ChangePassword.test.tsx > Change Password > enforces the password rules and confirmation
 ✓ tests/lab-03/ChangePassword.test.tsx > Change Password > submits a valid change and continues with the
updated user
 ✓ tests/lab-03/ChangePassword.test.tsx > Change Password > shows a safe API failure and keeps the form vi
sible
 ✓ tests/lab-02/ui-style.test.tsx > TokTickIT UI style and accessibility contract > uses labelled Bootstra
p controls, visible required markers, and action hierarchy
 ✓ tests/lab-03/Login.test.tsx > Login > shows field validation without calling the API
 ✓ tests/lab-03/Login.test.tsx > Login > shows a busy state and returns the authenticated user on success
 ✓ tests/lab-03/Login.test.tsx > Login > shows a safe message for invalid credentials
 ✓ tests/lab-02/RequesterSelection.test.tsx > Authenticated requester context > shows the requester from t
he authenticated session without a selector
Exit code: 0
song@MacBook-Air-khxng-Kirakit lab3-final-main-check %
```

Client output 5/6 - original log lines 53-64.



Client output 6/6 - original log lines 65-74.



Archived main f6dc6f2 E2E output in the native Terminal, enlarged view 1/2: test cases 1–8. The original SHA and command remain visible; this is a log view, not another run.

```
Viewing archived main verification output (not a new test run)
Viewer branch: docs/lab3-report-corrections
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm run test:e2e -- --config=playwright.report.config.ts --workers=1 --reporter=list
  ✓   9 [auth-tablet] › e2e/lab-03/authentication.spec.ts:11:5 › requires an initial password change and
 revokes the session on logout (656ms)
  ✓  10 [auth-mobile] › e2e/lab-03/authentication.spec.ts:11:5 › requires an initial password change and
 revokes the session on logout (720ms)
  ✓  11 [requester-desktop] › e2e/lab-03/requester-flow.spec.ts:24:5 › covers requester ownership, comme
nts, resolution, and Staff collaboration with attachments (2.2s)
  ✓  12 [accessibility] › e2e/lab-03/accessibility.spec.ts:31:5 › keyboard navigation, labels, active na
vigation, and feedback are accessible (777ms)
  ✓  13 [staff-desktop] › e2e/lab-03/staff-ticket-flow.spec.ts:24:5 › covers the staff queue, operationa
l detail, and responsive layouts (608ms)
  ✓  14 [staff-desktop] › e2e/lab-03/staff-ticket-flow.spec.ts:85:5 › claims, reassigns, unassigns, repr
ioritizes, and changes status with confirmation (879ms)
  ✓  15 [admin-desktop] › e2e/lab-03/user-administration.spec.ts:34:5 › Administrator can manage users,
reset an initial password, and use the responsive screen (1.2s)

  15 passed (20.0s)

Exit code: 0
```

Archived main f6dc6f2 E2E output in the native Terminal, enlarged view 2/2: test cases 9–15, 15 passed (20.0 s) and exit code 0. The original SHA and command remain visible; this is a log view, not another run.

```
Verification source: main
Commit: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm exec -- prisma migrate status
Prisma schema loaded from prisma/schema.prisma
Datasource "db": PostgreSQL database "lab3_report_f6dc6f2_20261003", schema "public" at "localhost:5432"

4 migrations found in prisma/migrations

Database schema is up to date!

Exit code: 0

FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm run prisma:seed

> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeded 5 categories, 7 related systems, and 5 requesters.
Seeded Lab 3 users, realistic tickets, comments, and internal notes.

Exit code: 0

FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm run build

> toktickit-server@1.0.0 build
> tsc


Exit code: 0

FINAL MAIN: f6dc6f249e75d11829d9ec9a0640ef6a884e902e
Branch: main (isolated checkout)
Command: npm run build

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
transforming...
✓ 37 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.39 kB │ gzip:  0.27 kB
dist/assets/index-BL5QWnkG.css  231.14 kB │ gzip: 30.69 kB
dist/assets/index-Du6sG0xC.js   229.25 kB │ gzip: 63.08 kB
✓ built in 327ms

Exit code: 0
```

Main f6dc6f2: migration status, seed, TypeScript server build and Vite client build; every command exited 0.

Answer Part 4

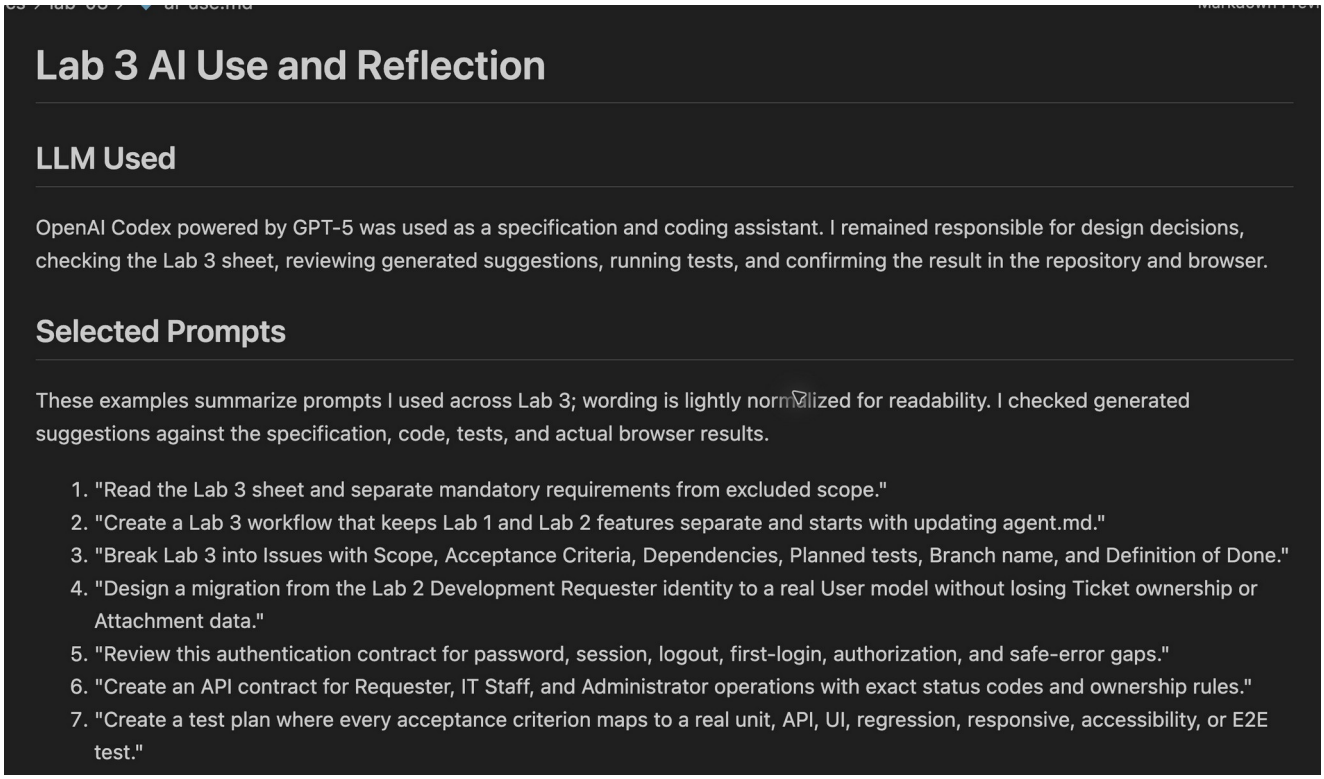
AI Use with Reflection

OpenAI Codex powered by GPT-5 was used for specification analysis, implementation support, reviews, and test planning. Generated suggestions were checked against the Lab 3 sheet, repository, database schema, automated tests, peer review, and browser behavior.

Link

AI use and reflection: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/ai-use.md

Screenshots



Lab 3 AI Use and Reflection

LLM Used

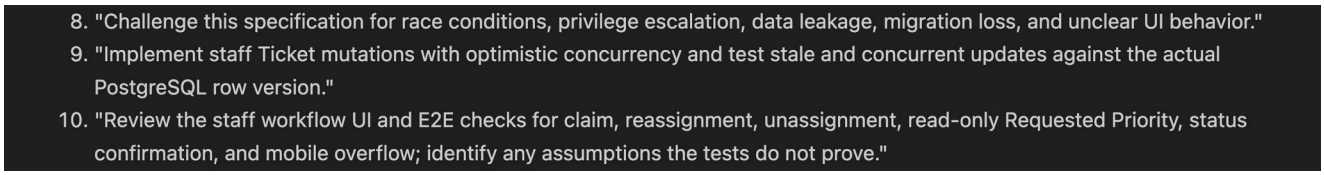
OpenAI Codex powered by GPT-5 was used as a specification and coding assistant. I remained responsible for design decisions, checking the Lab 3 sheet, reviewing generated suggestions, running tests, and confirming the result in the repository and browser.

Selected Prompts

These examples summarize prompts I used across Lab 3; wording is lightly normalized for readability. I checked generated suggestions against the specification, code, tests, and actual browser results.

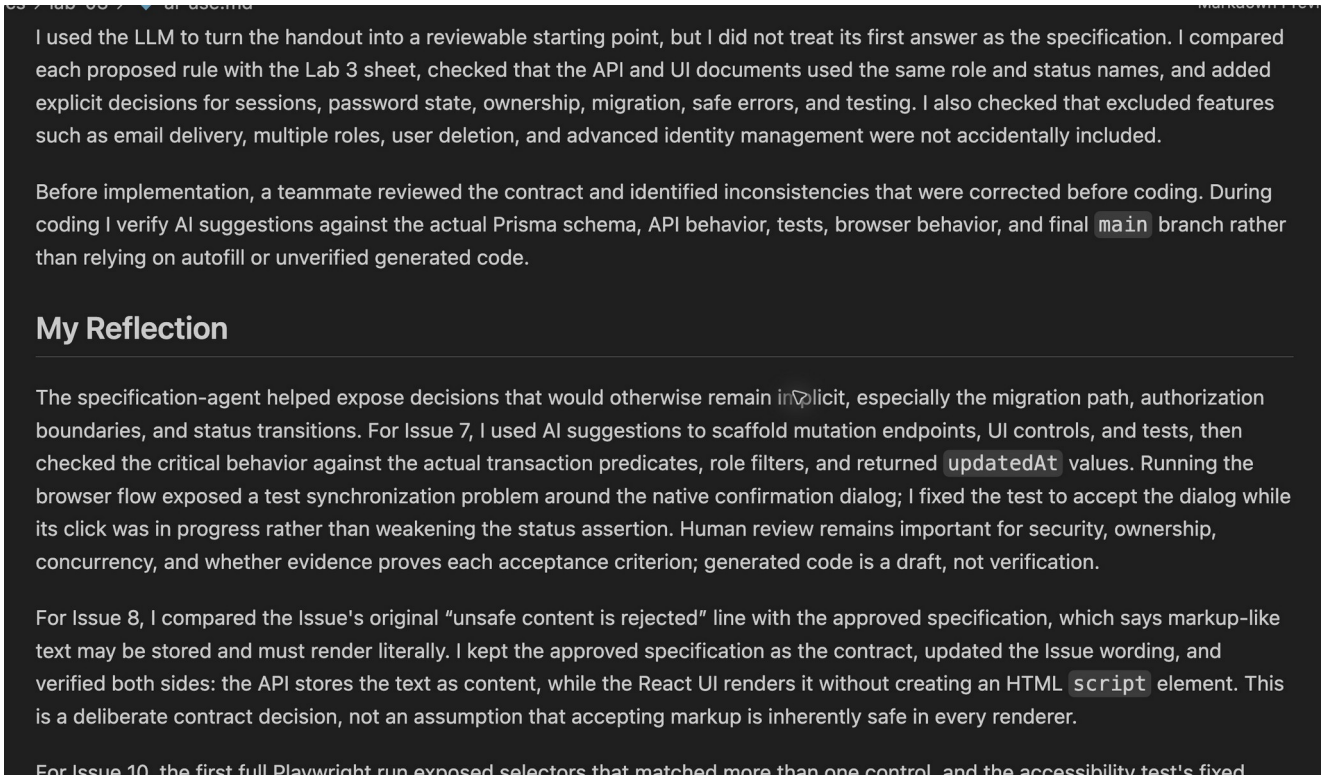
1. "Read the Lab 3 sheet and separate mandatory requirements from excluded scope."
2. "Create a Lab 3 workflow that keeps Lab 1 and Lab 2 features separate and starts with updating agent.md."
3. "Break Lab 3 into Issues with Scope, Acceptance Criteria, Dependencies, Planned tests, Branch name, and Definition of Done."
4. "Design a migration from the Lab 2 Development Requester identity to a real User model without losing Ticket ownership or Attachment data."
5. "Review this authentication contract for password, session, logout, first-login, authorization, and safe-error gaps."
6. "Create an API contract for Requester, IT Staff, and Administrator operations with exact status codes and ownership rules."
7. "Create a test plan where every acceptance criterion maps to a real unit, API, UI, regression, responsive, accessibility, or E2E test."

Actual ai-use.md Markdown Preview: LLM name and selected prompts 1-7



8. "Challenge this specification for race conditions, privilege escalation, data leakage, migration loss, and unclear UI behavior."
9. "Implement staff Ticket mutations with optimistic concurrency and test stale and concurrent updates against the actual PostgreSQL row version."
10. "Review the staff workflow UI and E2E checks for claim, reassignment, unassignment, read-only Requested Priority, status confirmation, and mobile overflow; identify any assumptions the tests do not prove."

Actual Markdown Preview: selected prompts 8-10



I used the LLM to turn the handout into a reviewable starting point, but I did not treat its first answer as the specification. I compared each proposed rule with the Lab 3 sheet, checked that the API and UI documents used the same role and status names, and added explicit decisions for sessions, password state, ownership, migration, safe errors, and testing. I also checked that excluded features such as email delivery, multiple roles, user deletion, and advanced identity management were not accidentally included.

Before implementation, a teammate reviewed the contract and identified inconsistencies that were corrected before coding. During coding I verify AI suggestions against the actual Prisma schema, API behavior, tests, browser behavior, and final `main` branch rather than relying on autofill or unverified generated code.

My Reflection

The specification-agent helped expose decisions that would otherwise remain implicit, especially the migration path, authorization boundaries, and status transitions. For Issue 7, I used AI suggestions to scaffold mutation endpoints, UI controls, and tests, then checked the critical behavior against the actual transaction predicates, role filters, and returned `updatedAt` values. Running the browser flow exposed a test synchronization problem around the native confirmation dialog; I fixed the test to accept the dialog while its click was in progress rather than weakening the status assertion. Human review remains important for security, ownership, concurrency, and whether evidence proves each acceptance criterion; generated code is a draft, not verification.

For Issue 8, I compared the Issue's original “unsafe content is rejected” line with the approved specification, which says markup-like text may be stored and must render literally. I kept the approved specification as the contract, updated the Issue wording, and verified both sides: the API stores the text as content, while the React UI renders it without creating an HTML `script` element. This is a deliberate contract decision, not an assumption that accepting markup is inherently safe in every renderer.

For Issue 10, the first full Playwright run exposed selectors that matched more than one control, and the accessibility test's fixed

Actual Markdown Preview: How I Used AI Critically and My Reflection

text may be stored and must render literally. I kept the approved specification as the contract, updated the Issue wording, and verified both sides: the API stores the text as content, while the React UI renders it without creating an HTML `script` element. This is a deliberate contract decision, not an assumption that accepting markup is inherently safe in every renderer.

For Issue 10, the first full Playwright run exposed selectors that matched more than one control, and the accessibility test's fixed number of Tab presses was not reliable after navigation. I checked the rendered page and test flow, used specific control IDs where labels were duplicated, and anchored the keyboard check at a known form field before pressing Tab. I reran the failing test and the complete E2E suite, then linked the screenshot evidence to the viewport names. I left final `main` verification pending until the release is merged and did not generate a PDF, as requested.

Actual Markdown Preview: reflection on contract decisions, browser checks and test reliability

Answer Part 5

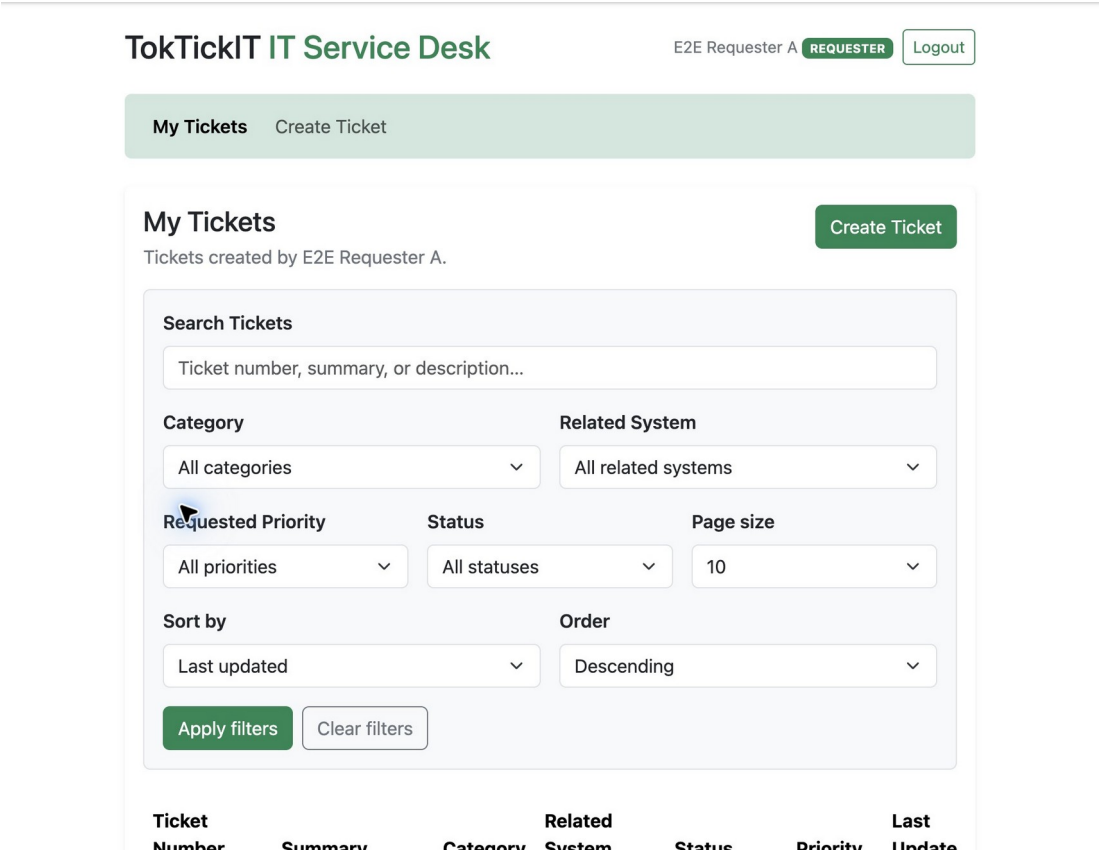
Working Login and Password Change UI

Authentication uses the real User identity and one role. The evidence set covers valid and invalid login, inactive accounts, busy/safe failure states, mandatory first-password change, authenticated role display, logout, and blocked direct access after logout.

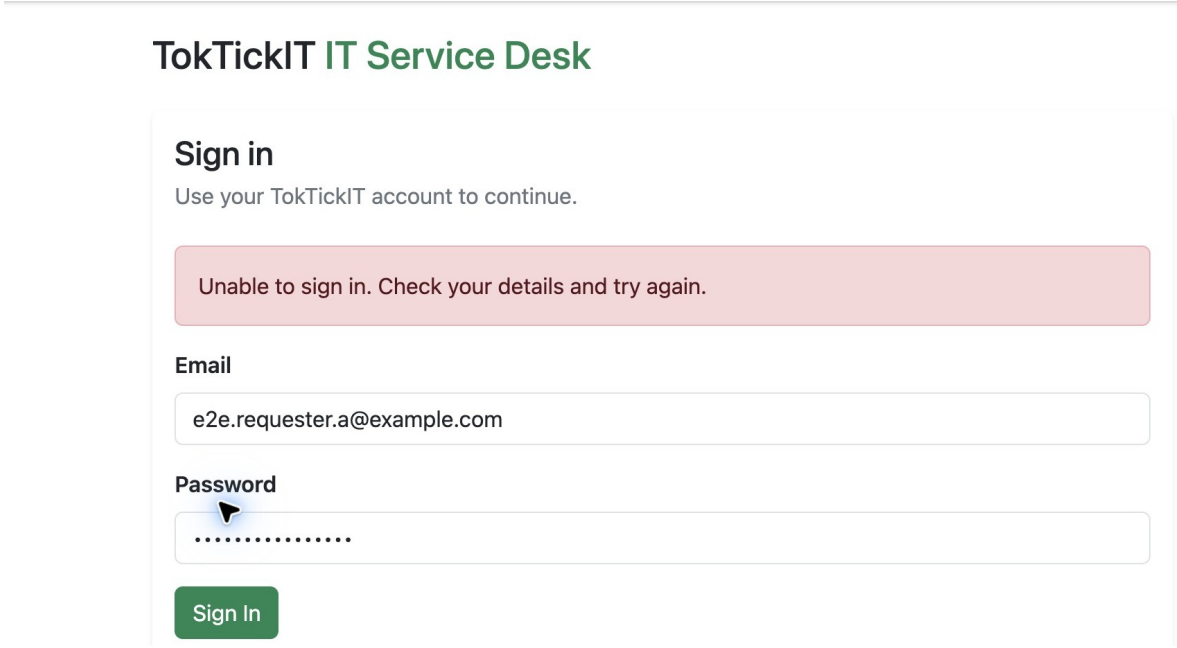
Links

UI specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/ui-spec.md#3-login-screen
Authentication E2E: https://github.com/songt888/toktickit/blob/f6dc6f2/e2e/lab-03/authentication.spec.ts
Authentication API tests: https://github.com/songt888/toktickit/blob/f6dc6f2/server/tests/lab-03/auth.api.test.ts

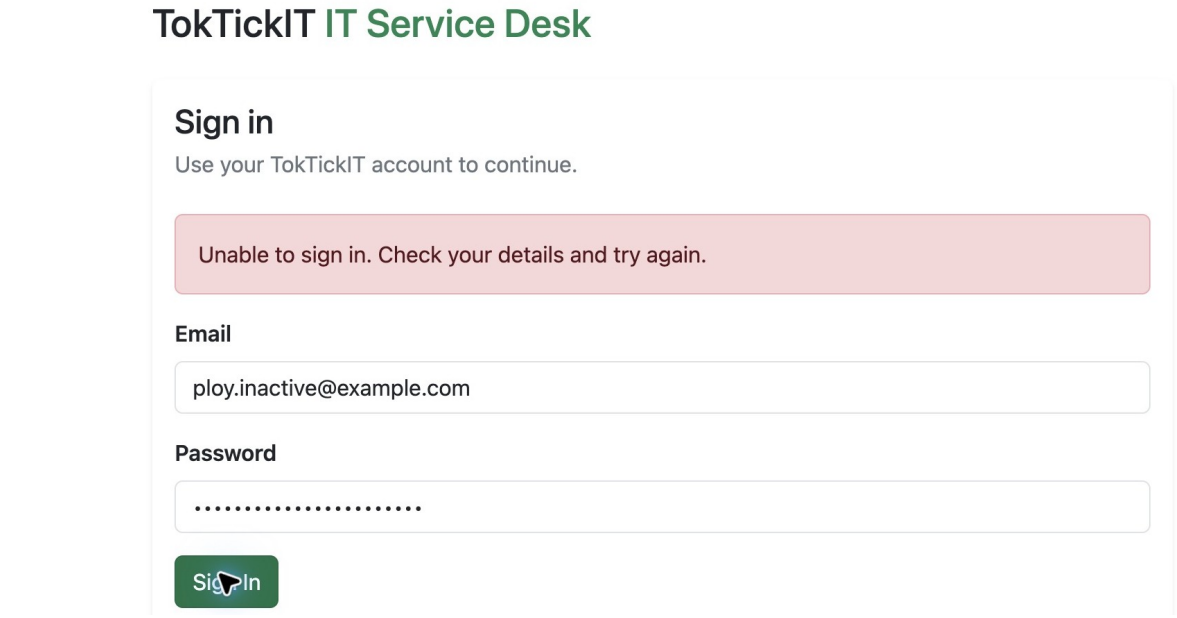
Screenshots



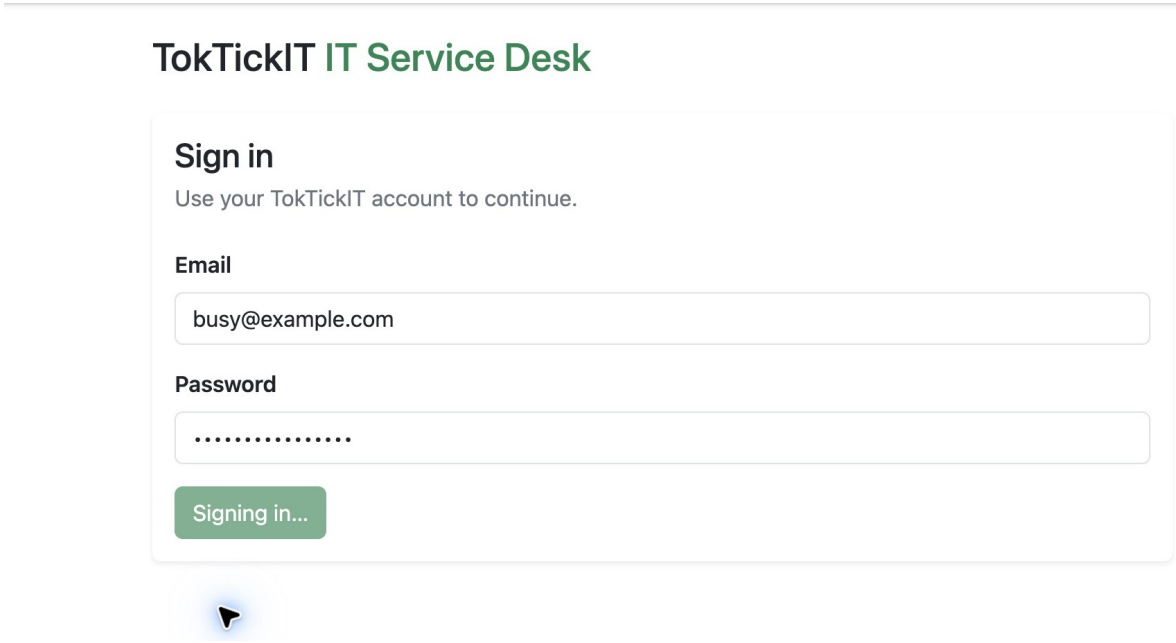
valid active-user login continuing to the permitted role screen



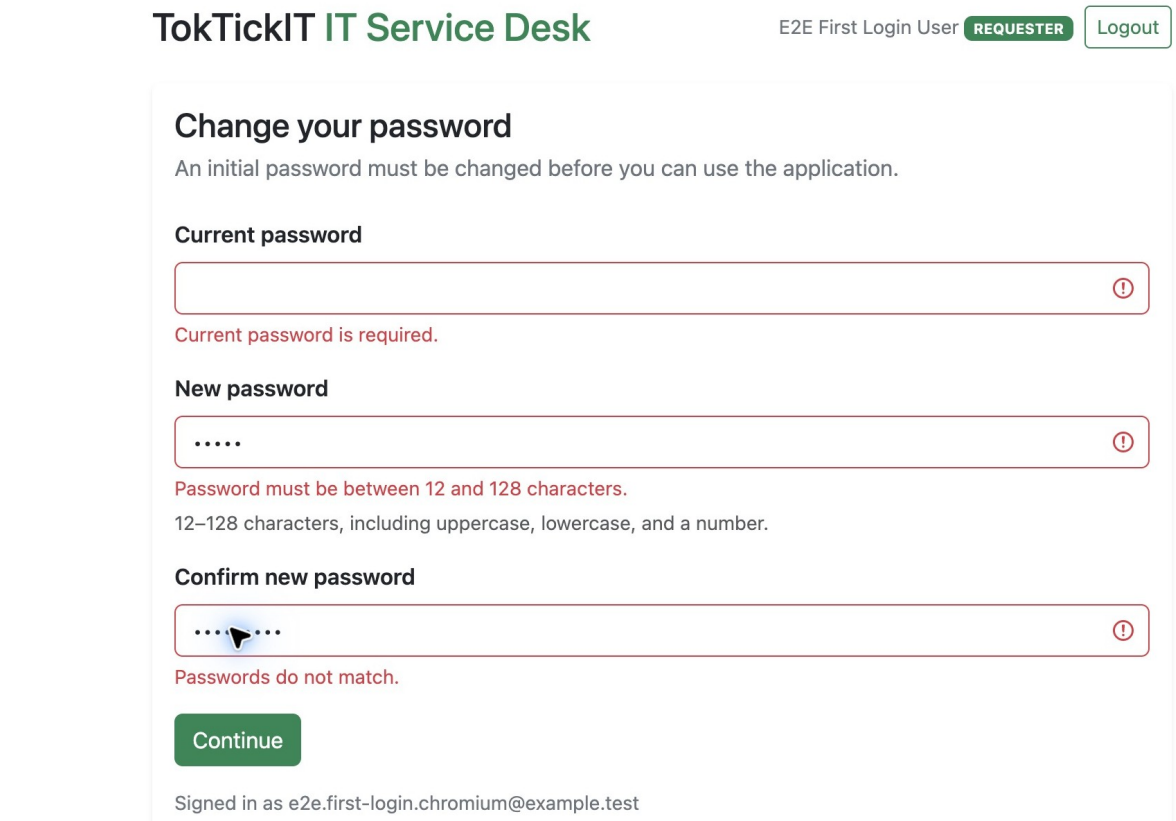
invalid credentials with a safe message



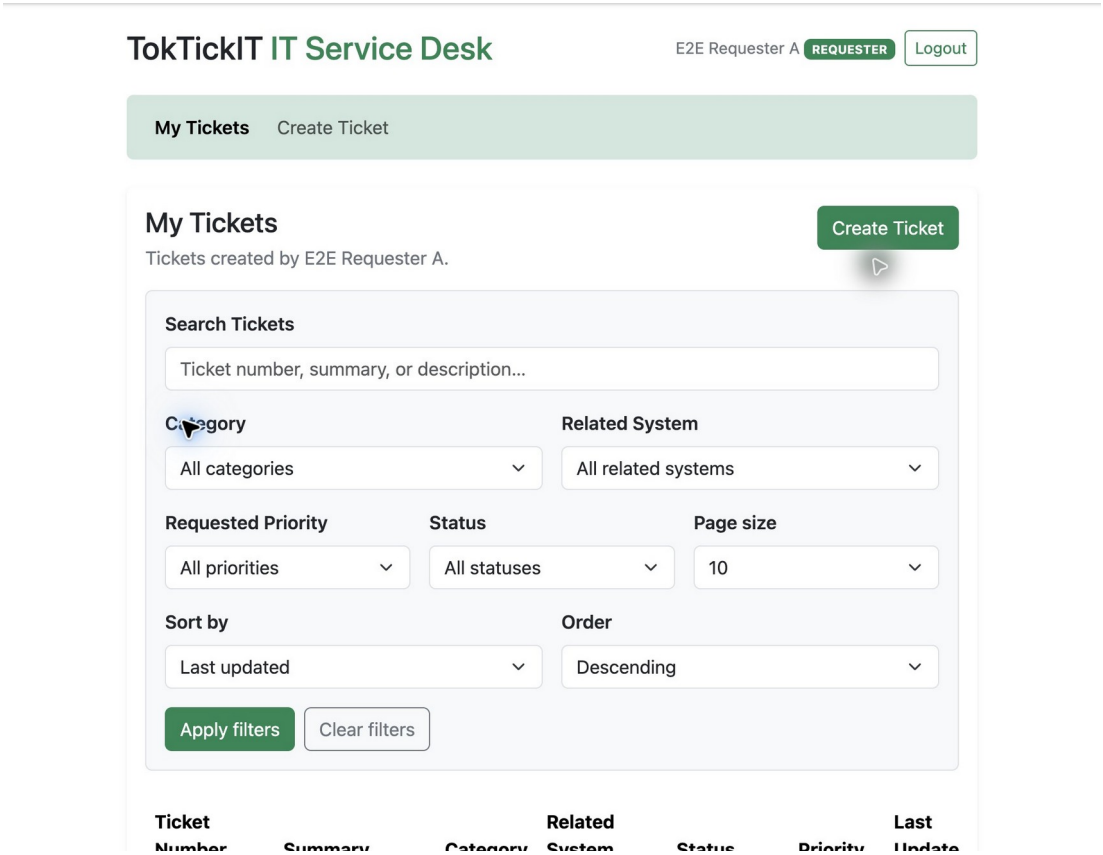
inactive account handled without revealing account details



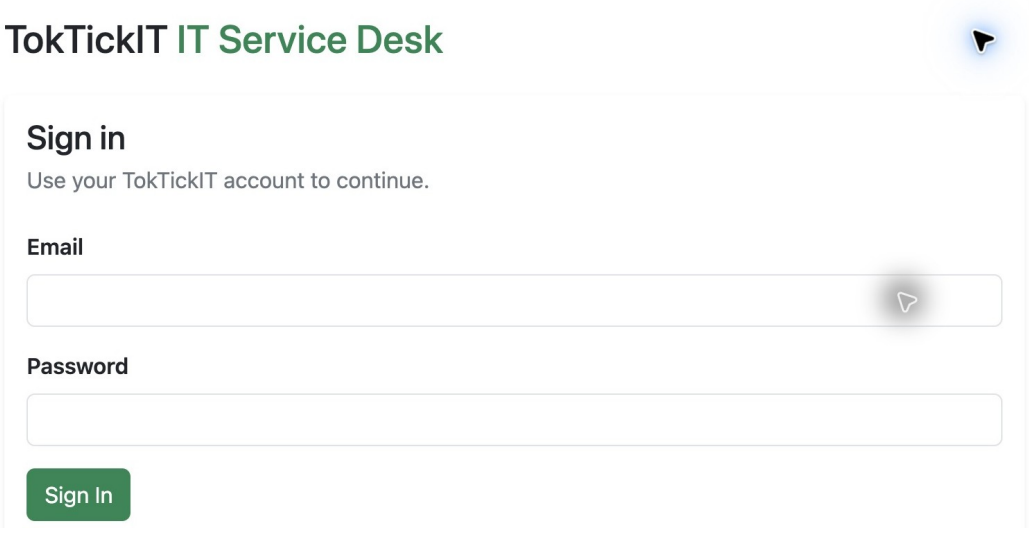
Sign In is disabled while the request is in progress



Change Password field validation



current user name, role, role navigation, and Logout



Login screen after logout; session revocation is separately covered by the authentication tests

Initial Login and mandatory Change Password screens at desktop/tablet/mobile sizes are shown once in Part 9.

Answer Part 6

Working IT Staff Ticket Queue UI

The IT Staff queue presents realistic Tickets with search, filtering, sorting, pagination, ownership, status, requested/IT priority, and an Open action. Separate states show loading, empty/no results, safe failure, and responsive behavior.

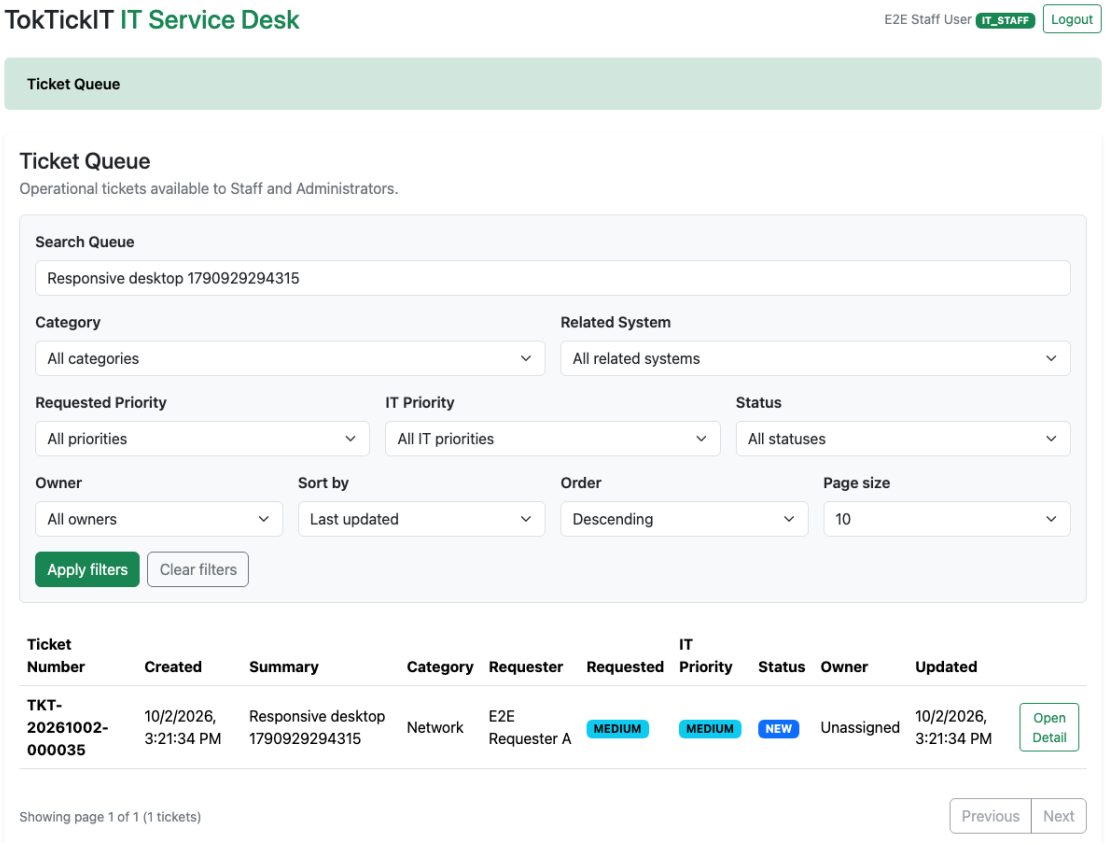
Links

Queue UI specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/ui-spec.md#6-it-staff-ticket-queue

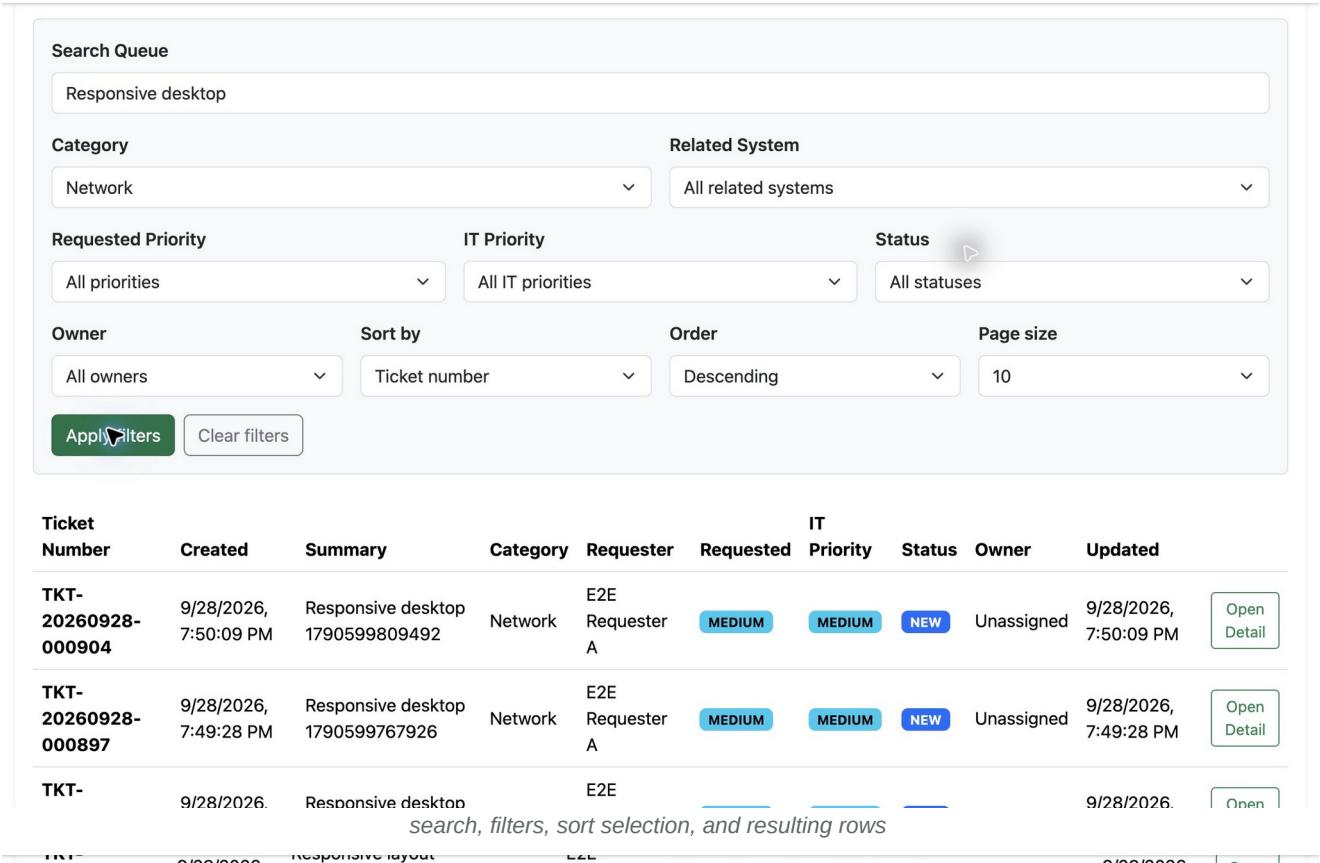
Queue API tests: https://github.com/songt888/toktickit/blob/f6dc6f2/server/tests/lab-03/staff-queue.api.test.ts

Queue UI tests: https://github.com/songt888/toktickit/blob/f6dc6f2/client/tests/lab-03/StaffTicketQueue.test.tsx

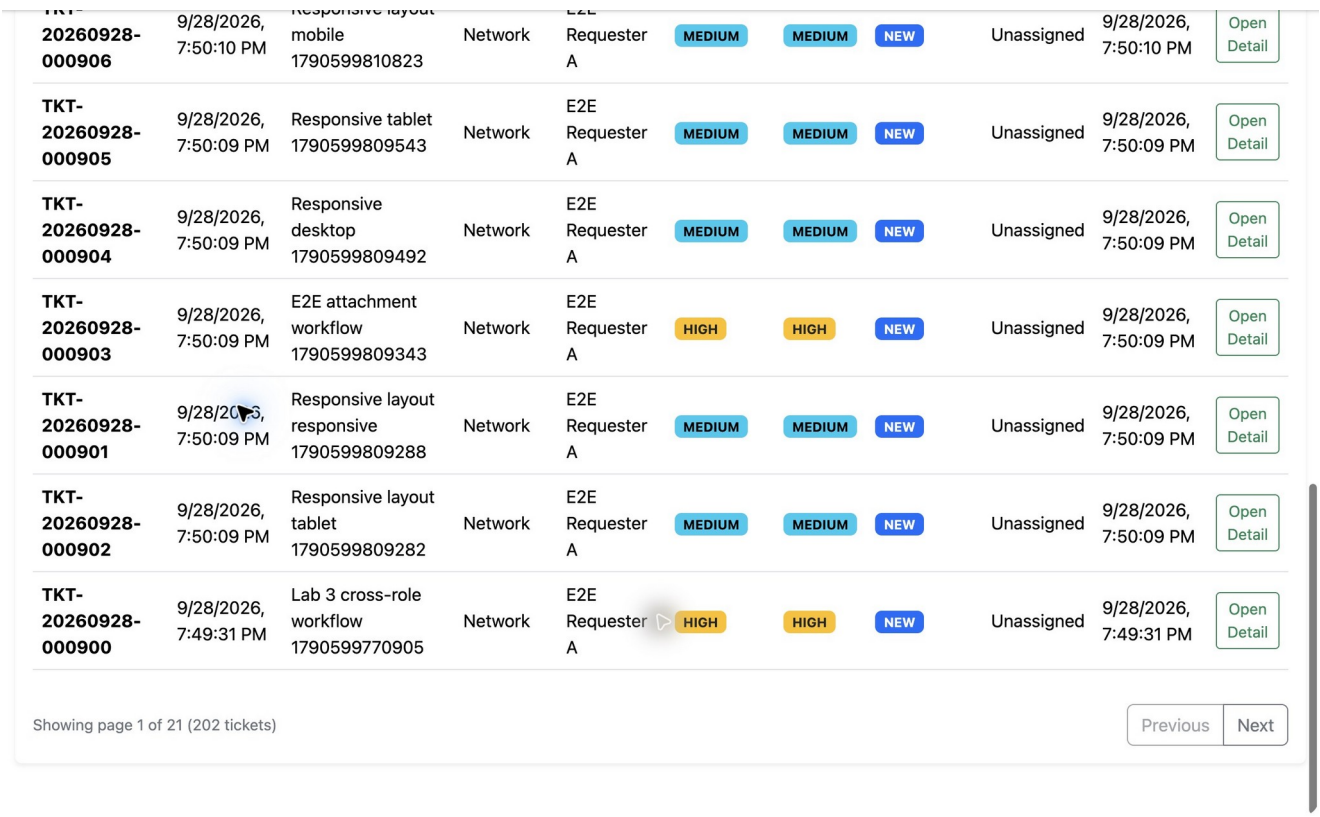
Screenshots



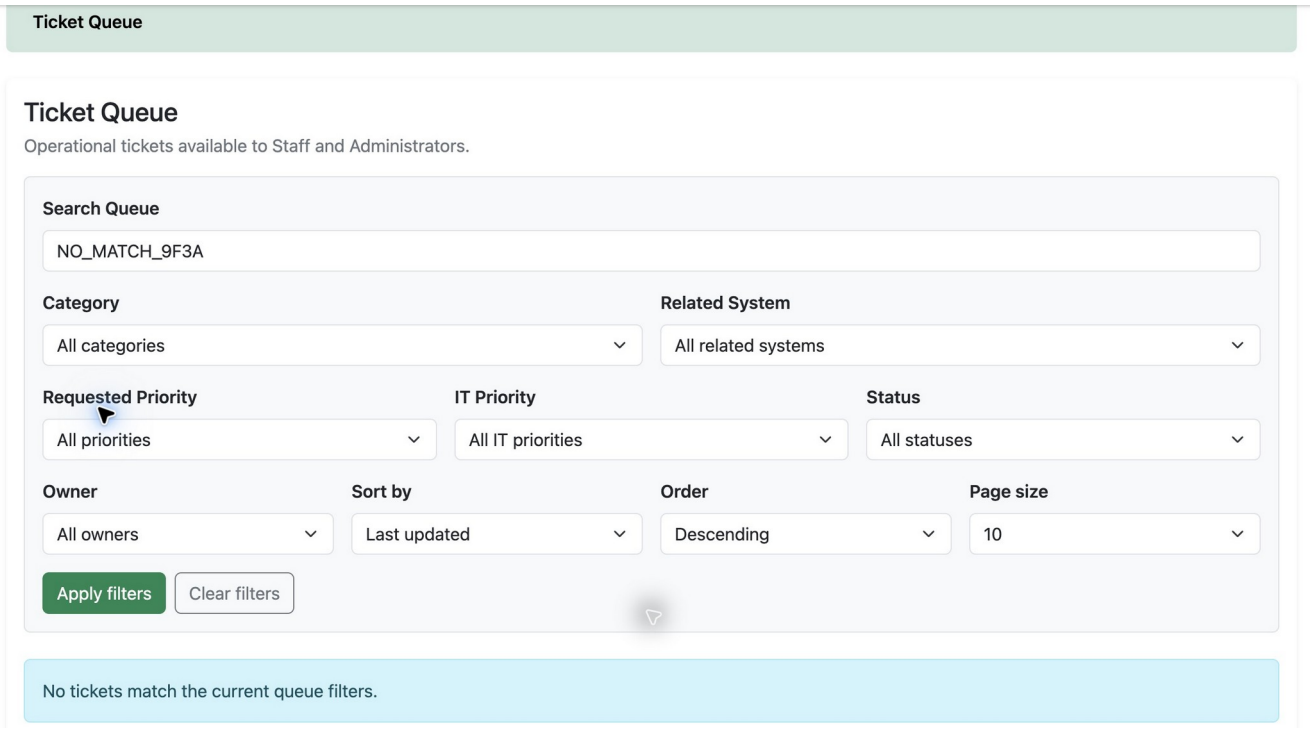
desktop queue with realistic Ticket rows and Open actions



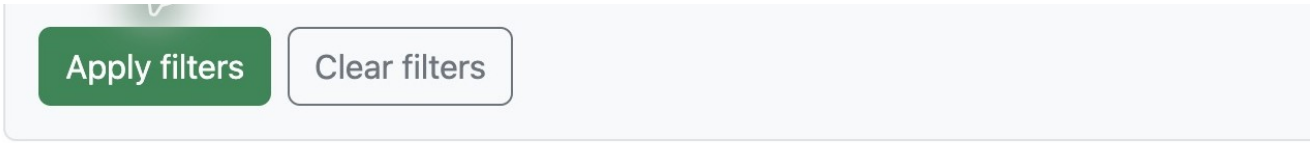
search, filters, sort selection, and resulting rows

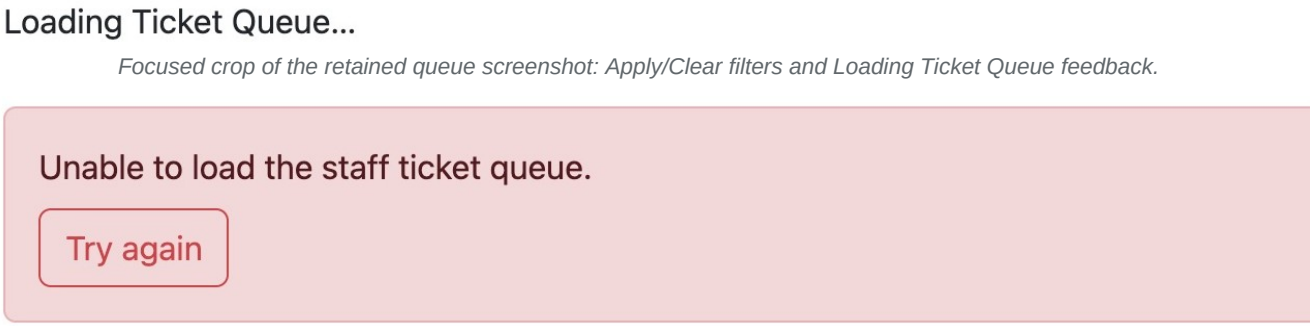


pagination plus assigned/unassigned, status, and priority badges



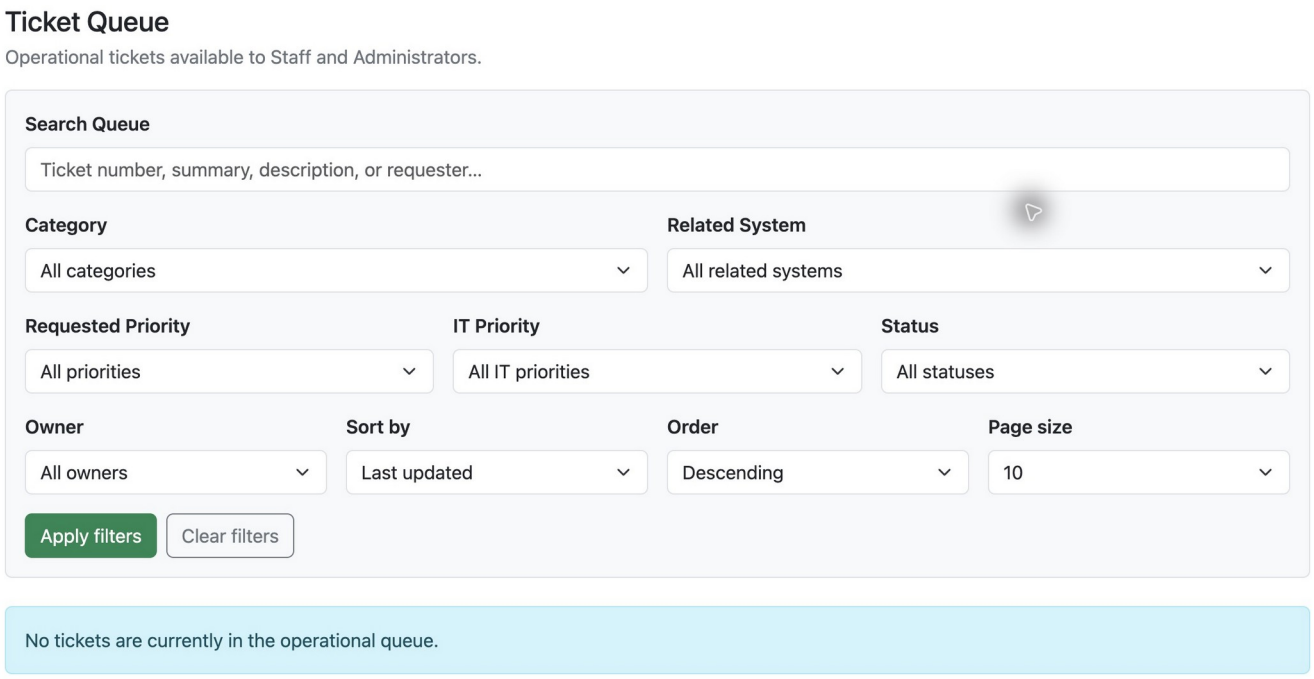
No-results feedback for the current queue filters





Focused crop of the retained queue screenshot: Apply/Clear filters and Loading Ticket Queue feedback.

Focused crop of the retained failure screenshot: Unable to load the staff ticket queue and the complete Try again action. Repeated filter fields are shown above, not duplicated here.



Actual main f6dc6f2 UI on a separate migrated database with zero Tickets: blank search, unrestricted filters and unfiltered empty queue feedback. No existing Tickets were deleted.

Answer Part 7

Working IT Staff Ticket Detail UI

The Staff detail workflow supports ownership, IT Priority, permitted status transitions, Public Comments, private Internal Notes, attachment continuity, and Requester resolution indication. Role restrictions, validation, safe failures, and direct API authorization are verified separately.

Links

Detail UI specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/ui-spec.md#7-it-staff-ticket-detail

Staff detail API tests: https://github.com/songt888/toktickit/blob/f6dc6f2/server/tests/lab-03/staff-ticket-detail.api.test.ts

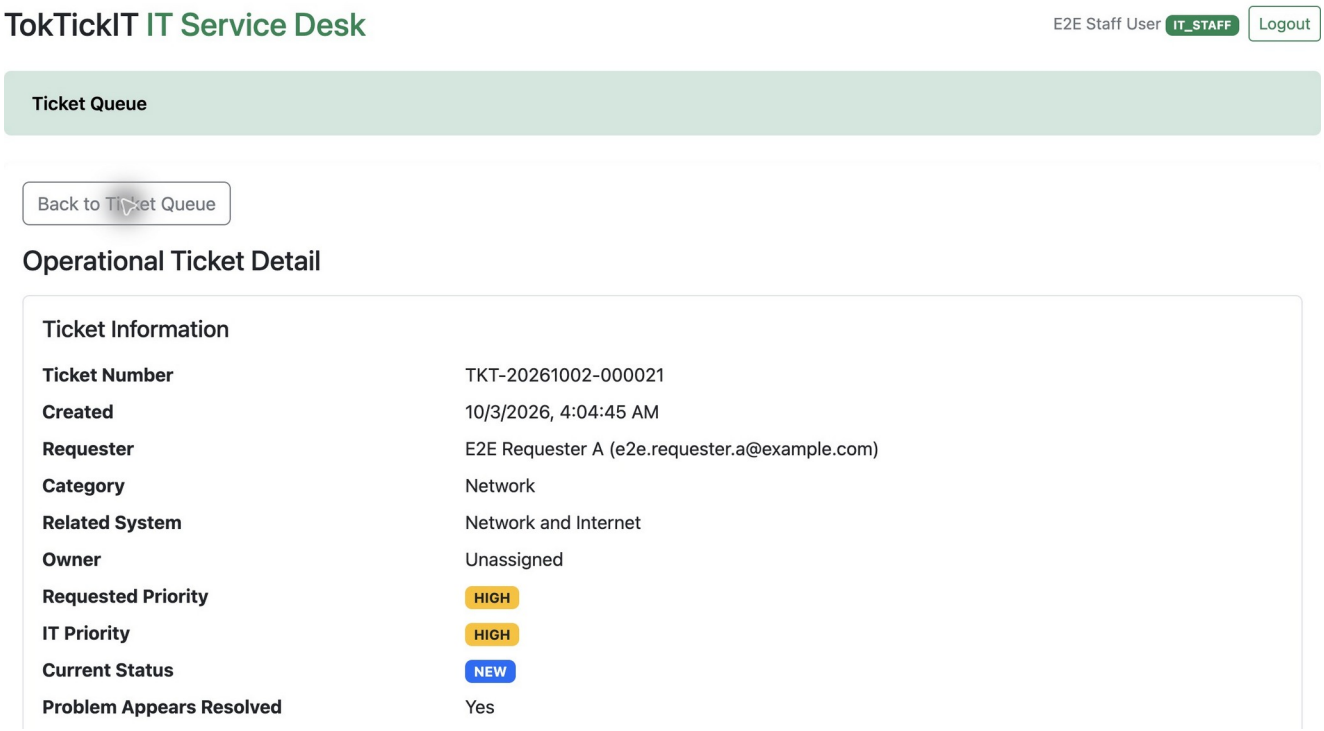
Comments and notes API tests: https://github.com/songt888/toktickit/blob/f6dc6f2/server/tests/lab-03/comments-notes.api.test.ts

Staff workflow E2E: https://github.com/songt888/toktickit/blob/f6dc6f2/e2e/lab-03/staff-ticket-flow.spec.ts

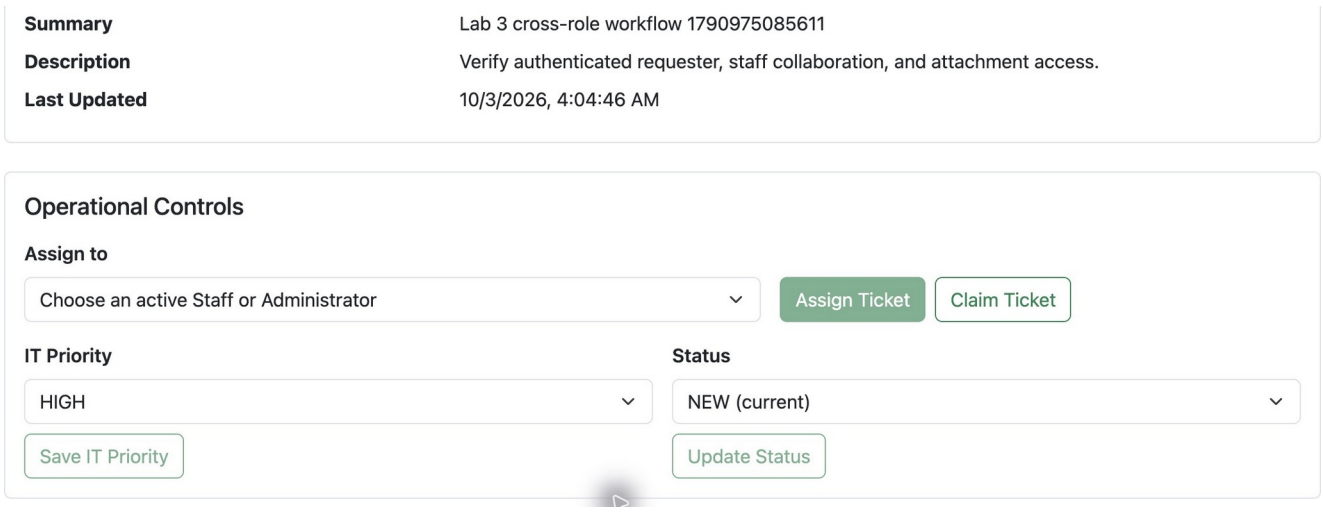
Screenshots

Complete desktop Staff Detail — main f6dc6f2

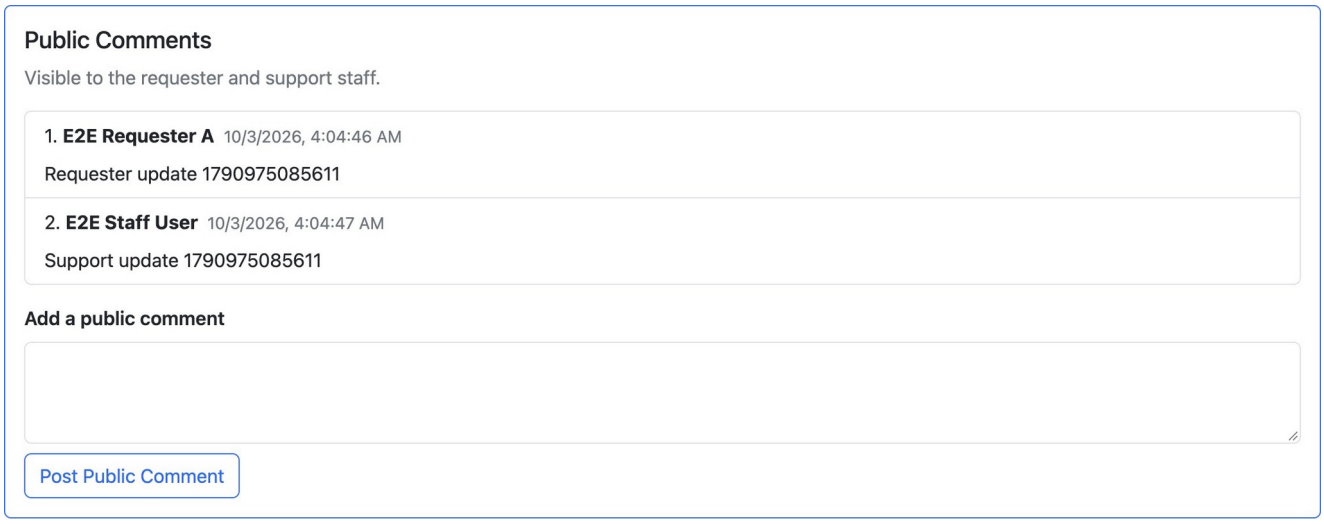
Consecutive native Edge views at 100% browser zoom, 1176 px wide, show the same Ticket TKT-20261002-000021. Each crop ends between complete rows or sections; the lower views continue the same page.



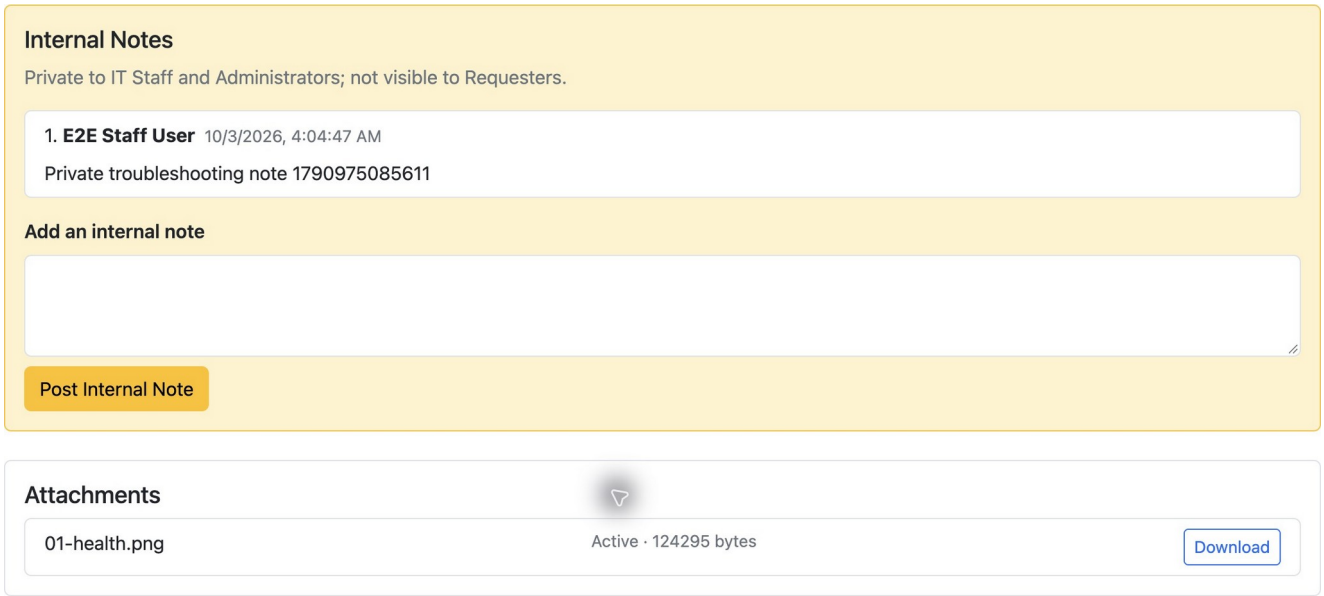
Desktop 1/4: full authenticated shell and Ticket Information through Problem Appears Resolved.



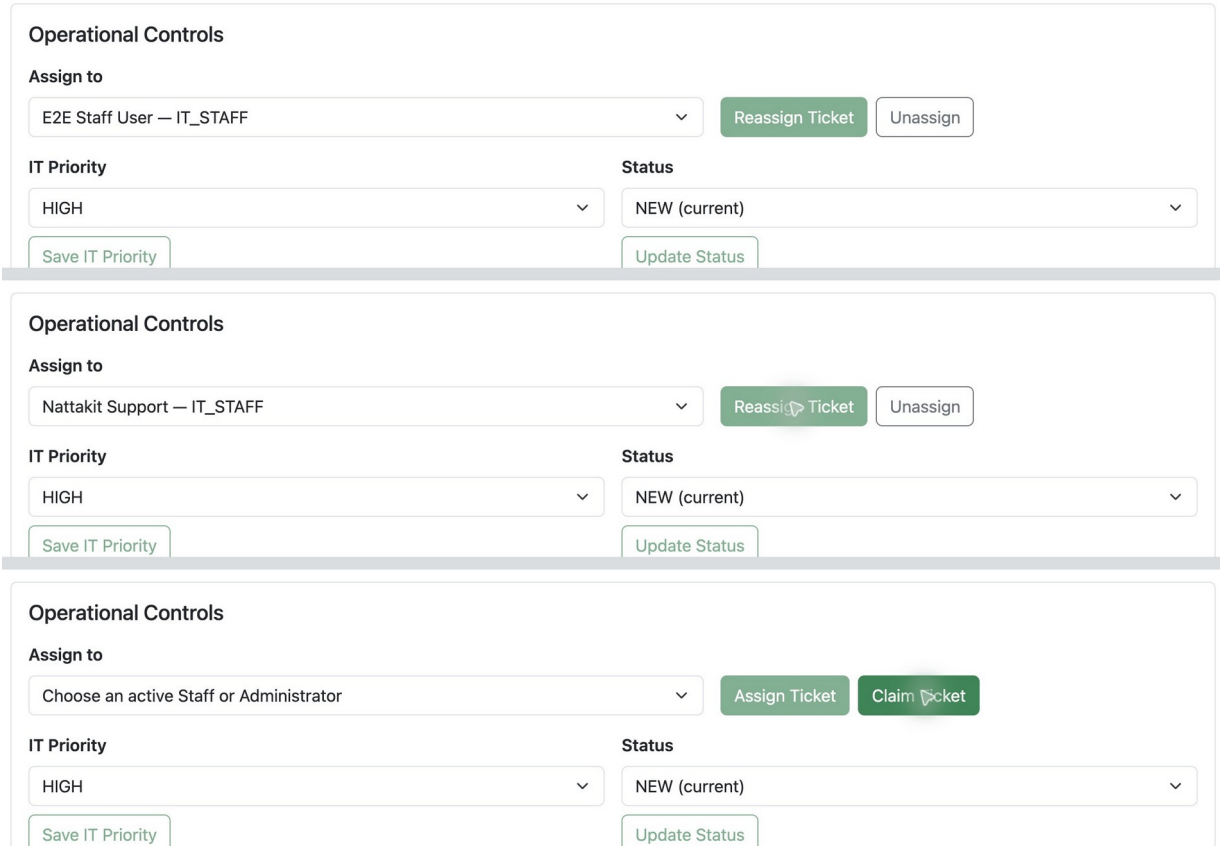
Desktop 2/4: remaining Summary, Description and Last Updated, followed by complete Operational Controls.



Desktop 3/4: complete Public Comments, authors/times, comment input and Post Public Comment action.

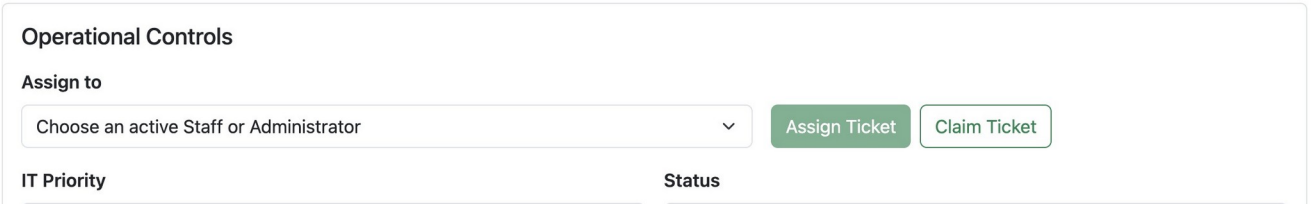


Desktop 4/4: complete Internal Notes, note input/Post Internal Note and attachment metadata/Download action.



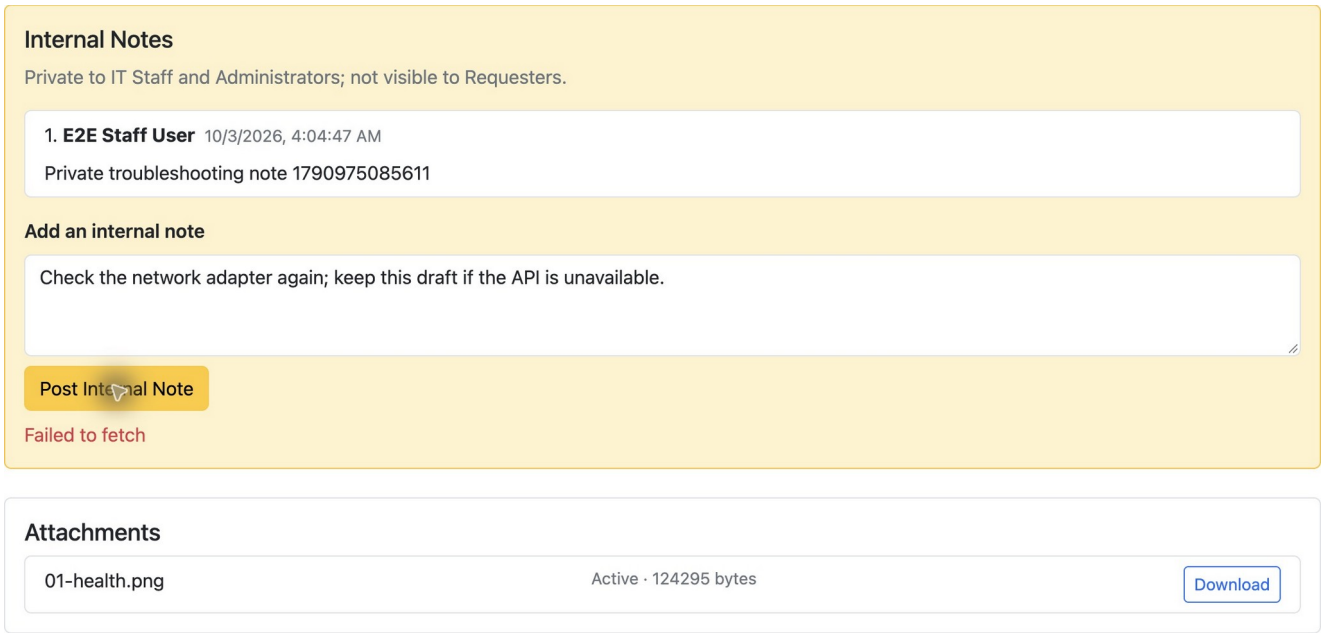
claim, reassign, and unassign results

Owner	Unassigned
Requested Priority	HIGH
IT Priority	URGENT
Current Status	IN_PROGRESS
Problem Appears Resolved	Yes
Summary	Lab 3 cross-role workflow 1790599812625
Description	Verify authenticated requester, staff collaboration, and attachment access.
Last Updated	9/29/2026, 12:07:34 AM

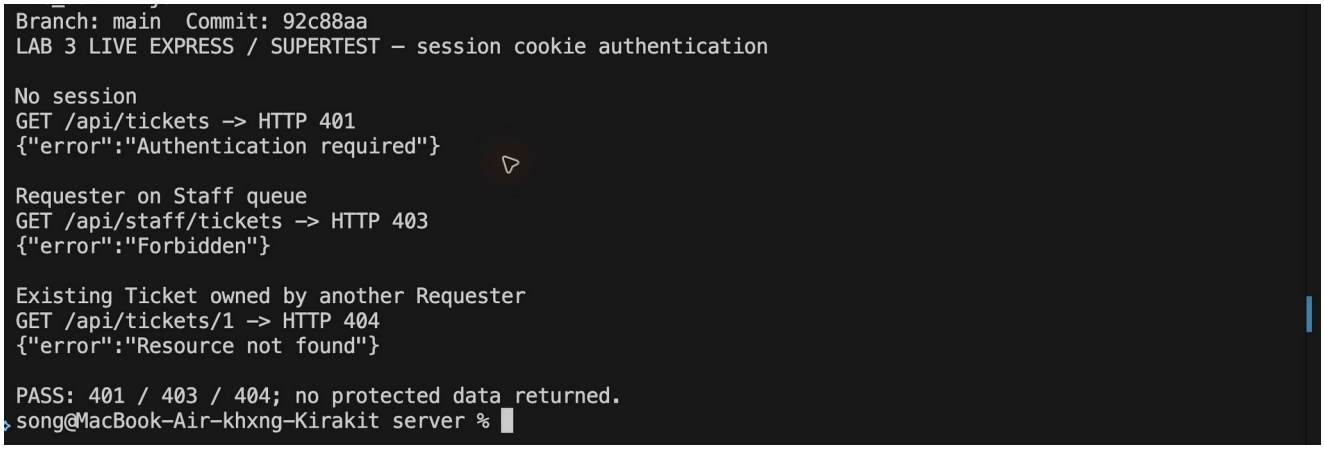


Updated IT Priority URGENT and Current Status IN_PROGRESS

Separate blank-comment validation: message is placed next to the Public Comment action.



Actual network failure after stopping only the isolated capture API: Post Internal Note shows Failed to fetch. The entire draft, existing note and attachment metadata remain visible. No note was saved; this is separate from field validation.



Retained historical main 92c88aa Lab 3 Express/Supertest capture: no session = 401, Requester on Staff queue = 403, another Requester's existing Ticket = 404. No protected data is returned; session-cookie authentication, not X-Requester-Id. Current main authorization tests also pass in Part 3.

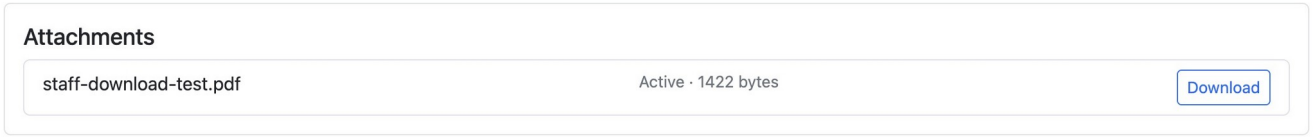
The complete Requester Ticket Detail screenshots in Part 9 show the authenticated shell, owned Ticket, Public Comment and Problem Appears Resolved = Yes. The runtime 404 above demonstrates ownership isolation. The screens show the resulting resolved state, not a before/after action sequence.

Staff attachment continuity and download authorization are covered by the requester-flow E2E test listed in Part 3. The following local browser check additionally demonstrates a completed Staff download.

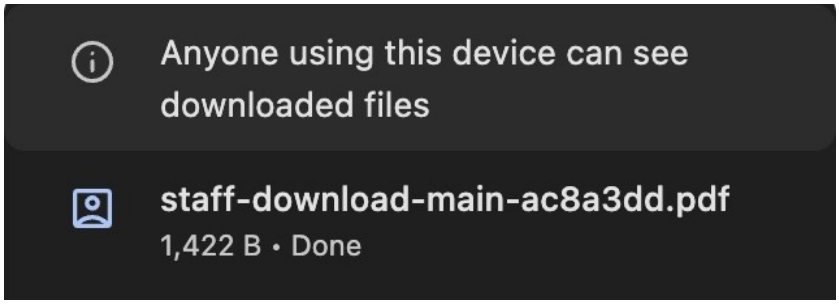
Staff attachment download in Chrome

Local browser verification from the main ac8a3dd application snapshot after approved PR #60 merged. Nattakit Support (IT_STAFF) downloaded the existing active staff-download-test.pdf attached to TKT-20260919-900002. It was saved as staff-download-main-ac8a3dd.pdf. Chrome shows 1,422 B and Done; its SHA-256 matches the previously verified original. No password or database reset was performed.

Actual Chrome screenshot: authenticated Nattakit Support with IT_STAFF role.



Actual Chrome screenshot: the active PDF attachment and Staff Download action.



Actual Chrome download panel: staff-download-main-ac8a3dd.pdf, 1,422 B, Done.

Answer Part 8

Working Administrator User Management UI

The minimalist Administrator screen supports listing, search, role filtering, create/edit, one-role assignment, activation, and initial-password reset. It also proves duplicate/invalid input handling, required next-login password change, Administrator safety rules, forbidden access, safe failure, and responsive behavior.

Links

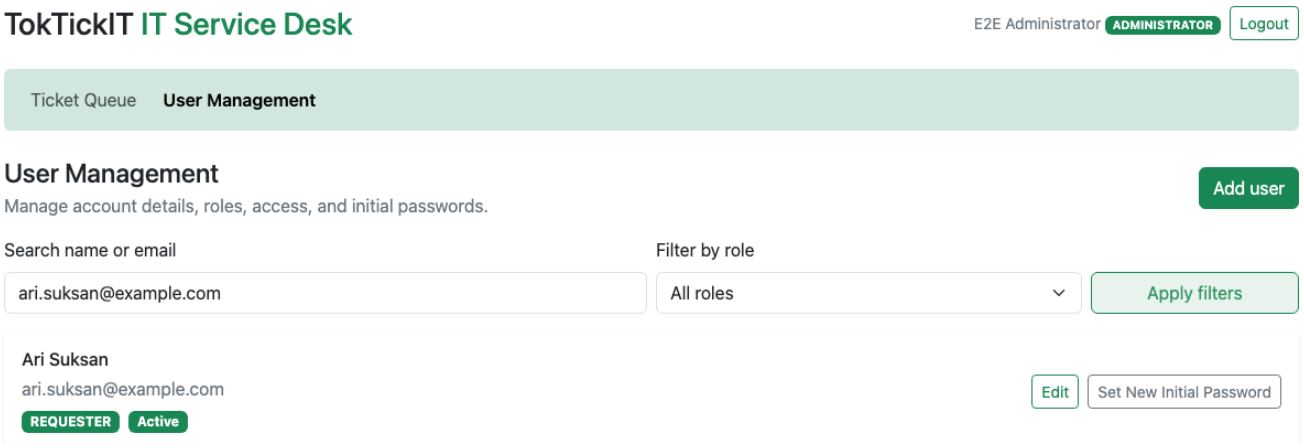
Admin UI specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/ui-spec.md#8-administrator-user-management

Admin API tests: https://github.com/songt888/toktickit/blob/f6dc6f2/server/tests/lab-03/users-admin.api.test.ts

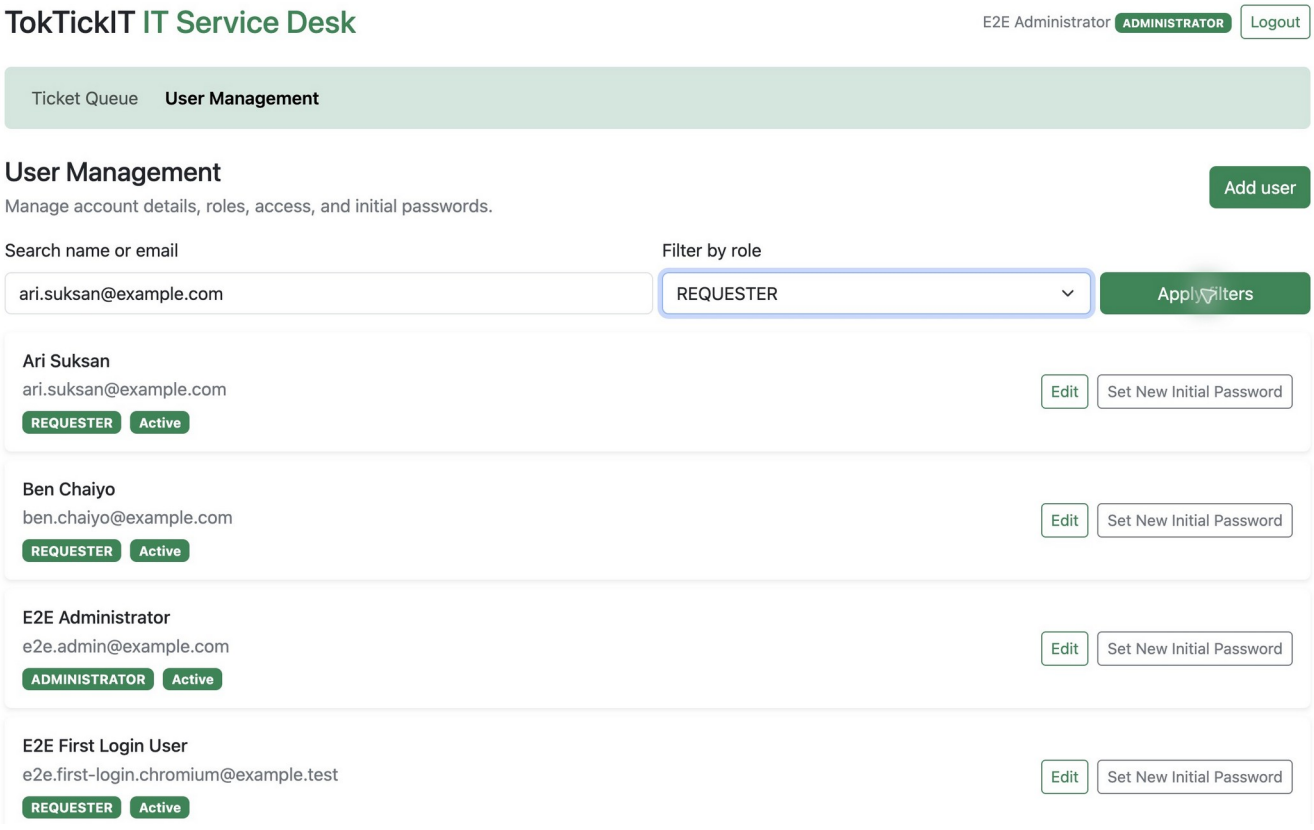
Admin UI tests: https://github.com/songt888/toktickit/blob/f6dc6f2/client/tests/lab-03/UserManagement.test.tsx

Admin E2E: https://github.com/songt888/toktickit/blob/f6dc6f2/e2e/lab-03/user-administration.spec.ts

Screenshots



Name, Email, Role, Status, and Edit action



name/email search and optional role filtering

TokTickIT IT Service Desk

E2E Administrator ADMINISTRATOR Logout

Ticket Queue **User Management**

User Management

Manage account details, roles, access, and initial passwords.

Add user

Create user

Name: Lab 3 Evidence Staff

Email: lab3.evidence.staff@example.test

Role: IT STAFF

Active account

Initial password: ••••••••••••••••••

12–128 characters, including uppercase, lowercase, and a number.

Save user | Cancel

Search name or email: ari.suksan@example.com

Filter by role: REQUESTER

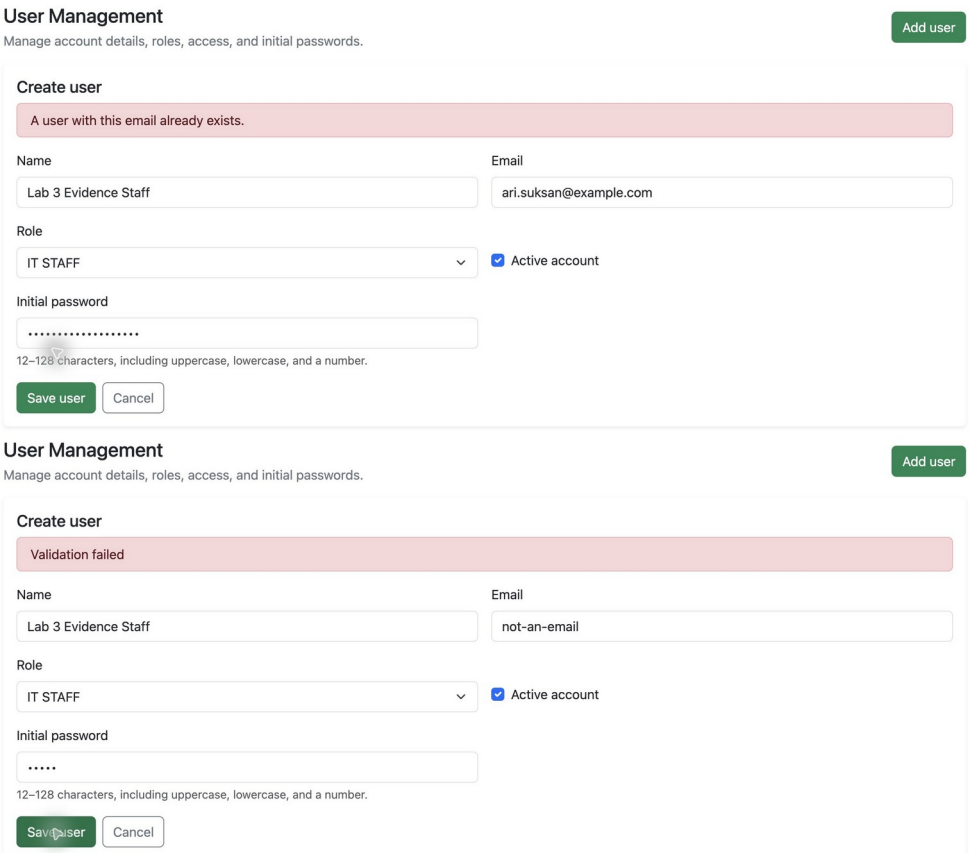
Apply filters

Ari Suksan

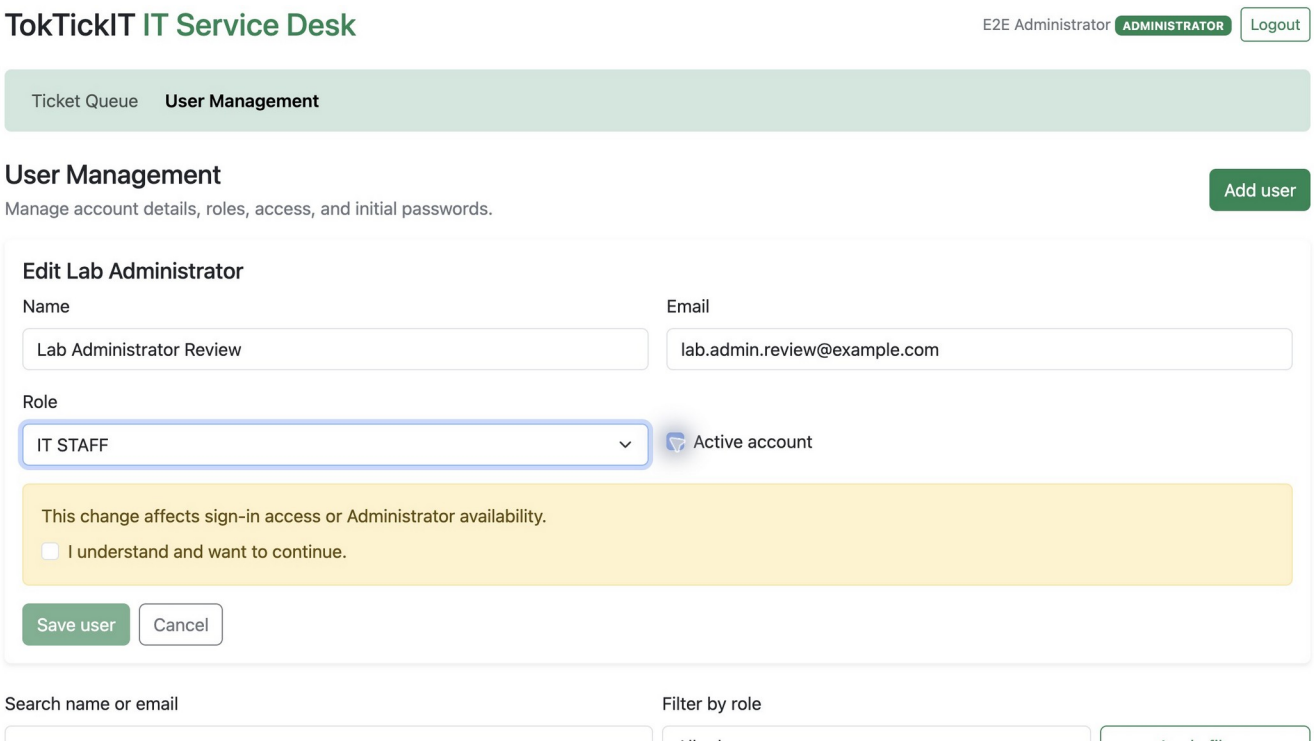
ari.suksan@example.com

Edit | Set New Initial Password

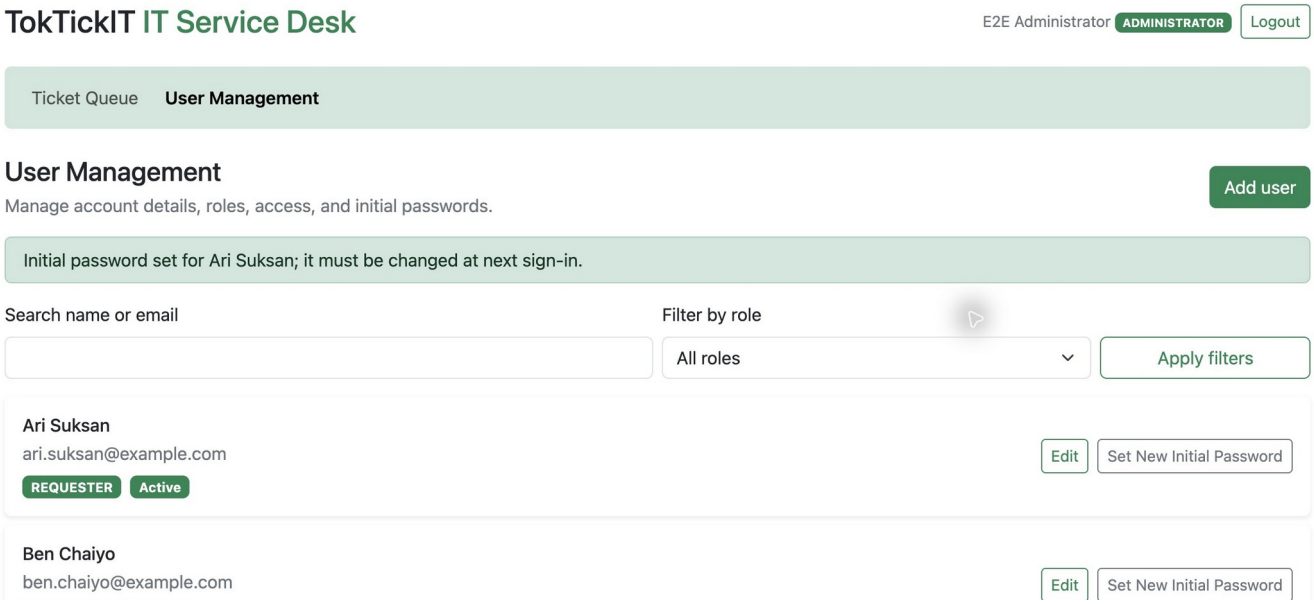
create user with one role, activation state, and initial password



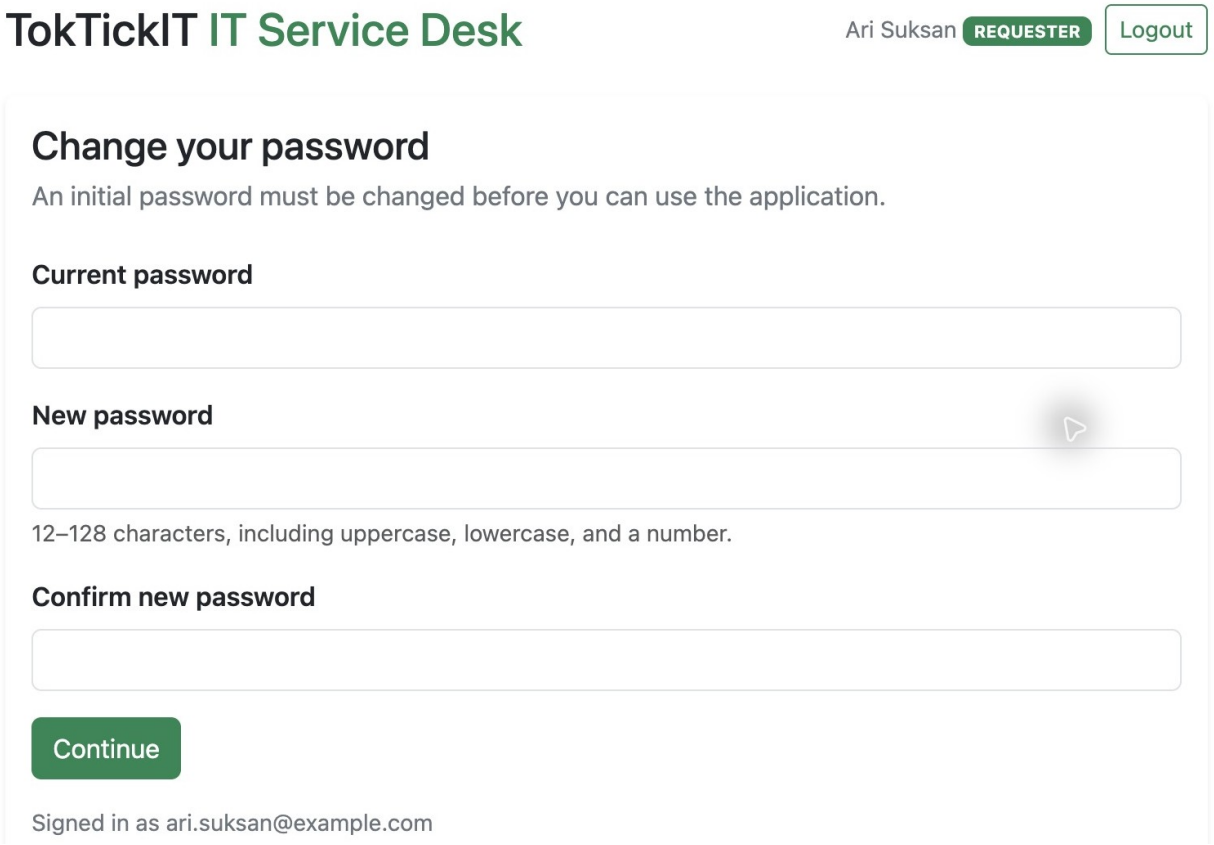
duplicate email and invalid-input validation



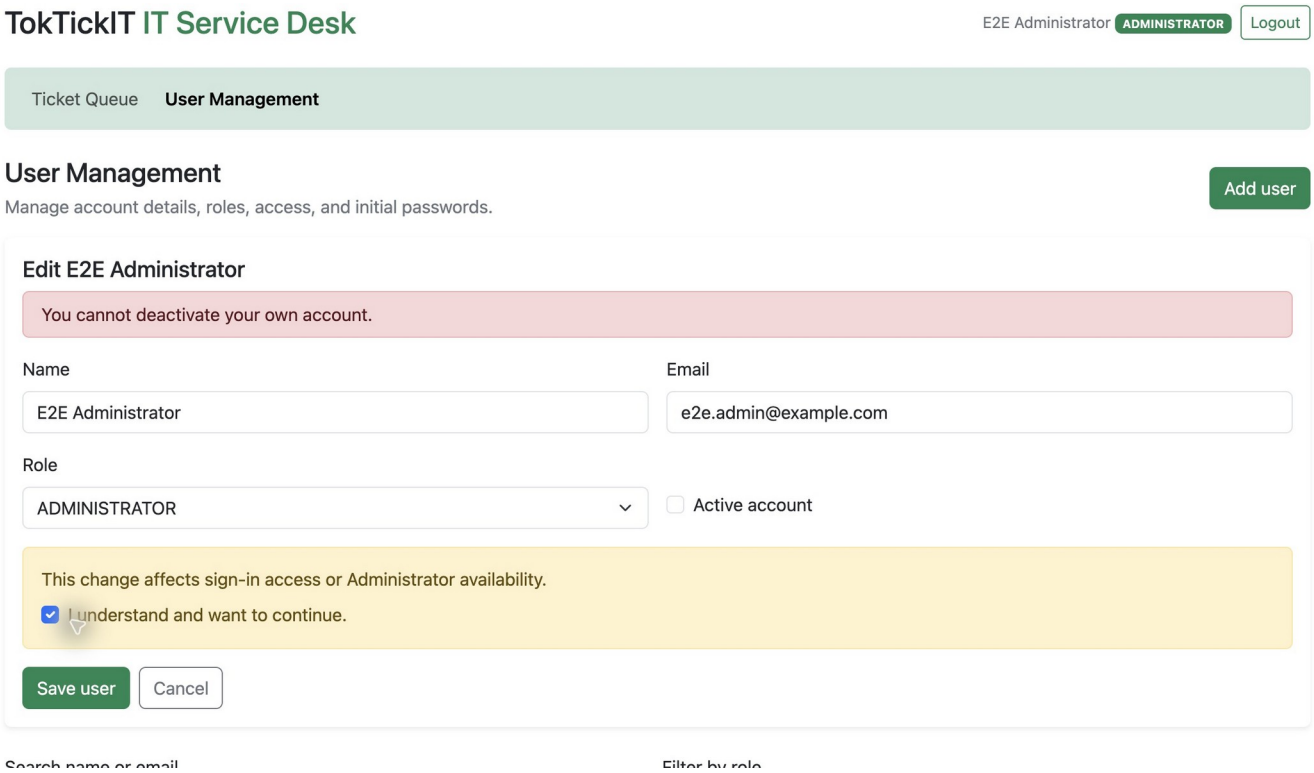
edit name, email, role, and activation state



Administrator sets a new initial password



reset user is forced to Change Password at next login



self-deactivation is prevented

TokTickIT IT Service Desk

E2E Administrator ADMINISTRATOR Logout

Ticket Queue **User Management**

User Management

Manage account details, roles, access, and initial passwords.

Add user

Edit E2E Administrator

The last active Administrator cannot be deactivated or demoted.

Name: E2E Administrator

Email: e2e.admin@example.com

Role: IT STAFF

Active account

This change affects sign-in access or Administrator availability.

I understand and want to continue.

Save user | Cancel

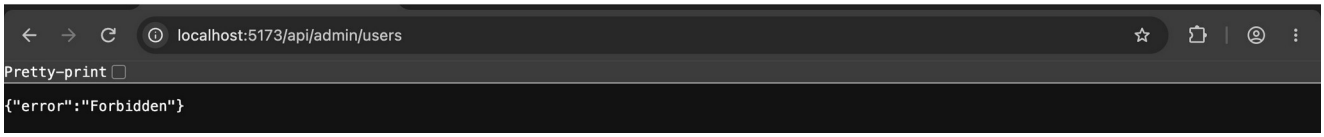
last active Administrator cannot be deactivated or demoted

TokTickIT IT Service Desk

E2E Staff User IT_STAFF Logout

Ticket Queue

Retained Staff shell: IT_STAFF role and Ticket Queue navigation, without User Management. Repeated queue filters omitted.



Retained real browser API response: /api/admin/users returns Forbidden for that non-Administrator session, without user data.

Edit E2E Requester A

Failed to fetch

Name: E2E Requester A - pending update

Email: e2e.requester.a@example.com

Role: REQUESTER

Active account

Save user | Cancel

Search name or email

Filter by role: All roles

Apply filters

Ari Suksan

ari.suksan@example.com

REQUESTER Active

Edit | Set New Initial Password

Actual main f6dc6f2 Administrator network failure, not duplicate-email rejection: Failed to fetch appears after Save user while the isolated capture API is stopped. Name, email, role, Active account, controls and loaded user cards remain. No edit was saved.

Answer Part 9

Zen Green UI and Responsive Evidence

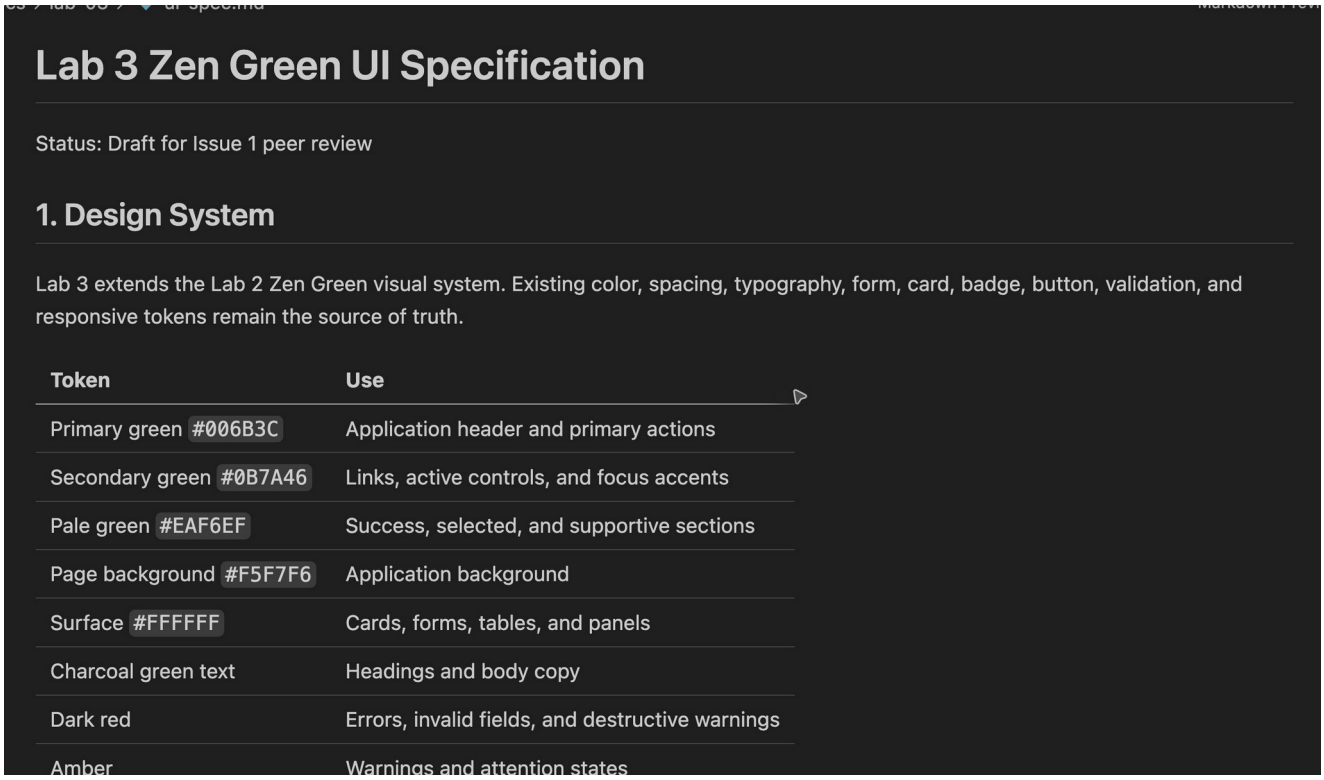
The UI specification records the Zen Green design and visual checklist. Responsive E2E measures document/body width; My Tickets uses an internal table scroll area. Complete desktop Staff Detail appears in Part 7; tablet/mobile views follow below.

These are browser screenshots with outer whitespace trimmed. Numbered segments continue the same capture, divided between whole fields or cards; adjacent panels run from left to right. Captions identify the screen and viewport.

Links

UI specification: https://github.com/songt888/toktickit/blob/f6dc6f2/docs/lab-03/ui-spec.md
Screenshot folder: https://github.com/songt888/toktickit/tree/f6dc6f2/artifacts/lab-03/screenshots
Responsive E2E: https://github.com/songt888/toktickit/blob/f6dc6f2/e2e/lab-03/responsive.spec.ts
Accessibility E2E: https://github.com/songt888/toktickit/blob/f6dc6f2/e2e/lab-03/accessibility.spec.ts

Screenshots



Lab 3 Zen Green UI Specification

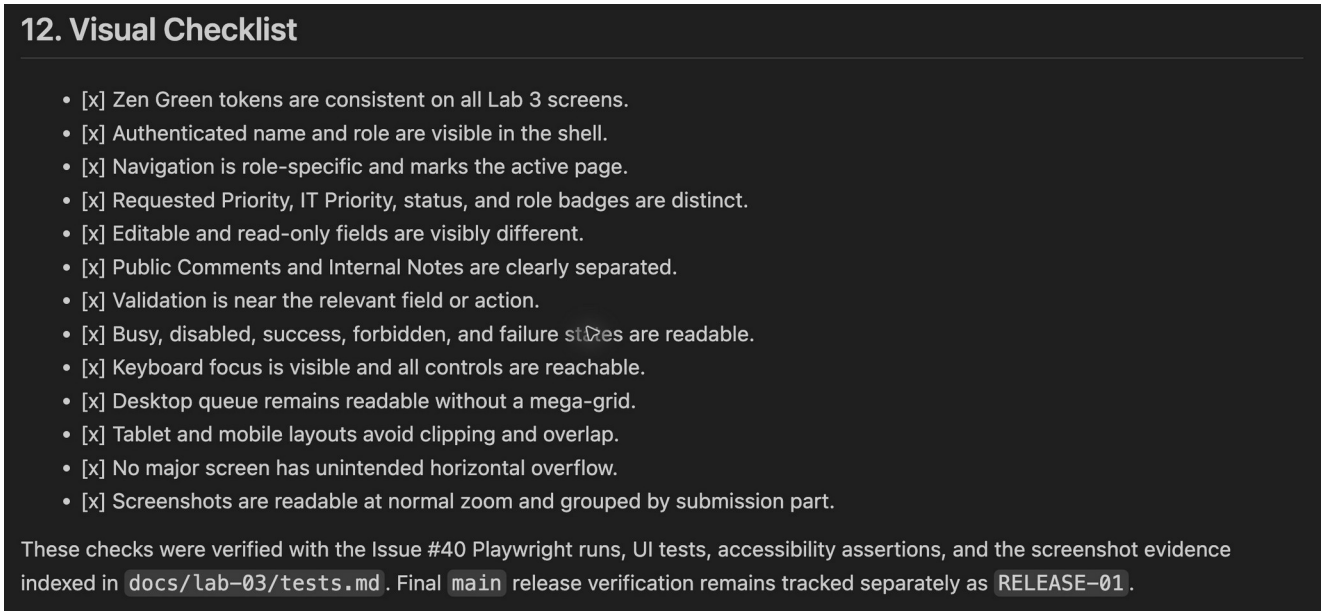
Status: Draft for Issue 1 peer review

1. Design System

Lab 3 extends the Lab 2 Zen Green visual system. Existing color, spacing, typography, form, card, badge, button, validation, and responsive tokens remain the source of truth.

Token	Use
Primary green `#006B3C`	Application header and primary actions
Secondary green `#0B7A46`	Links, active controls, and focus accents
Pale green `#EAF6EF`	Success, selected, and supportive sections
Page background `#F5F7F6`	Application background
Surface `#FFFFFF`	Cards, forms, tables, and panels
Charcoal green text	Headings and body copy
Dark red	Errors, invalid fields, and destructive warnings
Amber	Warnings and attention states

Actual ui-spec.md Markdown Preview: the shared Zen Green design system



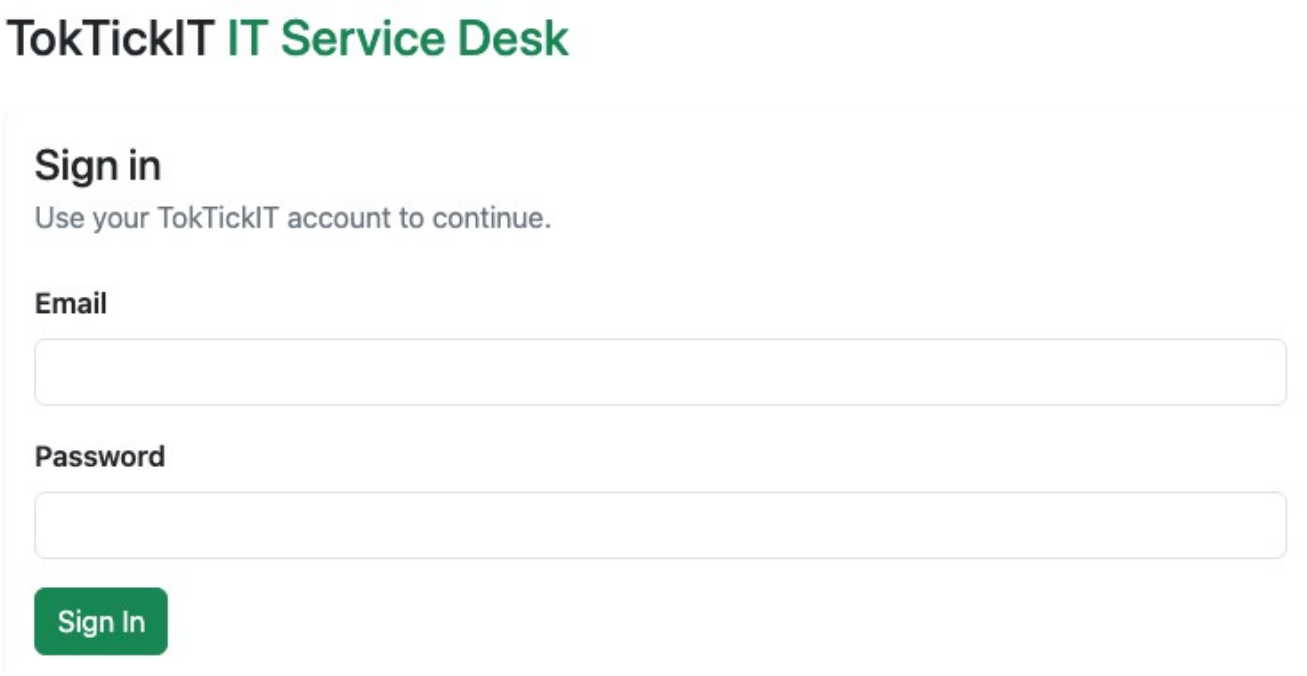
12. Visual Checklist

- [x] Zen Green tokens are consistent on all Lab 3 screens.
- [x] Authenticated name and role are visible in the shell.
- [x] Navigation is role-specific and marks the active page.
- [x] Requested Priority, IT Priority, status, and role badges are distinct.
- [x] Editable and read-only fields are visibly different.
- [x] Public Comments and Internal Notes are clearly separated.
- [x] Validation is near the relevant field or action.
- [x] Busy, disabled, success, forbidden, and failure states are readable.
- [x] Keyboard focus is visible and all controls are reachable.
- [x] Desktop queue remains readable without a mega-grid.
- [x] Tablet and mobile layouts avoid clipping and overlap.
- [x] No major screen has unintended horizontal overflow.
- [x] Screenshots are readable at normal zoom and grouped by submission part.

These checks were verified with the Issue #40 Playwright runs, UI tests, accessibility assertions, and the screenshot evidence indexed in `docs/lab-03/tests.md`. Final `main` release verification remains tracked separately as `RELEASE-01`.

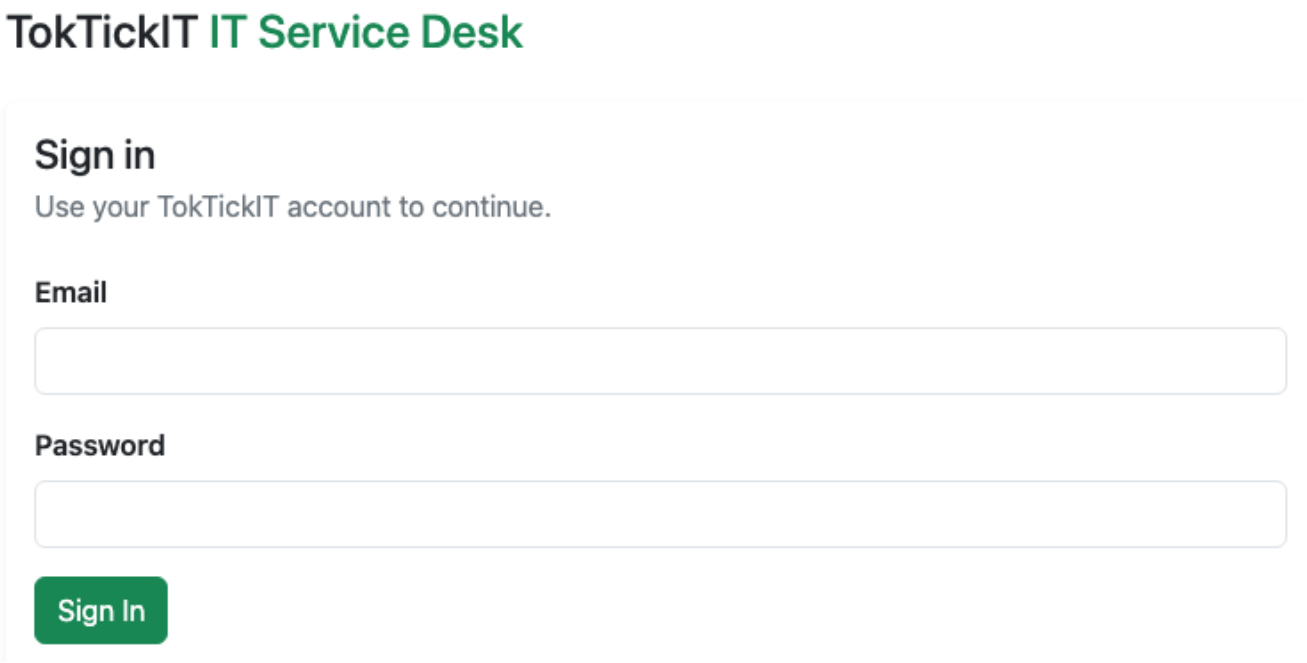
Actual ui-spec.md Markdown Preview: all 13 repository visual checks; unchanged on main

Login - desktop 1280 x 900



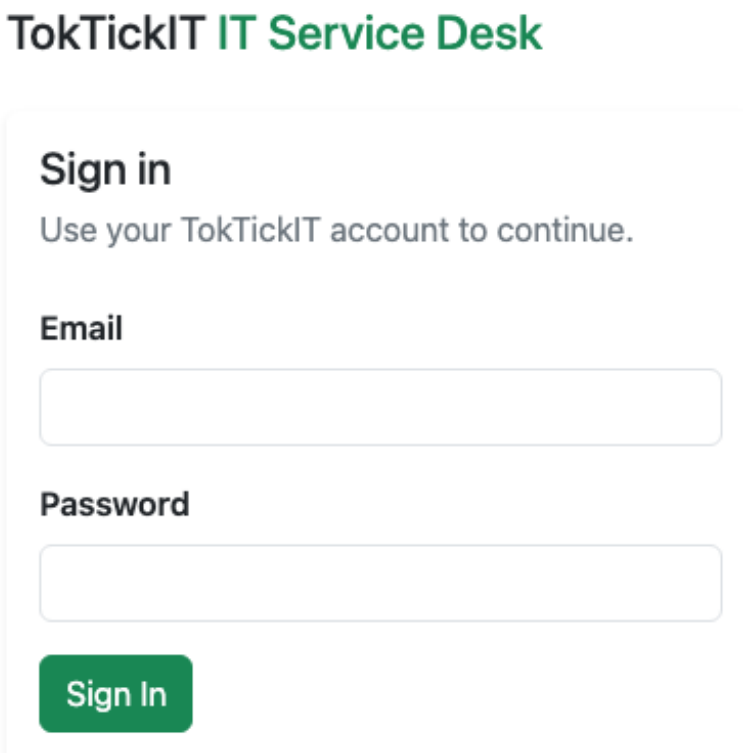
Login - desktop 1280 x 900

Login - tablet 820 x 1000



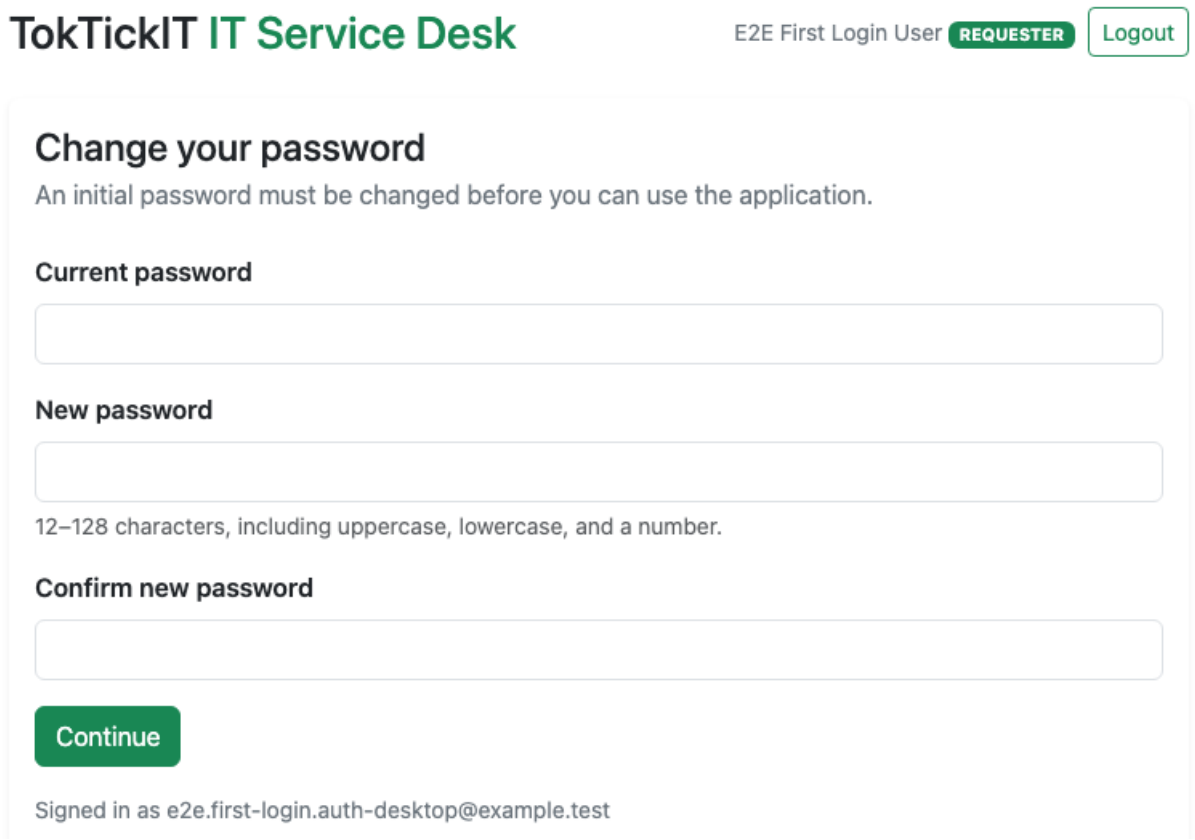
Login - tablet 820 x 1000

Login - mobile 390 x 844



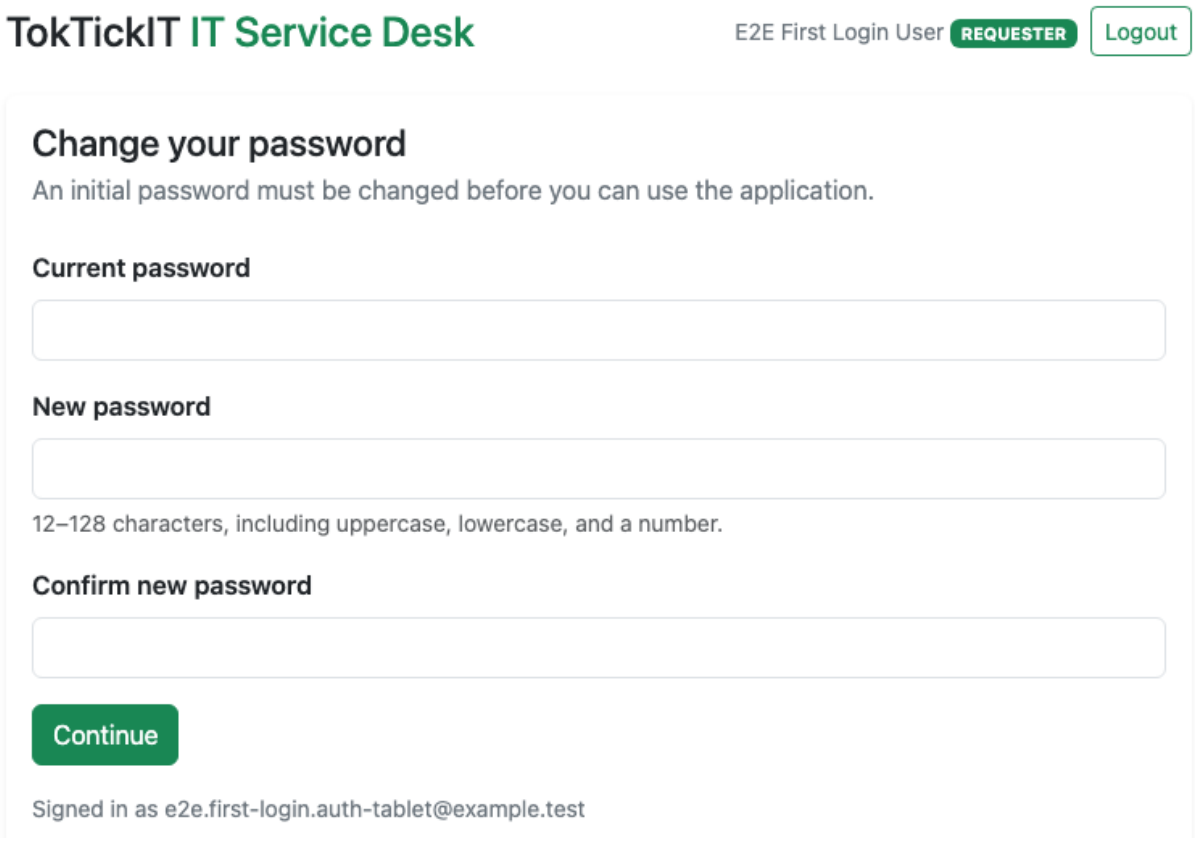
Login - mobile 390 x 844

Change Password - desktop 1280 x 900



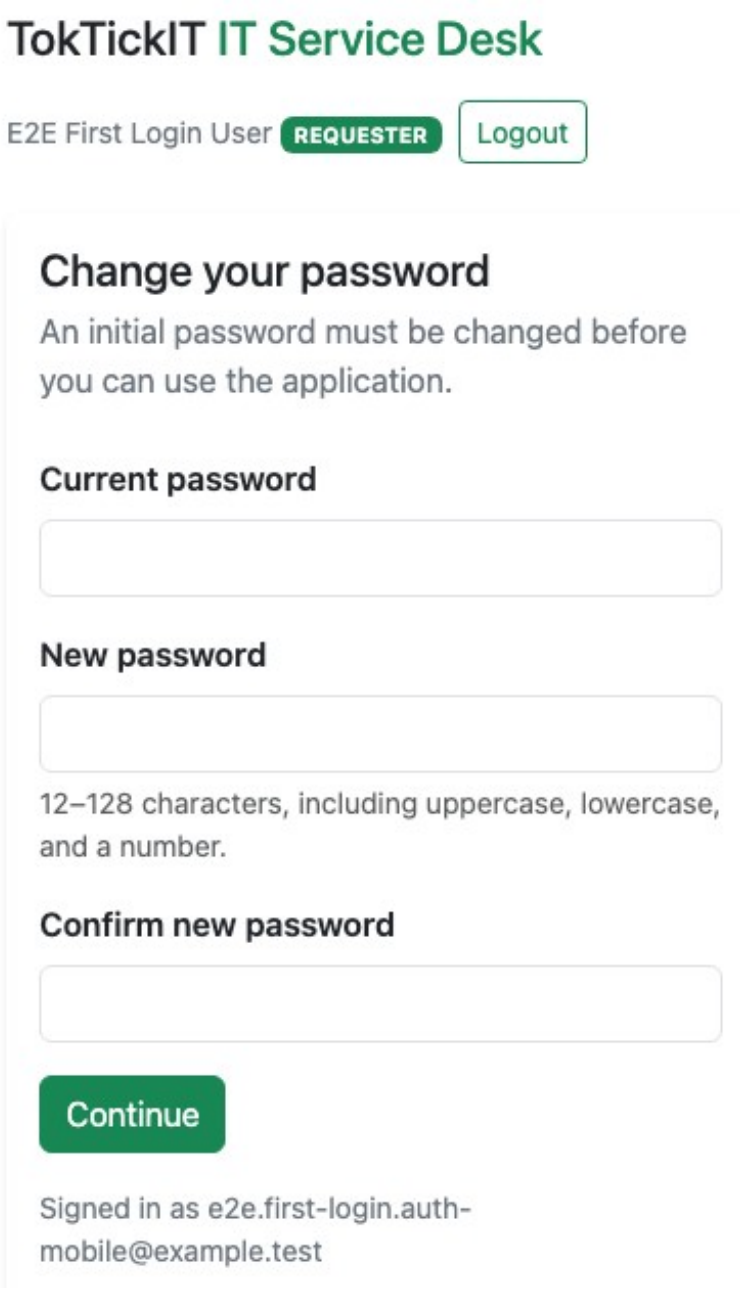
Change Password - desktop 1280 x 900

Change Password - tablet 820 x 1000



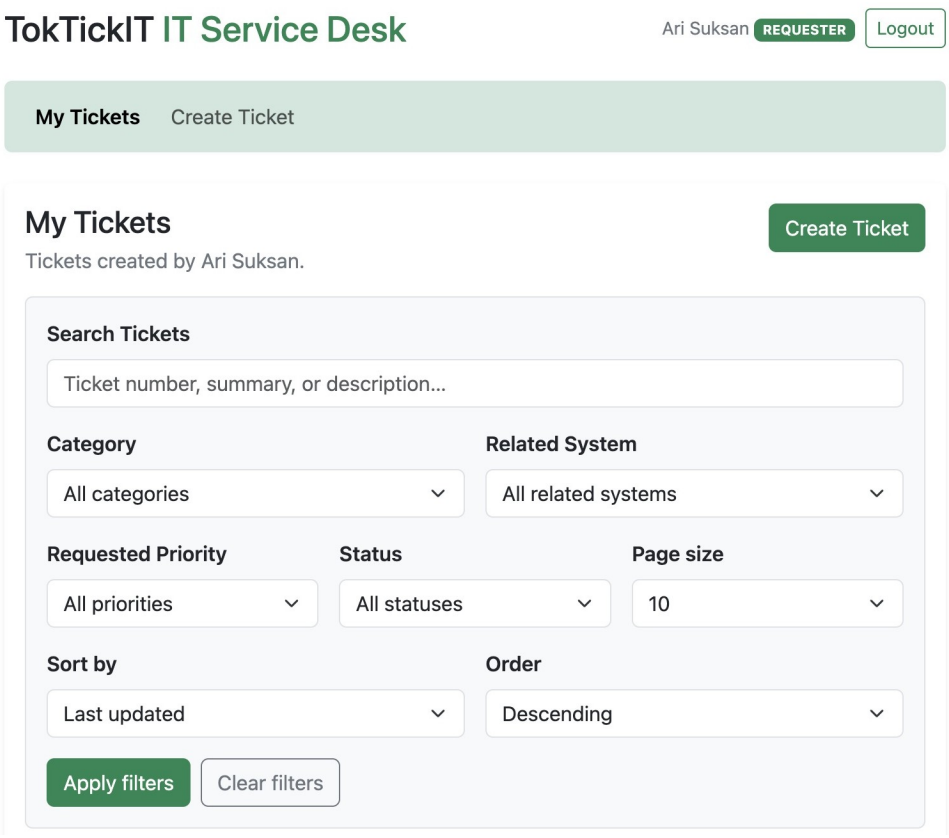
Change Password - tablet 820 x 1000

Change Password - mobile 390 x 844



Change Password - mobile 390 x 844

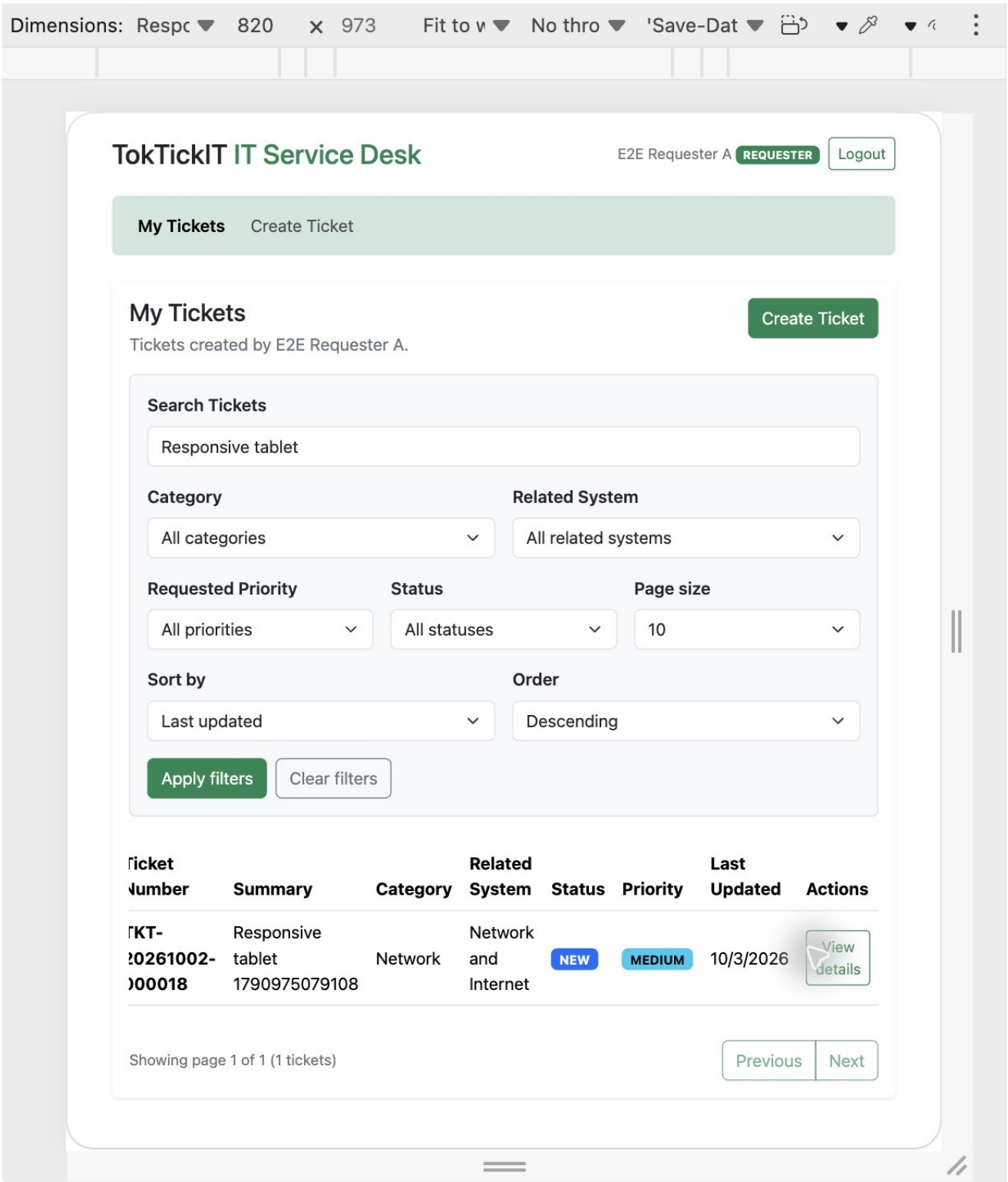
Requester My Tickets desktop



Requester My Tickets - desktop Edge 1176 x 804, controls; native Edge capture from the ac8a3dd application snapshot.

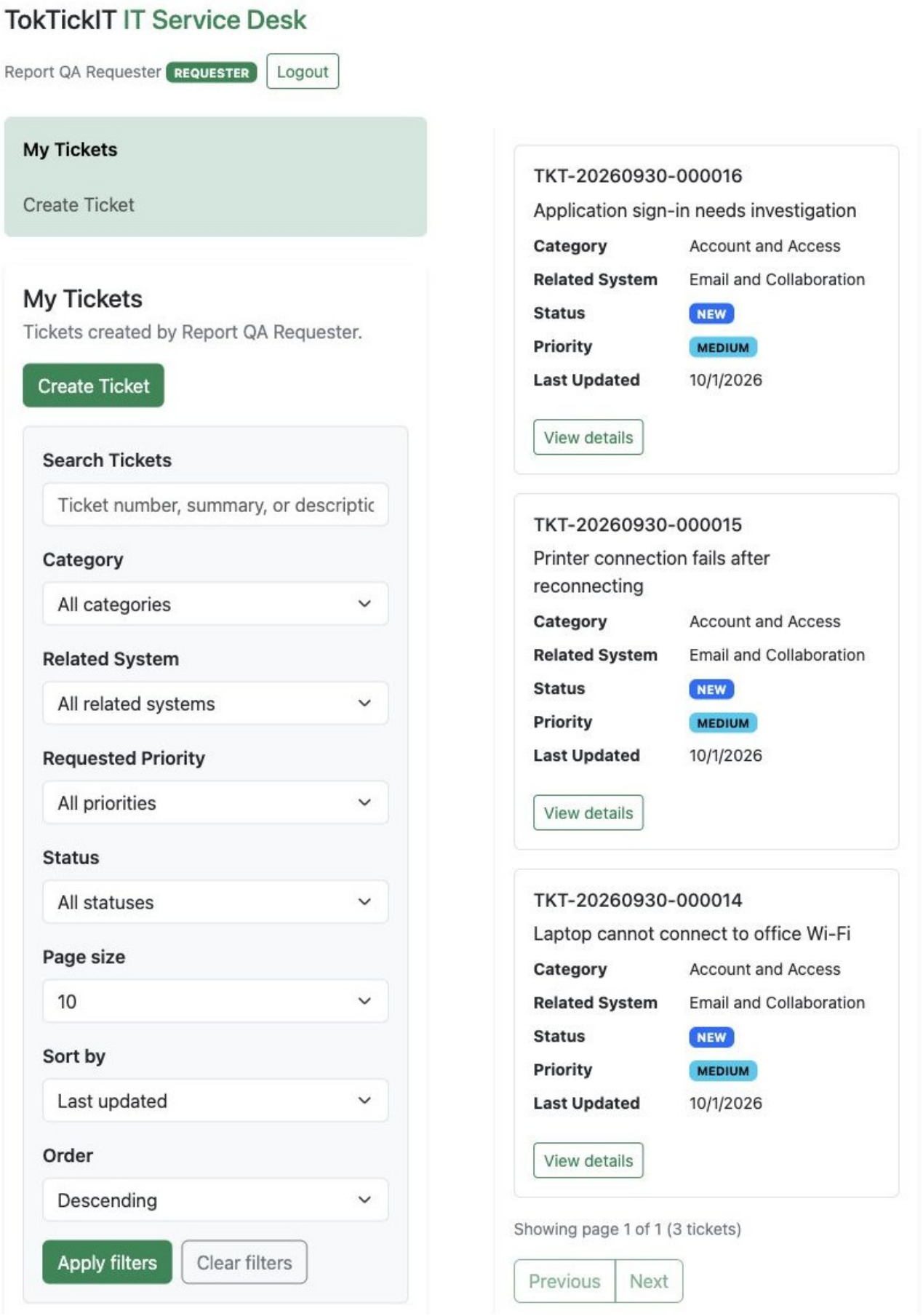
Requester My Tickets - desktop table, right position with complete View details action; native Edge capture from the ac8a3dd application snapshot.

Requester My Tickets - tablet 820 x 973, zoom 100%, table scrolled right (main f6dc6f2)



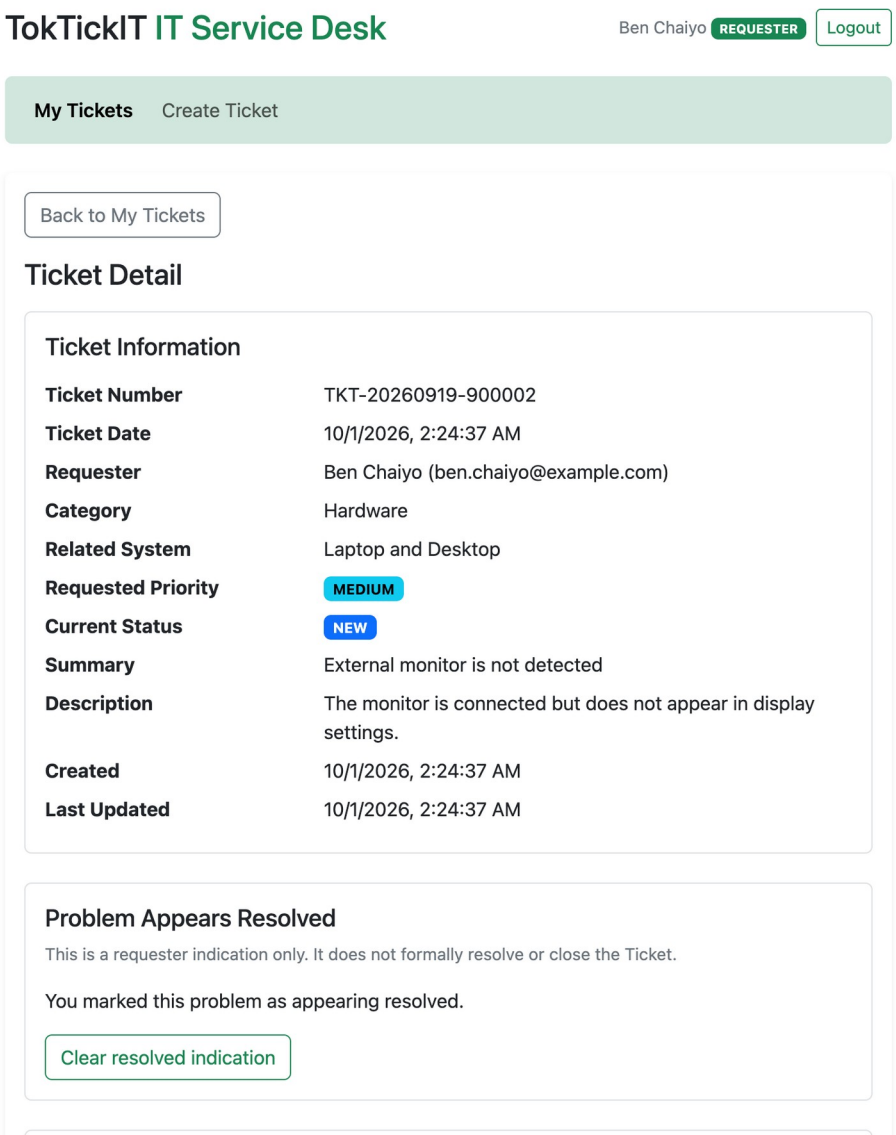
Requester My Tickets - tablet 820 x 973, zoom 100%, table scrolled right (main f6dc6f2)

Requester My Tickets - mobile 390 px full page



Full page: upper screen (left), continuation (right).

Requester Ticket Detail - desktop 1440 px viewport



Requester Ticket Detail - desktop 1440 px viewport

Requester Ticket Detail - tablet 768 px full page — 1/2

TokTickIT IT Service Desk

Ben Chaiyo REQUESTER Logout

My Tickets Create Ticket

Back to My Tickets

Ticket Detail

Ticket Information

Ticket Number	TKT-20260919-900002
Ticket Date	10/1/2026, 2:24:37 AM
Requester	Ben Chaiyo (ben.chaiyo@example.com)
Category	Hardware
Related System	Laptop and Desktop
Requested Priority	MEDIUM
Current Status	NEW
Summary	External monitor is not detected
Description	The monitor is connected but does not appear in display settings.
Created	10/1/2026, 2:24:37 AM
Last Updated	10/1/2026, 2:24:37 AM

Requester Ticket Detail - tablet 768 px full page - segment 1/2

Requester Ticket Detail - tablet 768 px full page — 2/2

Problem Appears Resolved

This is a requester indication only. It does not formally resolve or close the Ticket.

You marked this problem as appearing resolved.

Clear resolved indication

Public Comments

Comments are visible to permitted users and are saved as plain text.

1.
Ben Chaiyo (ben.chaiyo@example.com) 10/1/2026, 2:24:37 AM

I can reproduce this after reconnecting the cable.

Add Public Comment

1–4000 characters. Markup is displayed as text.

Add Public Comment

Attachments

Active files: 0/5

Upload attachment

Choose File | No file chosen | Upload

JPG, JPEG, PNG, WEBP, or PDF; up to 5 MB each and 5 active files.

No attachments for this ticket.

Requester Ticket Detail - tablet 768 px full page - segment 2/2

Requester Ticket Detail - mobile 440 px full page

TokTickIT IT Service Desk

Ben Chaiyo REQUESTER Logout

My Tickets

Create Ticket

Back to My Tickets

Ticket Detail

Ticket Information

Ticket Number
TKT-20260919-900002

Ticket Date
10/1/2026, 2:24:37 AM

Requester
Ben Chaiyo (ben.chaiyo@example.com)

Category
Hardware

Related System
Laptop and Desktop

Requested Priority
MEDIUM

Current Status
NEW

Summary
External monitor is not detected

Description
The monitor is connected but does not appear in display settings.

Created
10/1/2026, 2:24:37 AM

Last Updated
10/1/2026, 2:24:37 AM

Problem Appears Resolved

This is a requester indication only. It does not formally resolve or close the Ticket.

You marked this problem as appearing resolved.

Clear resolved indication

Public Comments

Comments are visible to permitted users and are saved as plain text.

1.
Ben Chaiyo (ben.chaiyo@example.com)
10/1/2026, 2:24:37 AM
I can reproduce this after reconnecting the cable.

Add Public Comment

1–4000 characters. Markup is displayed as text.

Add Public Comment

Attachments

Active files: 0/5

Upload attachment

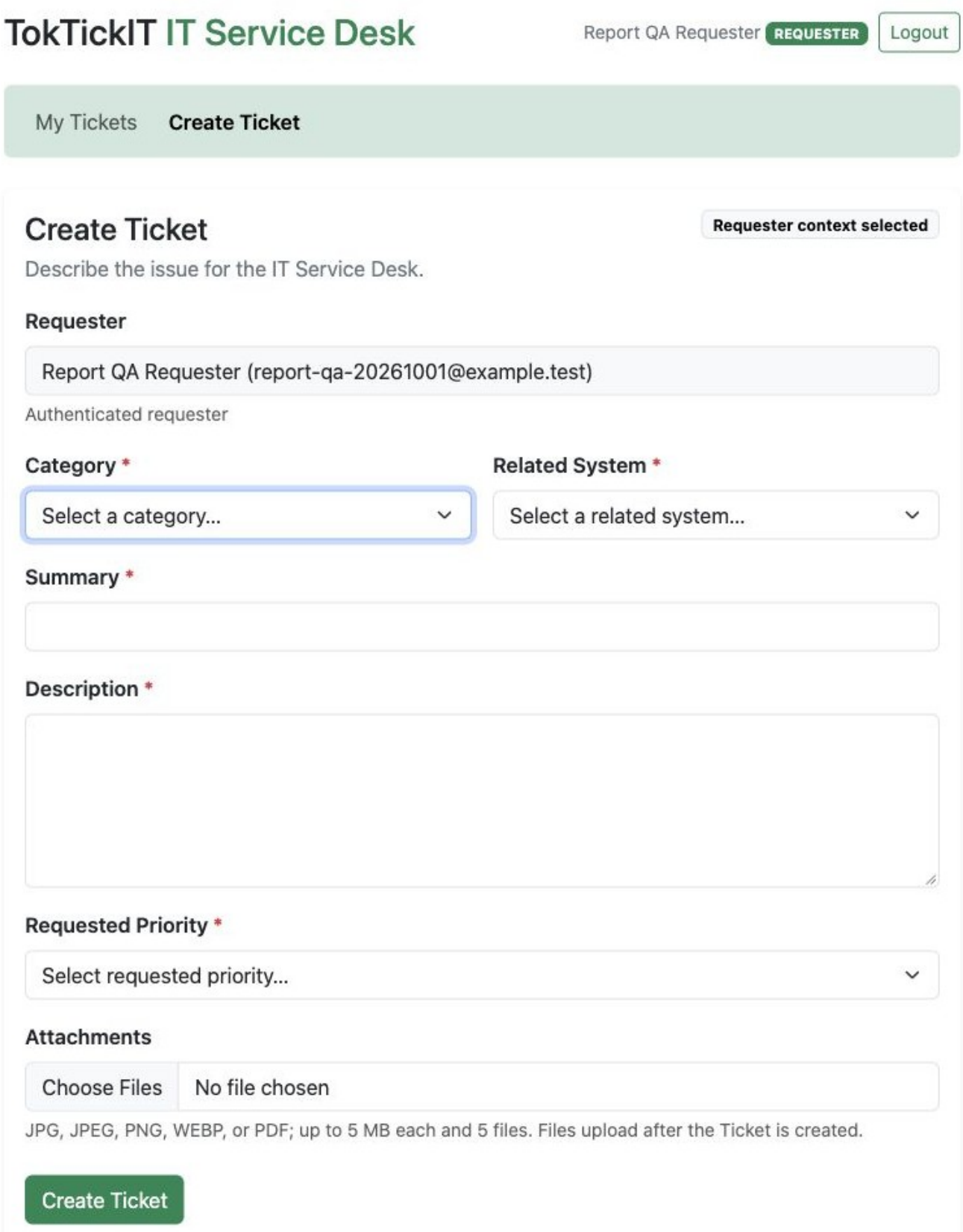
Choose File | No file chosen | Upload

JPG, JPEG, PNG, WEBP, or PDF; up to 5 MB each and 5 active files.

No attachments for this ticket.

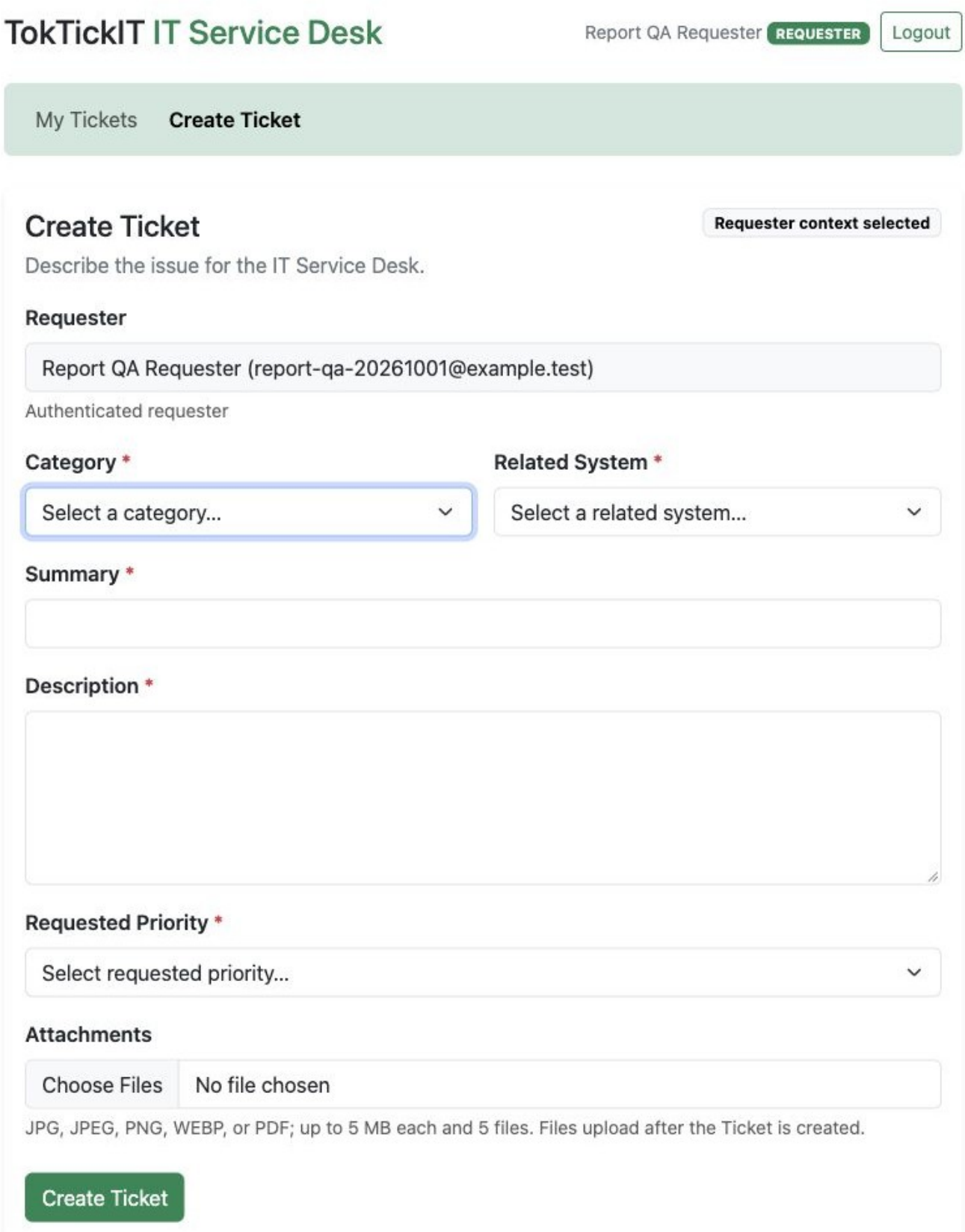
Full page: upper screen (left), continuation (right).

Requester Create Ticket - desktop 1280 px



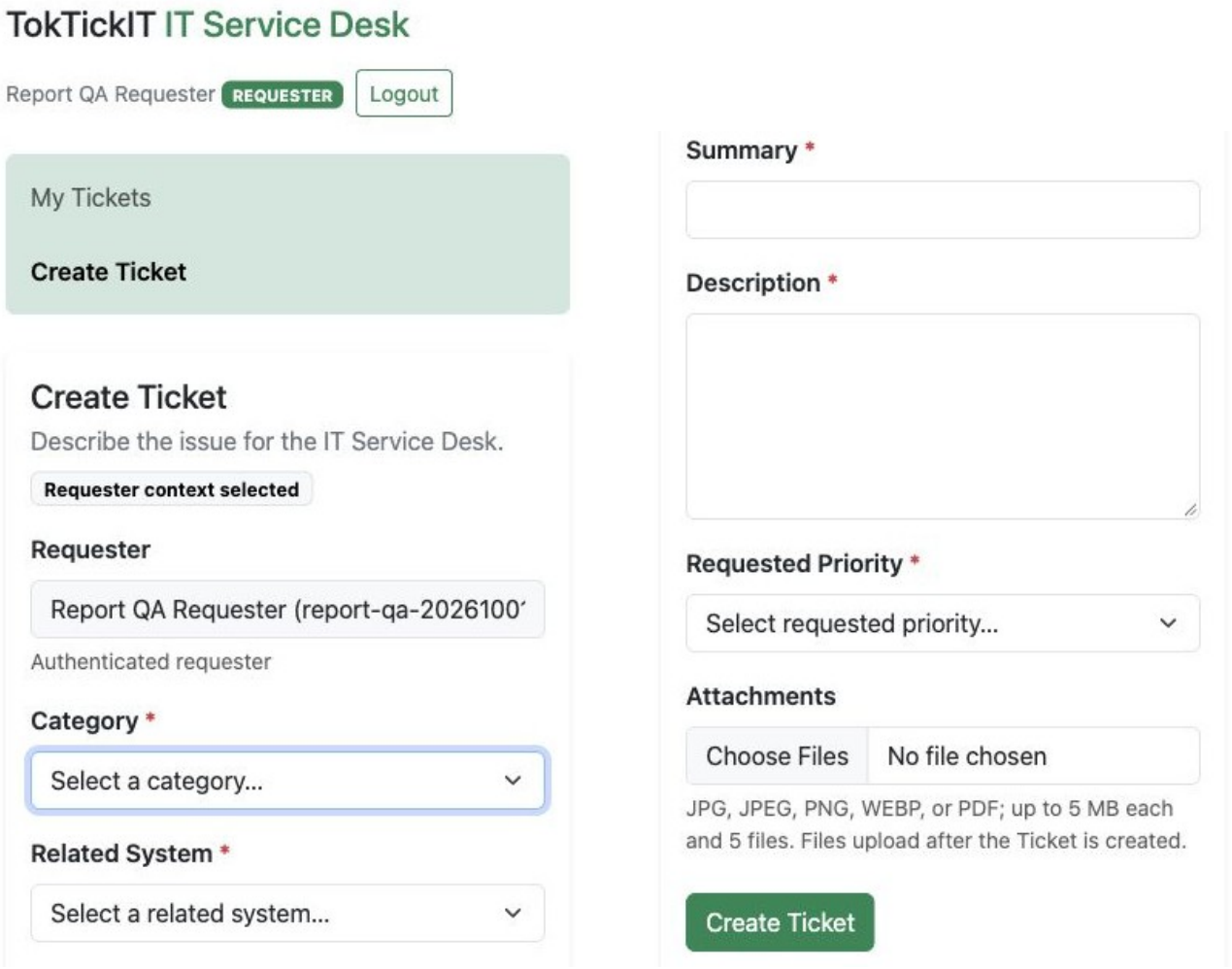
Requester Create Ticket - desktop 1280 px

Requester Create Ticket - tablet 820 px



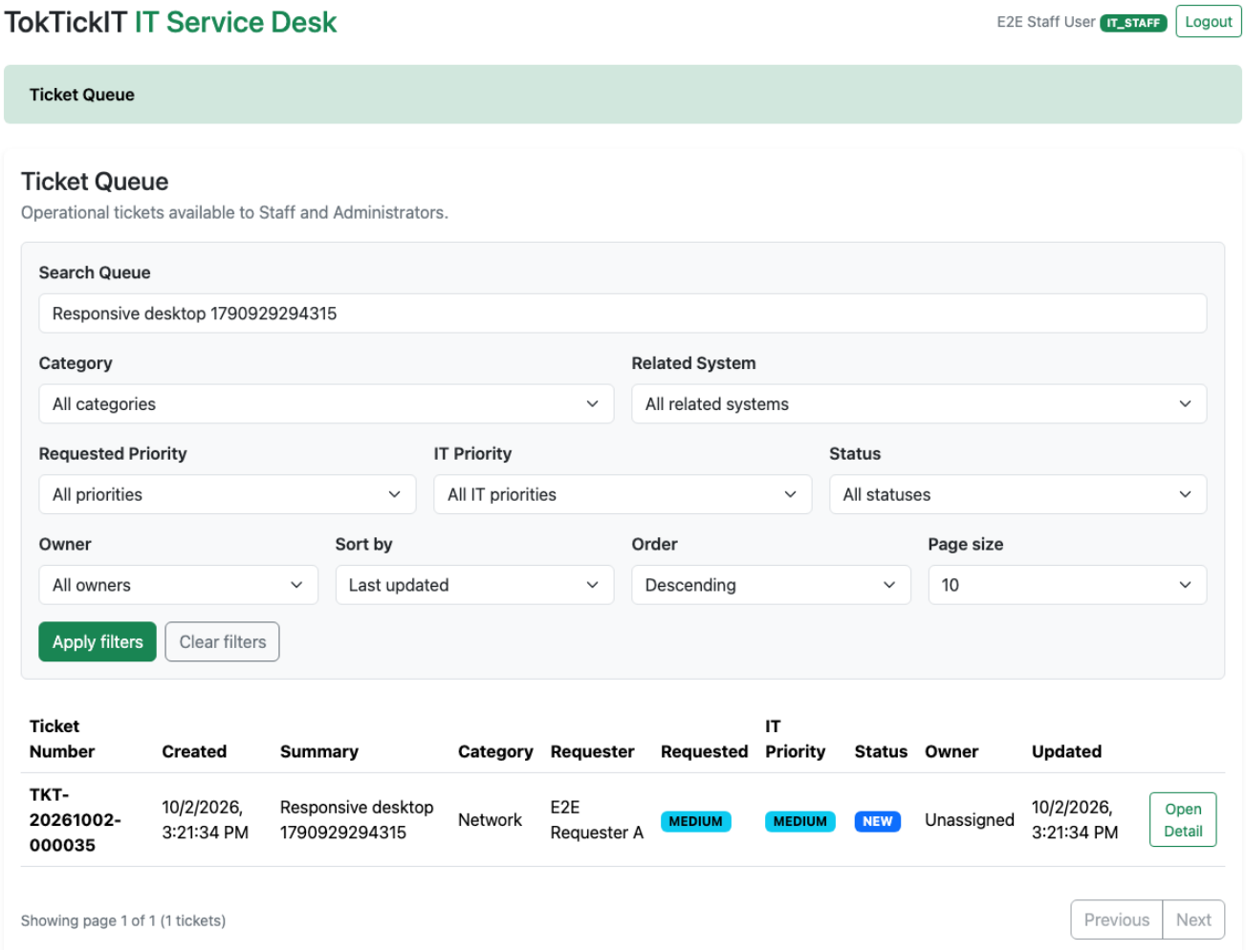
Requester Create Ticket - tablet 820 px

Requester Create Ticket - mobile 390 px full page



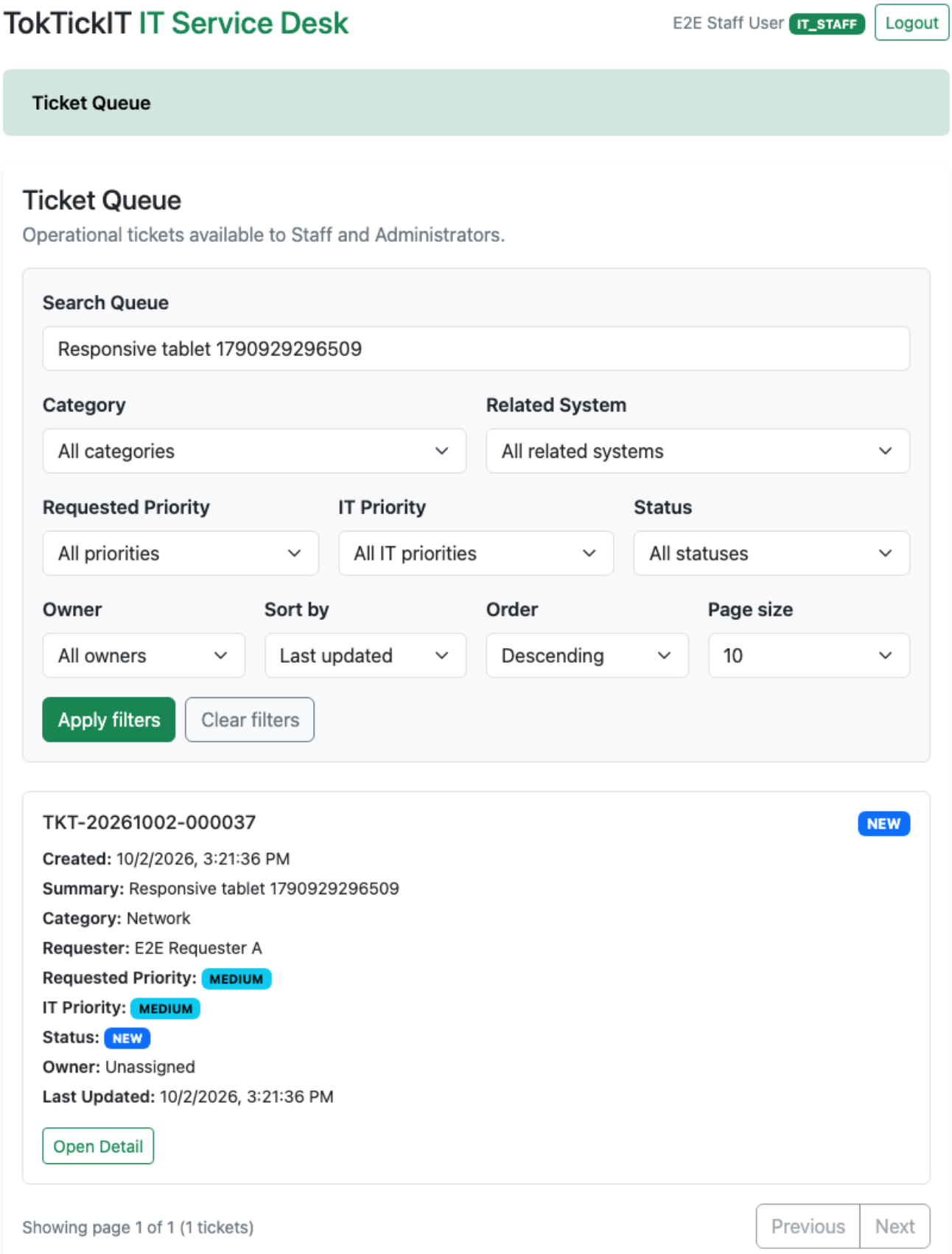
Full page: upper screen (left), continuation (right).

Staff Ticket Queue - desktop 1280 px



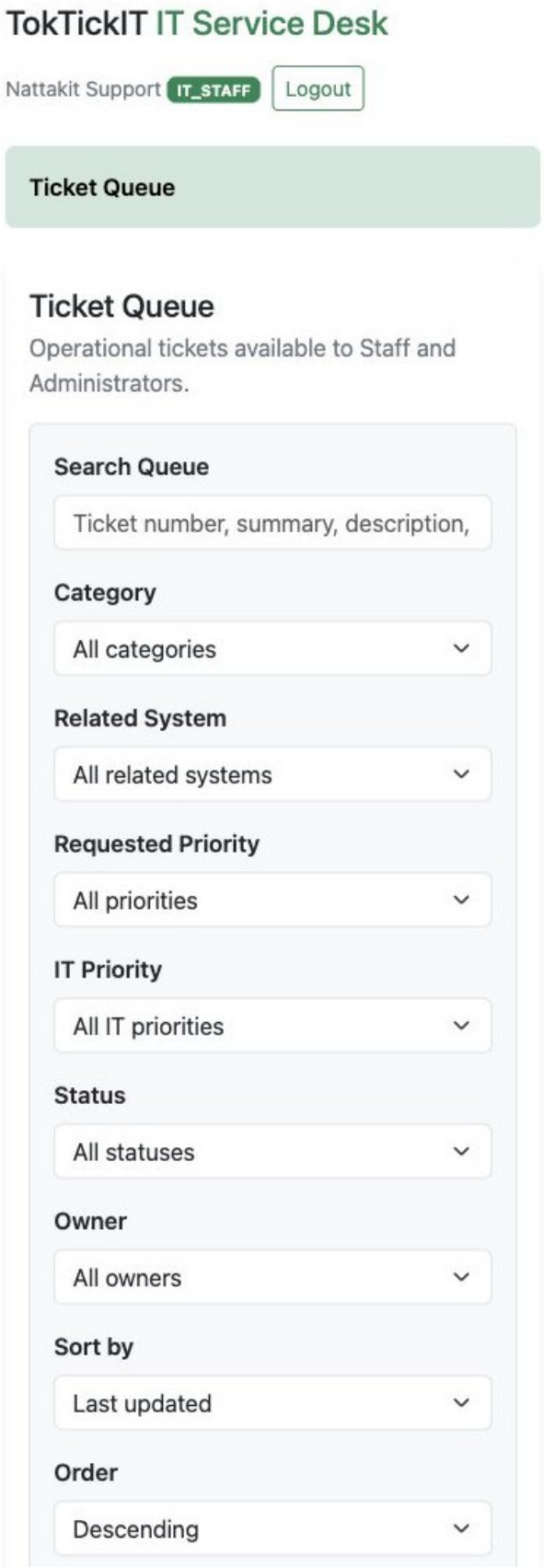
Staff Ticket Queue - desktop 1280 px

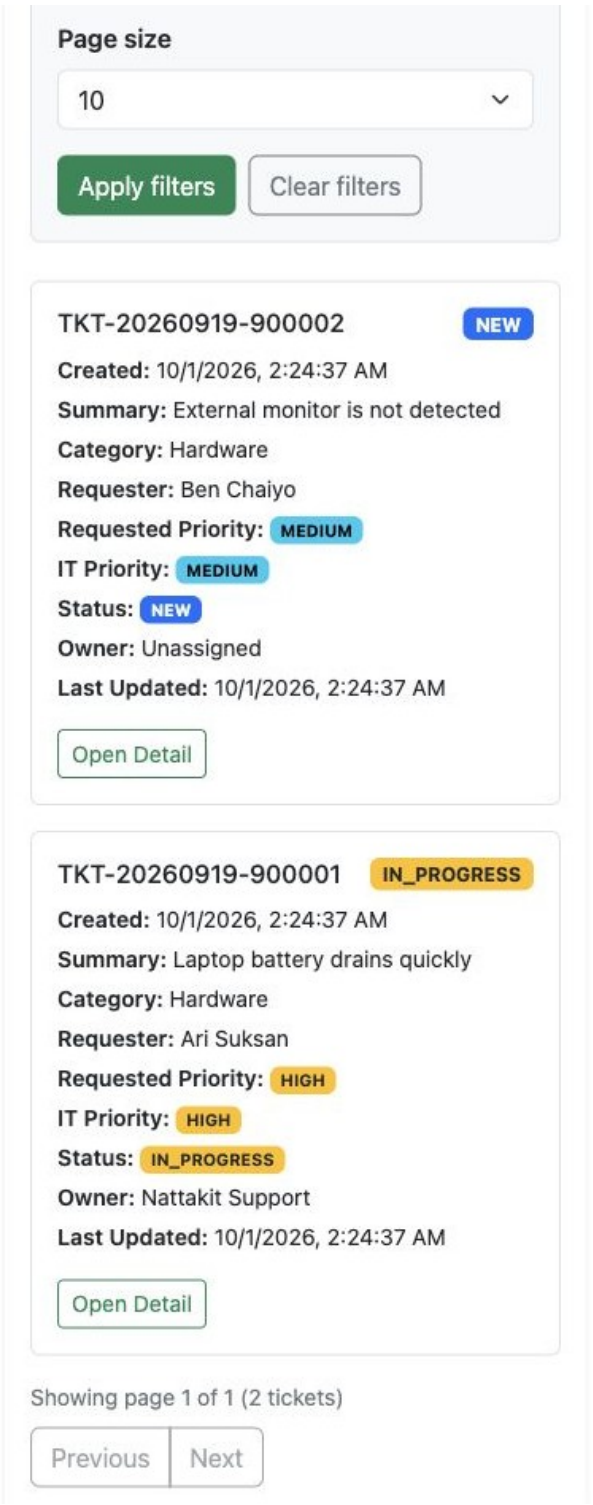
Staff Ticket Queue - tablet 820 px



Staff Ticket Queue - tablet 820 px

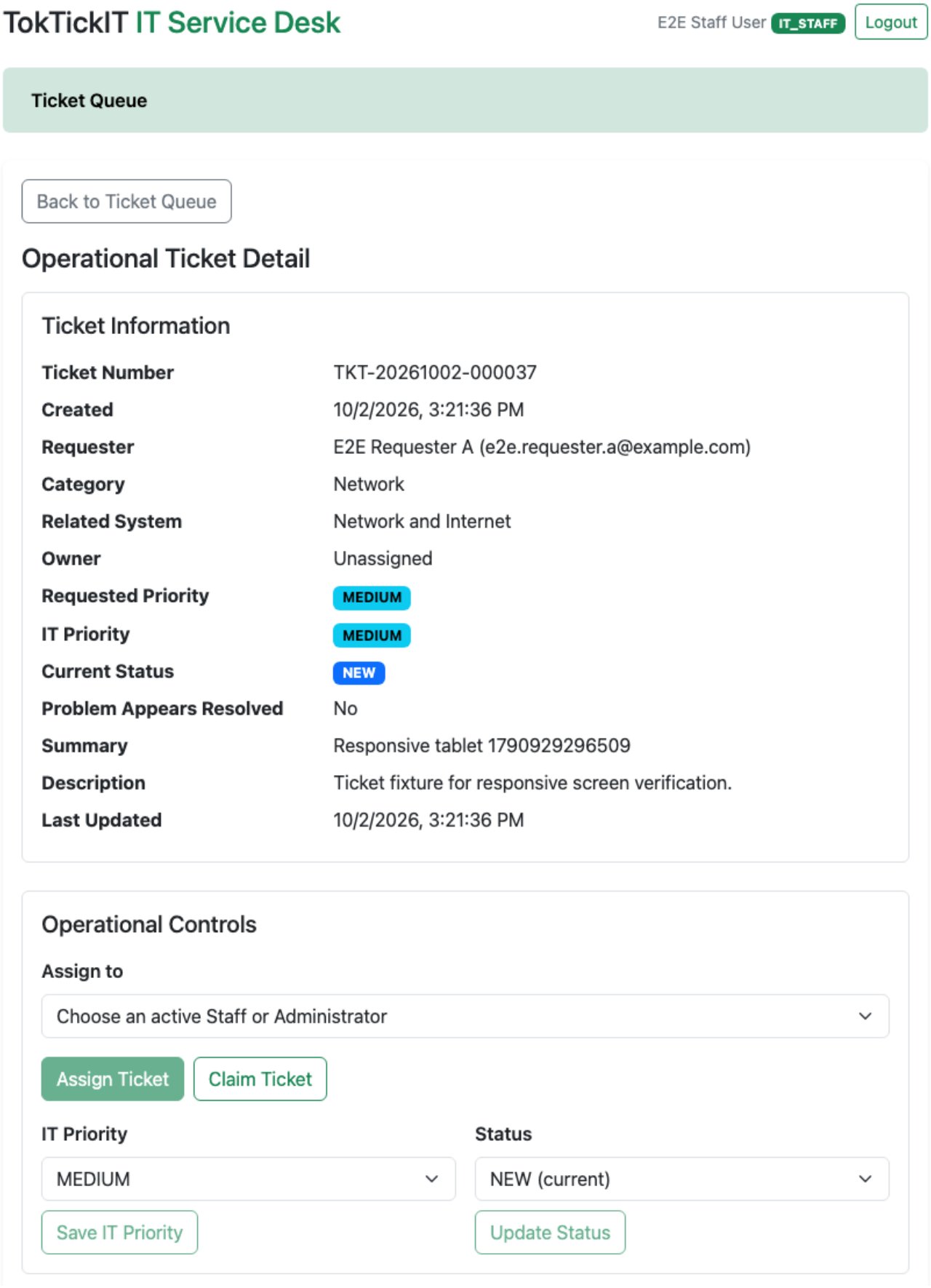
Staff Ticket Queue - mobile 390 px full-page local capture





Full page: upper screen (left), continuation (right).

Staff Ticket Detail - tablet 820 px full page — 1/2



Staff Ticket Detail - tablet 820 px full page - segment 1/2

Staff Ticket Detail - tablet 820 px full page — 2/2

Public Comments

Visible to the requester and support staff.

No public comments yet.

Add a public comment

Post Public Comment

Internal Notes

Private to IT Staff and Administrators; not visible to Requesters.

No internal notes yet.

Add an internal note

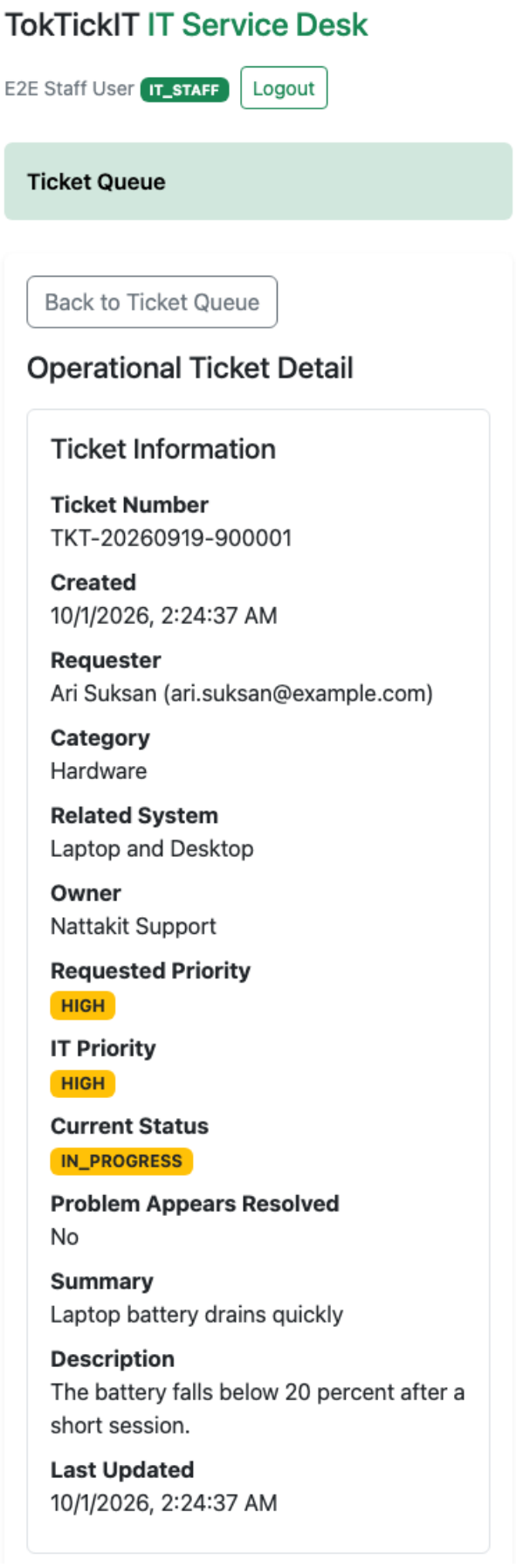
Post Internal Note

Attachments

No attachments for this Ticket.

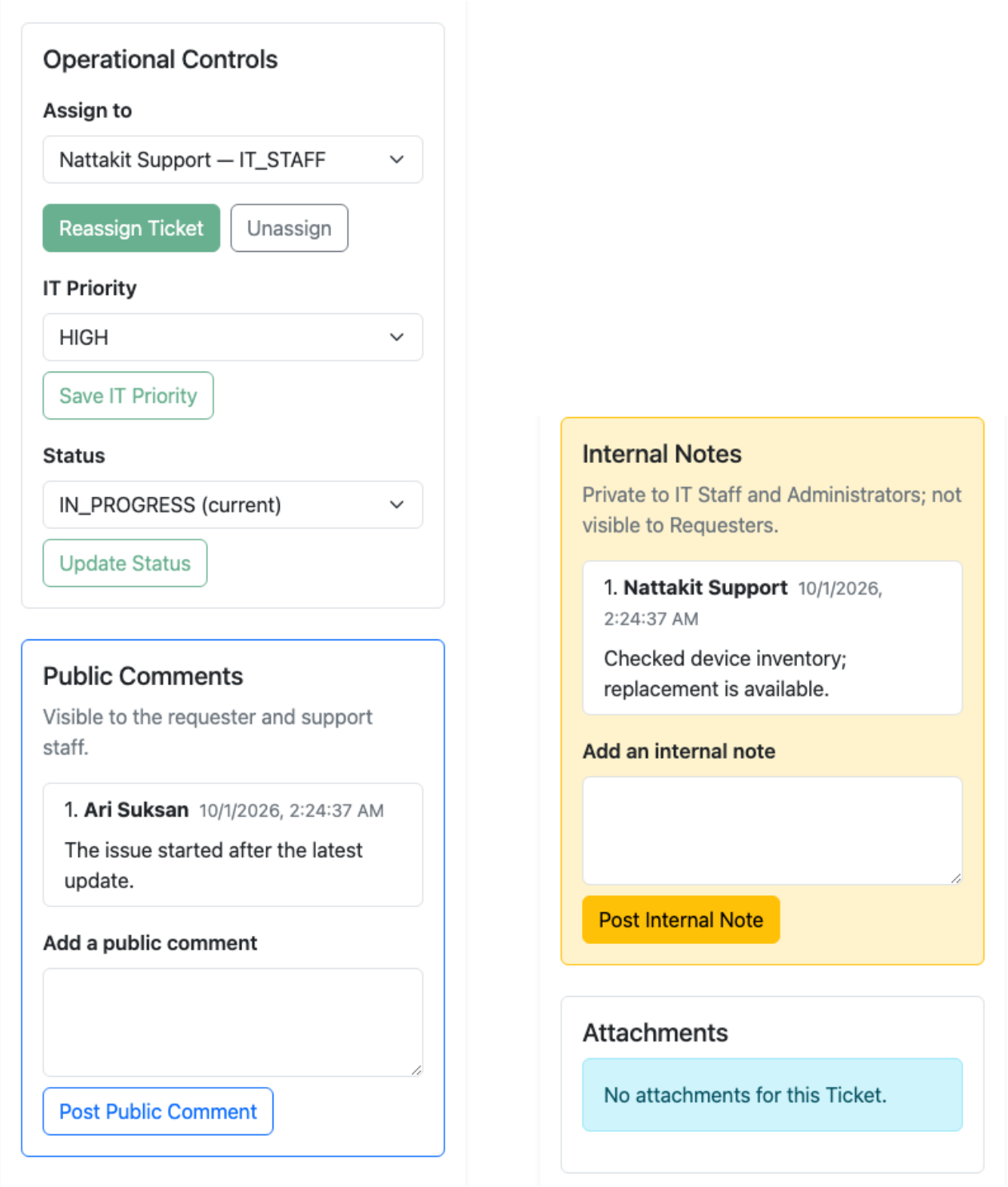
Staff Ticket Detail - tablet 820 px full page - segment 2/2

Staff Ticket Detail - mobile 390 px full page — upper screen



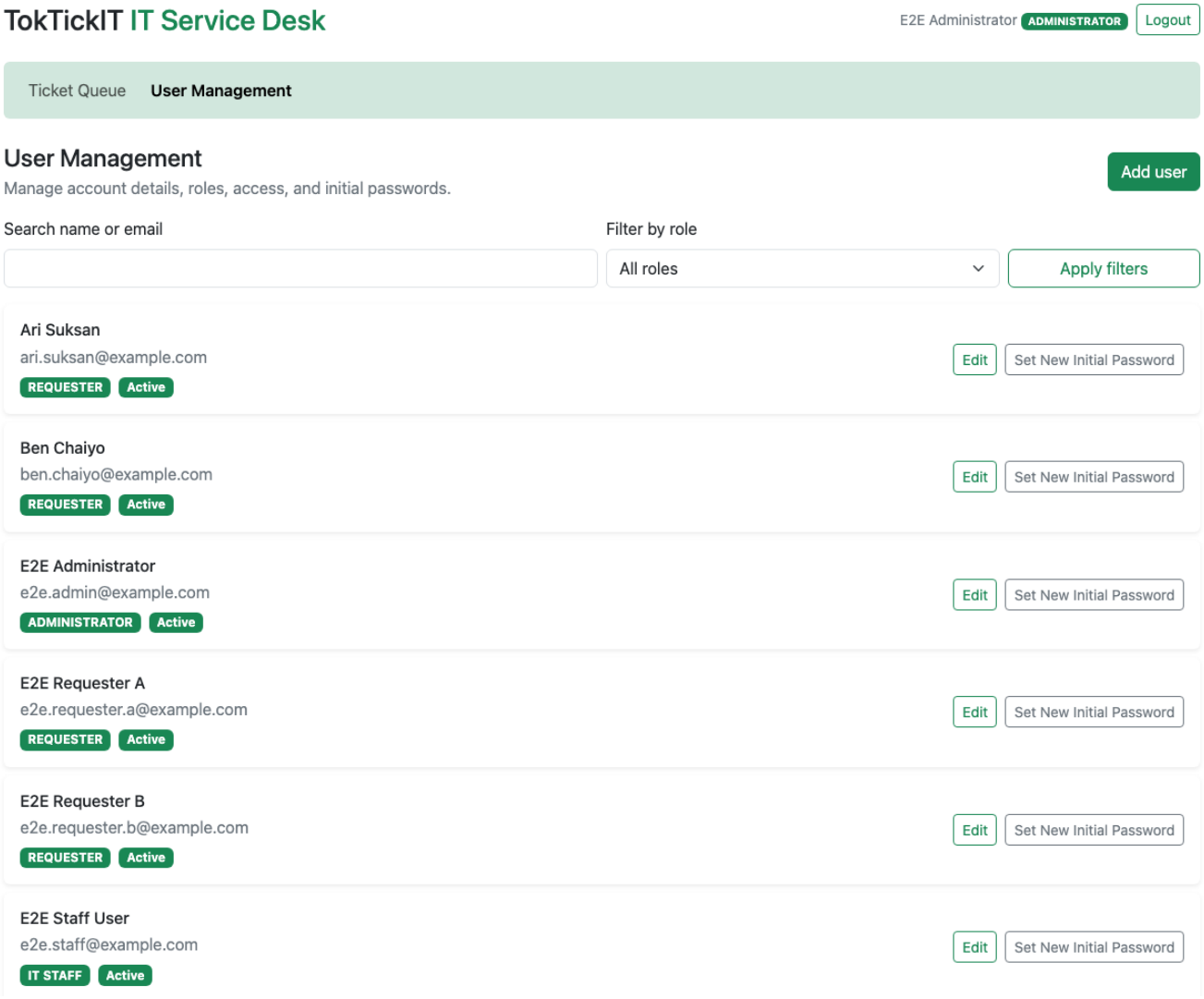
Same actual mobile capture: shell and complete Ticket information.

Staff Ticket Detail - mobile 390 px full page — continued, left then right



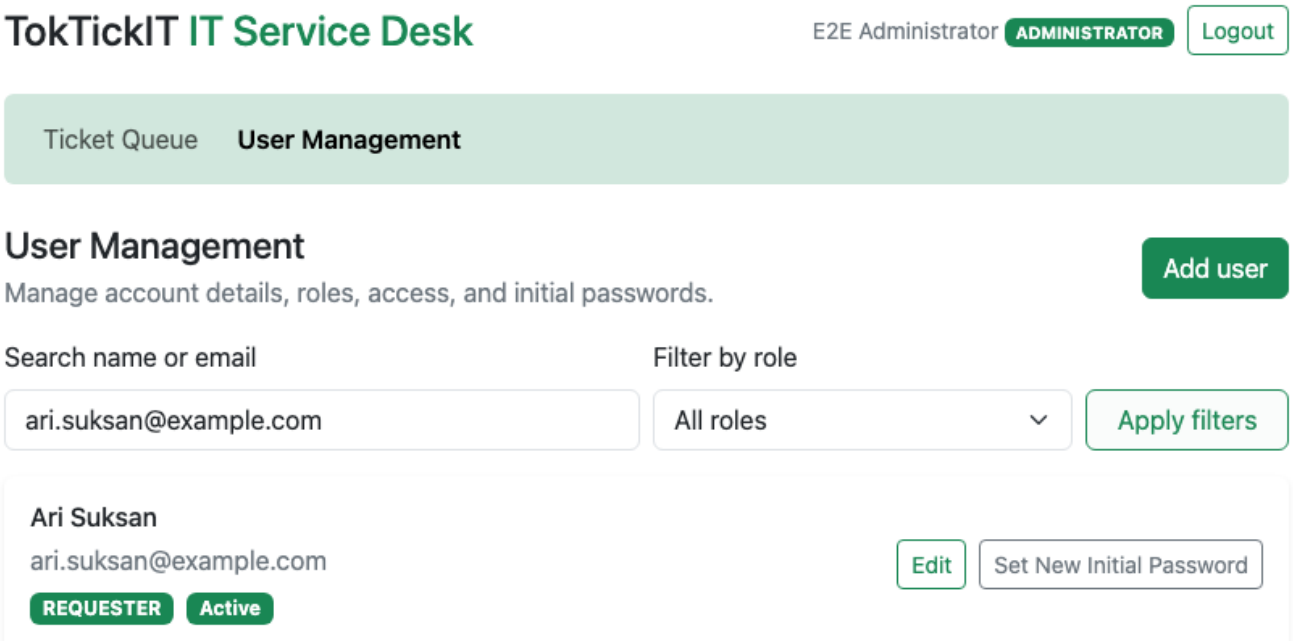
Continuation of the same capture: full Operational Controls and Public Comments (left); complete Internal Note input, Post Internal Note button and Attachments (right). No capture pixels are omitted between panels.

Administrator User Management - desktop viewport



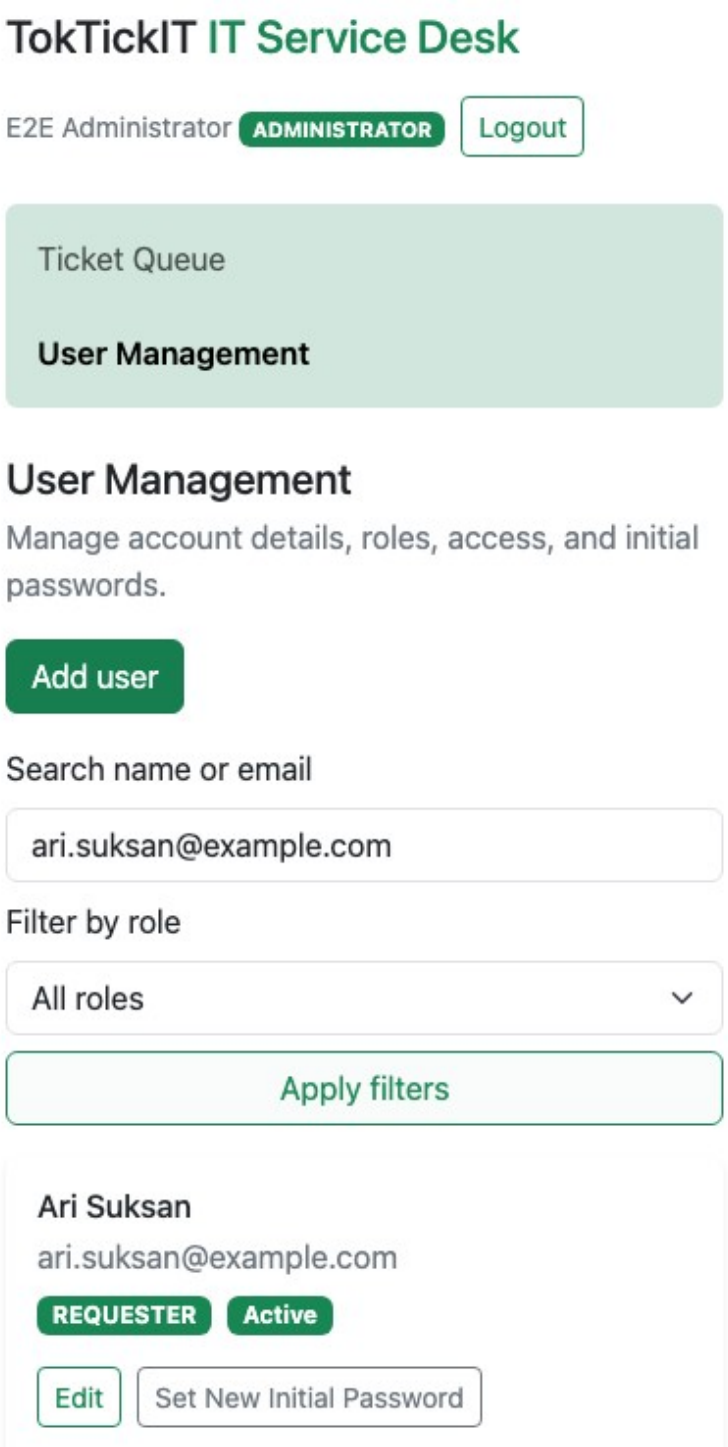
Administrator User Management - desktop viewport

Administrator User Management - tablet 820 px



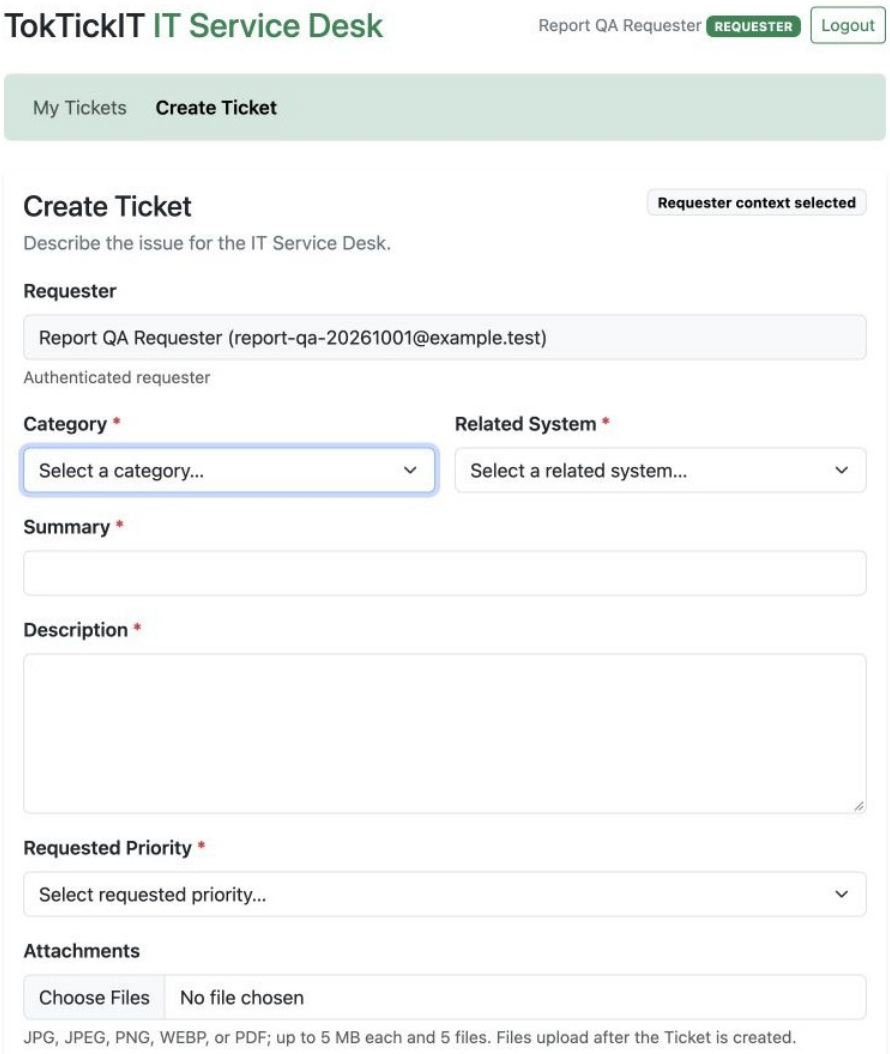
Administrator User Management - tablet 820 px

Administrator User Management - mobile 390 px full page



Administrator User Management - mobile 390 px full page

Actual keyboard focus



Actual browser capture after pressing Tab: the Category field has the native blue focus ring. No annotation is added.

Responsive automated assertions compare document.documentElement.scrollWidth and document.body.scrollWidth with the viewport width. These assertions demonstrate no page-level horizontal overflow; they do not by themselves prove the absence of clipping, overlap or hidden controls. The full-page screenshots above provide a separate visual record. The wider My Tickets table uses an internal horizontal scroll area.